

ISC Class XII Mathematics

Chapter 4 Determinants

167 Solved Problems – Numericals & Proofs

Step-by-Step ISC Board Solutions

Topics Covered

- I. 4.1 – Evaluation of Determinants: Minors and Cofactors
- II. 4.2 – Properties of Determinants and Proving Identities
- III. 4.3 – Applications of Determinants: Area, Collinearity and Equation of a Line
- IV. 4.4 – Adjoint and Inverse of a Matrix
- V. 4.5 – Solving Systems of Linear Equations Using Matrices

Compiled Answer Key – Step-by-step ISC board-style solutions

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Index of Problems

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
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Exercise 4.1 – Evaluation of Determinants: Minors and Cofactors

1	(see parts below)	1	Numerical	4.1
(i)	Evaluate the determinant: ...	1	Numerical	4.1
(ii)	Evaluate the determinant: ...	1	Numerical	4.1
(iii)	Evaluate the determinant: ...	2	Numerical	4.1
2	(see parts below)	1	Numerical	4.1
(i)	Evaluate the determinant: ...	1	Numerical	4.1
(ii)	Evaluate the determinant: ...	3	Numerical	4.1
(iii)	Evaluate the determinant: ...	3	Numerical	4.1
3	(see parts below)	2	Numerical	4.1
(i)	Find the value of $ \sin A \quad -\sin B \cos A \quad \cos B $, where ...	2	Numerical	4.1
(ii)	Find the value of $ \cos A \quad \sin A - \sin B \quad \cos B $, where ...	2	Numerical	4.1
4	(see parts below)	4	Numerical	4.1
(i)	If $A = \begin{bmatrix} 5 & 6 & -3 & -4 & 3 & 2 & -4 & -7 & 3 \end{bmatrix}$, then write the ...	4	Numerical	4.1

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(ii)	If A_{ij} is the cofactor of the element a_{ij} of the determinant ...	4	Numerical	4.1
5	(see parts below)	5	Numerical	4.1
(i)	Find the value of x if: $ 2x + 5 \quad 35x + 2 \quad 9 = 0$	5	Numerical	4.1
(ii)	Find the value of x if: $ x + 2 \quad 3x + 5 \quad 4 = 3$	5	Numerical	4.1
(iii)	Find the value of x if: ...	5	Numerical	4.1
6	(see parts below)	5	Numerical	4.1
(i)	If ...	5	Numerical	4.1
(ii)	If ...	5	Numerical	4.1
7	(see parts below)	5	Numerical	4.1
(i)	If $x \in \mathbf{N}$ and ...	5	Numerical	4.1
(ii)	If $x \in \mathbf{I}$ and ...	5	Numerical	4.1
8	If $A = \begin{bmatrix} 1 & 24 & 2 \end{bmatrix}$, then show that $ 2A = 4 A $.	6	Proof	4.1
9	(see parts below)	3	Numerical	4.1
(i)	Evaluate the determinant: $ 0 \quad -1 \quad 21 \quad 0 \quad 3 - 2 \quad -3 \quad 0 $	3	Numerical	4.1
(ii)	Evaluate the determinant: $ 3 \quad -4 \quad 51 \quad 1 \quad -22 \quad 3 \quad 1 $	3	Numerical	4.1
(iii)	Evaluate the determinant: ...	3	Numerical	4.1
10	Prove that: ...	3	Proof	4.1
11	Show that the value of the determinant ...	2	Proof	4.1
12	Using cofactors of elements of second row, evaluate ...	7	Numerical	4.1

Exercise 4.2 – Properties of Determinants and Proving Identities

1	(see parts below)	8	Numerical	4.2
(i)	Without expanding, find the value of: ...	8	Numerical	4.2
(ii)	Without expanding, find the value of: ...	8	Numerical	4.2
(iii)	Without expanding, find the value of: ...	8	Numerical	4.2
2	(see parts below)	8	Numerical	4.2
(i)	Without expanding, find the value of: ...	8	Numerical	4.2
(ii)	Without expanding, find the value of: ...	8	Numerical	4.2
3	(see parts below)	8	Numerical	4.2
(i)	Without expanding, find the value of: ...	8	Numerical	4.2

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(ii)	Without expanding, find the value of: ...	8	Numerical	4.2
4	(see parts below)	8	Numerical	4.2
(i)	Without expanding, find the value of: ...	8	Numerical	4.2
(ii)	Without expanding, find the value of: ...	8	Numerical	4.2
5	If A is a square matrix of order 2 and $ A = -5$, find the value of $ 3A $.	6	Numerical	4.2
6	If A is a square matrix of order 3 and $ A = 4$, then write the value of $ 2A $.	6	Numerical	4.2
7	If A is a square matrix of order 3 and $ A = 5$, find the value of $ 3A $.	6	Numerical	4.2
8	If A is a square matrix of order 3 and $ A = -2$, find the value of $ -5A $.	6	Numerical	4.2
9	If A is a matrix of order 3×3 and its determinant is 4, then find $ 3A $.	6	Numerical	4.2
10	If A is a square matrix of order 3 and $ 3A = k A $, then find the value of k .	6	Numerical	4.2
11	(see parts below)	9	Numerical	4.2
(i)	If A is a square matrix such that $ A = 5$, write the value of $ AA^T $.	9	Numerical	4.2
(ii)	If A is a square matrix satisfying $AA' = I$, write the value of $ A $.	9	Numerical	4.2
12	If A is a square matrix of order 3 such that $A^2 = 2A$, then find the value of $ A $.	9	Numerical	4.2
13	If $A = \begin{bmatrix} 1 & 23 & -1 \end{bmatrix}$ and ...	9	Numerical	4.2
14	If $A = \begin{bmatrix} x & 22 & x \end{bmatrix}$ and $ A^3 = 125$, find the value(s) of ...	9	Numerical	4.2
15	Using properties of determinants, solve the following equation for x : ...	5	Numerical	4.2
16	Show that one root of the equation ...	5	Proof	4.2
17	Without expanding, prove that ...	10	Proof	4.2
18	Without expanding, prove that ...	10	Proof	4.2
19	(see parts below)	5	Numerical	4.2
(i)	Use properties of determinants to solve for x : ...	5	Numerical	4.2
(ii)	Use properties of determinants to solve for x : ...	5	Numerical	4.2
20	If $s = a + b + c$, then prove that ...	10	Proof	4.2
21	If a, b and c are in A.P., then find the value of ...	8	Numerical	4.2

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
22	Using the properties of determinant, prove that ...	10	Proof	4.2
23	Using properties of determinants, prove that: ...	10	Proof	4.2
24	(see parts below)	10	Proof	4.2
(i)	Prove that $\begin{vmatrix} 1+a & b & ca & 1+b & ca & b & 1+c \end{vmatrix} = 1+a+b+c$	10	Proof	4.2
(ii)	Prove that ...	10	Proof	4.2
(iii)	Prove that $\begin{vmatrix} y+k & y & yy & y+k & yy & y & y+k \end{vmatrix} = k^2(3y+k)$	10	Proof	4.2
(iv)	Prove that ...	10	Proof	4.2
25	Show that: ...	10	Proof	4.2
26	(see parts below)	10	Proof	4.2
(i)	Prove that: ...	10	Proof	4.2
(ii)	Prove that: ...	10	Proof	4.2
27	(see parts below)	10	Proof	4.2
(i)	Prove that: ...	10	Proof	4.2
(ii)	Prove that: $\begin{vmatrix} 1 & x & y & 1 & x+y & y & 1 & x & x+y \end{vmatrix} = xy$	10	Proof	4.2
(iii)	Prove that: ...	10	Proof	4.2
28	Prove that: ...	10	Proof	4.2
29	Prove that: ...	10	Proof	4.2
30	Prove that: ...	10	Proof	4.2
31	Prove that: ...	10	Proof	4.2
32	Prove that: ...	10	Proof	4.2
33	Prove that: ...	10	Proof	4.2
34	(see parts below)	10	Proof	4.2
(i)	Prove that: ...	10	Proof	4.2
(ii)	Prove that: ...	10	Proof	4.2
35	Prove that: ...	10	Proof	4.2
36	Prove that: ...	10	Proof	4.2
37	Prove that: ...	10	Proof	4.2
38	If a, b, c are all non-zero and: ...	10	Proof	4.2
39	Without expanding any of the determinants given below, prove that: ...	10	Proof	4.2
40	(see parts below)	5	Numerical	4.2
(i)	Using properties of determinants, solve the following equation for x : ...	5	Numerical	4.2

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(ii)	Using properties of determinants, solve the following equation for x : ...	5	Numerical	4.2
(iii)	Using properties of determinants, solve the following equation for x : ...	5	Numerical	4.2
(iv)	Using properties of determinants, solve the following equation for x : ...	5	Numerical	4.2
41	If a, b, c are all different and: ...	10	Proof	4.2
42	In a triangle ABC , if: ...	10	Proof	4.2

Exercise 4.3 – Applications of Determinants: Area, Collinearity and Equation of a Line

1	(see parts below)	11	Numerical	4.3
(i)	Using determinants, find the area of the triangle whose vertices are $(3, 8)$, $(-4, 2)$...	11	Numerical	4.3
(ii)	Using determinants, find the area of the triangle whose vertices are $(2, 7)$, $(1, 1)$...	11	Numerical	4.3
2	Show that the points $(1, 0)$, $(6, 0)$, $(0, 0)$ are collinear.	12	Proof	4.3
3	Using determinants, determine whether the points $P(a, b + c)$, $Q(b, c + a)$ and ...	12	Numerical	4.3
4	Using determinants, find the value of λ so that the points $(\lambda, 7)$...	12	Numerical	4.3
5	Using determinants, find the equation of the line passing through the points $(1, 2)$ and ...	13	Numerical	4.3
6	Find the value(s) of k if the area of the triangle with vertices $(k, 0)$, $(4, 0)$ and ...	11	Numerical	4.3
7	Find the value(s) of x if the area of the triangle with vertices $(x, 4)$, $(2, -6)$ and ...	11	Numerical	4.3
8	If the points $(a, 0)$, $(0, b)$ and $(1, 1)$ are collinear, prove that $a + b = ab$.	12	Proof	4.3

Exercise 4.4 – Adjoint and Inverse of a Matrix

1	(see parts below)	14	Numerical	4.4
(i)	For what value of x , is the following matrix singular? ...	14	Numerical	4.4
(ii)	If $0 < x < \pi$ and the matrix $\begin{bmatrix} \cos x & 13 & 4 \cos x \end{bmatrix}$ is ...	14	Numerical	4.4

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
2	(see parts below)	14	Numerical	4.4
(i)	Determine if the following matrix is non-singular: ...	14	Numerical	4.4
(ii)	Determine if the following matrix is non-singular: ...	14	Numerical	4.4
3	(see parts below)	15	Numerical	4.4
(i)	Write the adjoint of the matrix: $\begin{bmatrix} 2 & -14 & 3 \end{bmatrix}$	15	Numerical	4.4
(ii)	Write the adjoint of the matrix: $\begin{bmatrix} 3 & 0 & -2 & -5 \end{bmatrix}$	15	Numerical	4.4
4	(see parts below)	14	Numerical	4.4
(i)	For what value of k , the matrix $\begin{bmatrix} 2 & k3 & 5 \end{bmatrix}$ has no ...	14	Numerical	4.4
(ii)	If $A = \begin{bmatrix} 3 & 102 & 7 \end{bmatrix}$, then write A^{-1} .	16	Numerical	4.4
(iii)	Write the inverse of the matrix $\begin{bmatrix} 3 & -11 & 2 \end{bmatrix}$.	16	Numerical	4.4
5	(see parts below)	17	Numerical	4.4
(i)	If $A = \begin{bmatrix} 8 & 23 & 2 \end{bmatrix}$, then find $ \text{adj } A $.	17	Numerical	4.4
(ii)	If A is a square matrix of order 3 and $ A = 5$, then find $ \text{adj } A $.	17	Numerical	4.4
(iii)	If A is a non-singular square matrix of order 3 such that $ \text{adj } A = 64$, find ...	17	Numerical	4.4
(iv)	If A is a square matrix of order 3 such that $ A = -2$, then find $ 3\text{adj } A $.	17	Numerical	4.4
(v)	If $ A^{-1} = 5$, then find the value of $ A $.	17	Numerical	4.4
(vi)	If $ A = -4$ and ...	17	Numerical	4.4
(vii)	If $A = \begin{bmatrix} 2 & 35 & -2 \end{bmatrix}$ be such that $A^{-1} = kA$, then find ...	16	Numerical	4.4
(viii)	If $A(\text{adj } A) = \begin{bmatrix} 3 & 0 & 00 & 3 & 00 & 0 & 3 \end{bmatrix}$, then ...	17	Numerical	4.4
(ix)	If the value of a determinant of third order is 12, then what is the value of the ...	17	Numerical	4.4
6	(see parts below)	17	Numerical	4.4
(i)	Without computing $\text{adj } A$, find $ \text{adj } A $ if: ...	17	Numerical	4.4
(ii)	Without computing $\text{adj } A$, find $ \text{adj } A $ if: ...	17	Numerical	4.4
(iii)	Without computing $\text{adj } A$, find $ \text{adj } A $ if: ...	17	Numerical	4.4

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
7	(see parts below)	17	Numerical	4.4
(i)	If $A = \begin{bmatrix} 2 & 17 & 5 \end{bmatrix}$, find $ A \text{adj } A $.	17	Numerical	4.4
(ii)	If $A = \begin{bmatrix} a & 0 & 00 & a & 00 & 0 & a \end{bmatrix}$, $a \neq 0$, find ...	17	Numerical	4.4
(iii)	If $A = \begin{bmatrix} 3 & -14 & 5 \end{bmatrix}$, find $\text{adj}(\text{adj } A)$.	17	Numerical	4.4
8	(see parts below)	18	Proof	4.4
(i)	If A is a non-singular matrix, then show that $ A^{-1} = \frac{1}{ A }$.	18	Proof	4.4
(ii)	If A is a matrix such that $A^3 = I$, then show that A is invertible.	18	Proof	4.4
(iii)	If A is a matrix such that $A^4 = I$, then show that $A^{-1} = A^3$.	18	Proof	4.4
9	If $A = \begin{bmatrix} 3 & -54 & 2 \end{bmatrix}$, find $A(\text{adj } A)$.	19	Numerical	4.4
10	If the matrix $\begin{bmatrix} x+4 & x & xx & x+4 & xx & x & x+4 \end{bmatrix}$ is ...	14	Numerical	4.4
11	(see parts below)	19	Numerical	4.4
(i)	Find the adjoint of the matrix: $A = \begin{bmatrix} -2 & 13 & 4 \end{bmatrix}$ Also ...	19	Numerical	4.4
(ii)	Find the adjoint of the matrix: $A = \begin{bmatrix} 2 & 3 & -4 & -6 \end{bmatrix}$ Also ...	19	Numerical	4.4
12	If $A = \begin{bmatrix} 3 & 1 & 22 & -3 & -11 & 2 & 1 \end{bmatrix}$, find ...	19	Numerical	4.4
13	Find the adjoint of the following matrices:	19	Numerical	4.4
(i)	$A = \begin{bmatrix} 1 & -1 & 22 & 3 & 5 & -2 & 0 & 1 \end{bmatrix}$	19	Numerical	4.4
(ii)	$A = \begin{bmatrix} 1 & -1 & 23 & 0 & -21 & 0 & 3 \end{bmatrix}$	19	Numerical	4.4
14	If ...	19	Numerical	4.4
15	For the matrix $A = \begin{bmatrix} 2 & -1 & 34 & 2 & 50 & 4 & -1 \end{bmatrix}$, verify ...	19	Proof	4.4
16	For the matrix $A = \begin{bmatrix} 5 & 2 & -3 & -1 \end{bmatrix}$, verify that ...	19	Proof	4.4
17	See parts below:	18	Proof	4.4
(i)	Given $A = \begin{bmatrix} 2 & -3 & -4 & 7 \end{bmatrix}$, compute A^{-1} and ...	18	Proof	4.4
(ii)	If $A = \begin{bmatrix} 4 & 52 & 1 \end{bmatrix}$, show that ...	18	Proof	4.4
18	Find the inverse of the matrix ...	18	Proof	4.4
19	If $A = \begin{bmatrix} 0 & 1 & 31 & 2 & x2 & 3 & 1 \end{bmatrix}$ and ...	18	Numerical	4.4
20	If $A = \begin{bmatrix} 2 & -35 & 1 \end{bmatrix}$ and ...	21	Proof	4.4
21	Find the inverse of each of the following matrices:	16	Numerical	4.4

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(i)	$A = \begin{bmatrix} 1 & 0 & 03 & 3 & 05 & 2 & -1 \end{bmatrix}$	16	Numerical	4.4
(ii)	$A = \begin{bmatrix} -1 & 1 & 23 & -1 & 1 & -1 & 3 & 4 \end{bmatrix}$	16	Numerical	4.4
22	Given $A = \begin{bmatrix} 5 & 0 & 42 & 3 & 21 & 2 & 1 \end{bmatrix}$...	21	Numerical	4.4
23	See parts below:	20	Numerical	4.4
(i)	Show that $A = \begin{bmatrix} 5 & 3 & -1 & -2 \end{bmatrix}$ satisfies the ...	20	Numerical	4.4
(ii)	If $A = \begin{bmatrix} 2 & -1 & -1 & 2 \end{bmatrix}$ and I is the identity ...	20	Numerical	4.4
(iii)	If $A = \begin{bmatrix} 3 & -24 & -2 \end{bmatrix}$, then find the value of ...	20	Numerical	4.4
(iv)	If $A = \begin{bmatrix} 1 & 3 & 22 & 0 & -11 & 2 & 3 \end{bmatrix}$, then show ...	20	Numerical	4.4
24	See parts below:	20	Numerical	4.4
(i)	If $A = \begin{bmatrix} 1 & 2 & 22 & 1 & 22 & 2 & 1 \end{bmatrix}$, prove that ...	20	Numerical	4.4
(ii)	Find A^{-1} if ...	20	Numerical	4.4
25	See parts below:	18	Proof	4.4
(i)	If ...	18	Proof	4.4
(ii)	If $A = \begin{bmatrix} 1 & -1 & 12 & -1 & 01 & 0 & 0 \end{bmatrix}$, prove ...	18	Proof	4.4
(iii)	If ...	18	Proof	4.4
26	Show that for the matrix ...	20	Numerical	4.4
27	See parts below:	20	Numerical	4.4
(i)	Find a 2×2 matrix B such that ...	20	Numerical	4.4
(ii)	Solve the matrix equation ...	20	Numerical	4.4
28	See parts below:	20	Numerical	4.4
(i)	Find the matrix A satisfying the matrix equation ...	20	Numerical	4.4
(ii)	Find the matrix P satisfying the matrix equation ...	20	Numerical	4.4

Exercise 4.5 – Solving Systems of Linear Equations Using Matrices

1	(see parts below)	22	Numerical	4.5
(i)	Use matrix method to solve the following system of equations: $5x + 2y = 4$ $7x + 3y = 5$	22	Numerical	4.5
(ii)	Use matrix method to solve the following system of equations: $4x - 3y = 3$ $3x - 5y = 7$	22	Numerical	4.5
2	(see parts below)	23	Numerical	4.5
(i)	Using matrix method, determine whether the following system of equations is consistent or ...	23	Numerical	4.5

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(ii)	Using matrix method, determine whether the following system of equations is consistent or ...	23	Numerical	4.5
(iii)	Using matrix method, determine whether the following system of equations is consistent or ...	23	Numerical	4.5
(iv)	Using matrix method, determine whether the following system of equations is consistent or ...	23	Numerical	4.5
3	Solve the following system of equations by using matrix method: $x + y = 1$ $7x + 7y = 7$	24	Numerical	4.5
4	(see parts below)	22	Numerical	4.5
(i)	Using matrix method, solve the following system of equations: $x - 2y + 3z = 6$...	22	Numerical	4.5
(ii)	Using matrix method, solve the following system of equations: $x + y + z = 6$...	22	Numerical	4.5
(iii)	Using matrix method, solve the following system of equations: $x + y + z = 9$...	22	Numerical	4.5
(iv)	Using matrix method, solve the following system of equations: $x - 2y = 10$...	22	Numerical	4.5
(v)	Using matrix method, solve the following system of equations: $3x + y + z = 1$...	22	Numerical	4.5
(vi)	Using matrix method, solve the following system of equations: $2x - 3y + 5z = 11$...	22	Numerical	4.5
5	Solve the following system of linear equations using matrix method: ...	25	Numerical	4.5
6	State the condition under which the following equations have a unique solution: ...	22	Numerical	4.5
7	The sum of three numbers is 6. If we multiply third number by 3 and add second to it we ...	26	Numerical	4.5
8	Garima owns a cafe that serves burgers, samosas, and juice bottles. On a particular ...	26	Numerical	4.5
(i)	Form a set of simultaneous equations for the above information.	26	Numerical	4.5
(ii)	Solve the set of equations formed in (i) by matrix method.	26	Numerical	4.5

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
9	If $A = \begin{bmatrix} 1 & 2 & -32 & 3 & 23 & -3 & -4 \end{bmatrix}$, find A^{-1} ...	27	Numerical	4.5
10	If $A = \begin{bmatrix} 2 & -3 & 53 & 2 & -41 & 1 & -2 \end{bmatrix}$, find A^{-1} ...	27	Numerical	4.5
11	If $A = \begin{bmatrix} 3 & 5 & -2 & 3 \end{bmatrix}$, find A^{-1} and use it to solve the ...	28	Numerical	4.5
12	If $A = \begin{bmatrix} 1 & -1 & 12 & 1 & -31 & 1 & 1 \end{bmatrix}$, find A^{-1} and ...	28	Numerical	4.5
13	Given two matrices A and B where ...	29	Numerical	4.5
14	If $A = \begin{bmatrix} -3 & -2 & -42 & 1 & 22 & 1 & 3 \end{bmatrix}$ and ...	29	Numerical	4.5
15	(see parts below)	30	Numerical	4.5
(i)	Using matrices, solve the following system of homogeneous equations: $5x + 5y + 2z = 0$...	30	Numerical	4.5
(ii)	Using matrices, solve the following system of homogeneous equations: $3x + y - 2z = 0$...	30	Numerical	4.5
16	(see parts below)	30	Numerical	4.5
(i)	Using matrices, solve the following system of homogeneous equations: $2x - 3y - z = 0$...	30	Numerical	4.5
(ii)	Using matrices, solve the following system of homogeneous equations: $x + y - z = 0$...	30	Numerical	4.5

Formula & Standard Results Sheet

I. Evaluation of Determinants: Minors and Cofactors (Ex 4.1)

2×2 Determinant

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

The basic expansion for a second-order determinant

3×3 Determinant – Expansion Along First Row

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - b_1 \begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1 \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}$$

Can be expanded along any row or column; signs alternate by the checkerboard pattern

Minor and Cofactor

$$M_{ij} = \text{determinant left after deleting row } i \text{ and column } j, \quad A_{ij} = (-1)^{i+j} M_{ij}$$

The cofactor is the minor with a sign attached based on position

Determinant via Cofactor Expansion (Any Row/Column)

$$\Delta = a_{i1}A_{i1} + a_{i2}A_{i2} + a_{i3}A_{i3}$$

Expanding along row i ; an analogous formula holds for any column

Effect of Scalar Multiplication on a Determinant

$$|kA| = k^n |A|, \quad \text{for an } n \times n \text{ matrix } A$$

Multiplying an $n \times n$ matrix by k scales its determinant by k^n , not k

II. Properties of Determinants and Proving Identities (Ex 4.2)

Row/Column Operations Preserve Value

$$R_i \rightarrow R_i + kR_j \quad (\text{or the analogous column operation}) \text{ leaves } |A| \text{ unchanged}$$

The standard tool for simplifying a determinant before expanding it

Scalar Factored From a Single Row/Column

If every entry of one row (or column) has a common factor k , then k can
Only one row/column at a time – not the whole determinant

Zero Determinant Conditions

$$|A| =$$

0 if any two rows (or columns) are identical or proportional
A quick way to prove a determinant vanishes without full expansion

Determinant of a Product

$$|AB| = |A| |B|$$

Also gives $|A^k| = |A|^k$ and $|A^{-1}| = \frac{1}{|A|}$

Determinant of a Transpose

$$|A^T| = |A|$$

The determinant is unaffected by transposing the matrix

III. Applications of Determinants: Area, Collinearity and Equation of a Line (Ex 4.3)

Area of a Triangle

$$\text{Area} = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

Take the absolute value, since area cannot be negative

Collinearity of Three Points

$$(x_1, y_1), (x_2, y_2), (x_3, y_3) \text{ are collinear} \iff \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$$

Equivalent to the triangle formed having zero area

Equation of a Line Through Two Points

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$

Setting the determinant to zero forces (x, y) to be collinear with the two given points

IV. Adjoint and Inverse of a Matrix (Ex 4.4)**Singular and Non-Singular Matrices**

$$A \text{ is singular} \iff |A| = 0; \quad A \text{ is non-singular} \iff |A| \neq 0$$

A matrix has an inverse only when it is non-singular

Adjoint of a 2×2 Matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \implies \text{adj } A = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Swap the leading diagonal entries and negate the off-diagonal entries

Adjoint via Cofactor Matrix

$$\text{adj } A = C', \quad \text{where } C = [C_{ij}] = [(-1)^{i+j} M_{ij}]$$

Adjoint is the transpose of the cofactor matrix

Inverse of a Matrix

$$A^{-1} = \frac{1}{|A|} \text{adj } A, \quad \text{defined only when } |A| \neq 0$$

Divide every entry of the adjoint by the determinant

Inverse of a 2×2 Matrix (Direct Form)

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \implies A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

The fast shortcut for 2×2 matrices, skipping the general cofactor method

Fundamental Adjoint Identity

$$A(\text{adj } A) = (\text{adj } A)A = |A| I_n$$

The identity that most adjoint/inverse proofs are built on

Determinant of the Adjoint

$$|\text{adj } A| = |A|^{n-1}, \quad \text{for an } n \times n \text{ matrix } A$$

Also gives $\text{adj}(\text{adj } A) = |A|^{n-2} A$

Reversal Law for Inverse and Adjoint

$$(AB)^{-1} = B^{-1}A^{-1}, \quad \text{adj}(AB) = (\text{adj } B)(\text{adj } A)$$

The order reverses, just as with the transpose of a product

Inverse from a Matrix's Own Characteristic Relation

If $A^2 - c_1A - c_0I = O$ with $c_0 \neq 0$, then $A^{-1} = \frac{1}{c_0}(A - c_1I)$

Lets A^{-1} be found algebraically without computing the adjoint, using a relation the matrix itself satisfies

V. Solving Systems of Linear Equations Using Matrices (Ex 4.5)**Matrix Form of a Linear System**

$$AX = B$$

A is the coefficient matrix, X the column of unknowns, B the column of constants

Solving a Consistent System

$$X = A^{-1}B = \frac{1}{|A|}(\text{adj } A) B, \quad \text{valid when } |A| \neq 0$$

Gives a unique solution; compute A^{-1} first, then multiply by B

Consistency of the System

$$\begin{aligned} |A| \neq 0 &\implies \text{unique solution}; & |A| = 0 \text{ and } (\text{adj } A)B \neq O &\implies \text{no solution}; \\ |A| = 0 \text{ and } (\text{adj } A)B = O &\implies \text{infinitely many solutions (or none, needing further check)} \end{aligned}$$

Always check $|A|$ first before attempting to solve or classify the system

Homogeneous System

$$AX = O \implies x = y = z = 0 \text{ (trivial solution) when } |A| \neq 0$$

A homogeneous system has a non-trivial solution only when $|A| = 0$

Reducible (Non-Linear) Systems via Substitution

$$\text{Substituting } u = \frac{1}{x}, v = \frac{1}{y}, w =$$

$\frac{1}{z}$ converts a reciprocal system into a linear one $AU = B$
Solve for u, v, w first via the matrix method, then invert the substitution to recover x, y, z

4.1 – Evaluation of Determinants: Minors and Cofactors

Question 1 (i)

Evaluate the determinant:

$$\begin{vmatrix} x^2 - x + 1 & x - 1 \\ x + 1 & x + 1 \end{vmatrix}$$

TYPE 1 Evaluation of 2x2 Determinants

Given: $\Delta = \begin{vmatrix} x^2 - x + 1 & x - 1 \\ x + 1 & x + 1 \end{vmatrix}$

Concept / Formula Used: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

Solution:

Expanding the 2×2 determinant:

$$\Delta = (x^2 - x + 1)(x + 1) - (x - 1)(x + 1)$$

Applying algebraic expansion and identity $(x - 1)(x + 1) = x^2 - 1$:

$$(x^2 - x + 1)(x + 1) = x^3 + 1$$

$$(x - 1)(x + 1) = x^2 - 1$$

Substituting these expressions back into the determinant expansion:

$$\begin{aligned} \Delta &= (x^3 + 1) - (x^2 - 1) \\ &= x^3 + 1 - x^2 + 1 \\ &= x^3 - x^2 + 2 \end{aligned}$$

Final Answer

$$\Delta = x^3 - x^2 + 2$$

Question 1 (ii)

Evaluate the determinant:

$$\begin{vmatrix} \cos 90^\circ & -\cos 45^\circ \\ \sin 90^\circ & \sin 45^\circ \end{vmatrix}$$

TYPE 1 Evaluation of 2x2 Determinants

Given: $\Delta = \begin{vmatrix} \cos 90^\circ & -\cos 45^\circ \\ \sin 90^\circ & \sin 45^\circ \end{vmatrix}$

Concept / Formula Used: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

Solution:

Evaluating standard trigonometric values:

$$\cos 90^\circ = 0$$

$$\sin 90^\circ = 1$$

$$\cos 45^\circ = \frac{1}{\sqrt{2}}$$

$$\sin 45^\circ = \frac{1}{\sqrt{2}}$$

Substituting these values into the determinant:

$$\begin{aligned} \Delta &= \begin{vmatrix} 0 & -\frac{1}{\sqrt{2}} \\ 1 & \frac{1}{\sqrt{2}} \end{vmatrix} \\ &= (0) \left(\frac{1}{\sqrt{2}} \right) - \left(-\frac{1}{\sqrt{2}} \right) (1) \\ &= 0 + \frac{1}{\sqrt{2}} \\ &= \frac{1}{\sqrt{2}} \end{aligned}$$

Final Answer

$$\Delta = \frac{1}{\sqrt{2}}$$

Question 1 (iii)

Evaluate the determinant:

$$\begin{vmatrix} 2 \cos \theta & -2 \sin \theta \\ \sin \theta & \cos \theta \end{vmatrix}$$

TYPE 2 Evaluation of Determinants Using Trigonometric Identities

Given: $\Delta = \begin{vmatrix} 2 \cos \theta & -2 \sin \theta \\ \sin \theta & \cos \theta \end{vmatrix}$

Concept / Formula Used: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$ $\sin^2 \theta + \cos^2 \theta = 1$

Solution:

Expanding the 2×2 determinant:

$$\begin{aligned} \Delta &= (2 \cos \theta)(\cos \theta) - (-2 \sin \theta)(\sin \theta) \\ &= 2 \cos^2 \theta + 2 \sin^2 \theta \\ &= 2(\cos^2 \theta + \sin^2 \theta) \end{aligned}$$

Using the standard Pythagorean identity $\cos^2 \theta + \sin^2 \theta = 1$:

$$\begin{aligned}\Delta &= 2(1) \\ &= 2\end{aligned}$$

Final Answer

$$\Delta = 2$$

Question 2 (i)

Evaluate the determinant:

$$\begin{vmatrix} \sin 30^\circ & \cos 30^\circ \\ -\sin 60^\circ & \cos 60^\circ \end{vmatrix}$$

TYPE 1 Evaluation of 2x2 Determinants

Given: $\Delta = \begin{vmatrix} \sin 30^\circ & \cos 30^\circ \\ -\sin 60^\circ & \cos 60^\circ \end{vmatrix}$

Concept / Formula Used: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

Solution:

Substituting standard angle trigonometric values:

$$\begin{aligned}\sin 30^\circ &= \frac{1}{2} \\ \cos 30^\circ &= \frac{\sqrt{3}}{2} \\ \sin 60^\circ &= \frac{\sqrt{3}}{2} \\ \cos 60^\circ &= \frac{1}{2}\end{aligned}$$

Evaluating the 2×2 determinant:

$$\begin{aligned}\Delta &= \left(\frac{1}{2}\right) \left(\frac{1}{2}\right) - \left(\frac{\sqrt{3}}{2}\right) \left(-\frac{\sqrt{3}}{2}\right) \\ &= \frac{1}{4} - \left(-\frac{3}{4}\right) \\ &= \frac{1}{4} + \frac{3}{4} \\ &= \frac{4}{4} \\ &= 1\end{aligned}$$

Final Answer

$$\Delta = 1$$

Question 2 (ii)

Evaluate the determinant:

$$\begin{vmatrix} -1 & 31 & 40 \\ 0 & 5 & -432 \\ 0 & 0 & 20 \end{vmatrix}$$

TYPE 3 Evaluation of 3x3 Determinants

Given: $\Delta = \begin{vmatrix} -1 & 31 & 40 \\ 0 & 5 & -432 \\ 0 & 0 & 20 \end{vmatrix}$

Concept / Formula Used: $\Delta = a_{11}a_{22}a_{33}$ **Solution:**Expanding along the first column C_1 :

$$\begin{aligned} \Delta &= -1 \cdot \begin{vmatrix} 5 & -432 \\ 0 & 20 \end{vmatrix} - 0 + 0 \\ &= -1 \cdot ((5)(20) - (-432)(0)) \\ &= -1 \cdot (100 - 0) \\ &= -1 \cdot 100 \\ &= -100 \end{aligned}$$

Final Answer

$$\Delta = -100$$

Question 2 (iii)

Evaluate the determinant:

$$\begin{vmatrix} 2 & 0 & 0 \\ -25 & 3 & 0 \\ 71 & 64 & 7 \end{vmatrix}$$

TYPE 3 Evaluation of 3x3 Determinants

Given: $\Delta = \begin{vmatrix} 2 & 0 & 0 \\ -25 & 3 & 0 \\ 71 & 64 & 7 \end{vmatrix}$

Concept / Formula Used: $\Delta = a_{11}a_{22}a_{33}$ **Solution:**

Expanding along the first row R_1 :

$$\begin{aligned}\Delta &= 2 \cdot \begin{vmatrix} 3 & 0 \\ 64 & 7 \end{vmatrix} - 0 + 0 \\ &= 2 \cdot ((3)(7) - (0)(64)) \\ &= 2 \cdot (21 - 0) \\ &= 2 \cdot 21 \\ &= 42\end{aligned}$$

Final Answer

$$\Delta = 42$$

Question 3 (i)

Find the value of $\begin{vmatrix} \sin A & -\sin B \\ \cos A & \cos B \end{vmatrix}$, where $A = 63^\circ$ and $B = 27^\circ$.

TYPE 2 Evaluation of Determinants Using Trigonometric Identities

Given: $\Delta = \begin{vmatrix} \sin A & -\sin B \\ \cos A & \cos B \end{vmatrix}$, $A = 63^\circ$, $B = 27^\circ$

Concept / Formula Used: $\sin(A + B) = \sin A \cos B + \cos A \sin B$

Solution:

Expanding the 2×2 determinant:

$$\begin{aligned}\Delta &= (\sin A)(\cos B) - (-\sin B)(\cos A) \\ &= \sin A \cos B + \cos A \sin B\end{aligned}$$

Applying the sine addition formula $\sin A \cos B + \cos A \sin B = \sin(A + B)$:

$$\Delta = \sin(A + B)$$

Substituting $A = 63^\circ$ and $B = 27^\circ$:

$$\begin{aligned}\Delta &= \sin(63^\circ + 27^\circ) \\ &= \sin(90^\circ) \\ &= 1\end{aligned}$$

Final Answer

$$\Delta = 1$$

Question 3 (ii)

Find the value of $\begin{vmatrix} \cos A & \sin A \\ -\sin B & \cos B \end{vmatrix}$, where $A = 75^\circ$ and $B = 45^\circ$.

TYPE 2 Evaluation of Determinants Using Trigonometric Identities

Given: $\Delta = \begin{vmatrix} \cos A & \sin A \\ -\sin B & \cos B \end{vmatrix}, A = 75^\circ, B = 45^\circ$

Concept / Formula Used: $\cos(A - B) = \cos A \cos B + \sin A \sin B$

Solution:

Expanding the 2×2 determinant:

$$\begin{aligned}\Delta &= (\cos A)(\cos B) - (\sin A)(-\sin B) \\ &= \cos A \cos B + \sin A \sin B\end{aligned}$$

Applying the cosine subtraction formula $\cos A \cos B + \sin A \sin B = \cos(A - B)$:

$$\Delta = \cos(A - B)$$

Substituting $A = 75^\circ$ and $B = 45^\circ$:

$$\begin{aligned}\Delta &= \cos(75^\circ - 45^\circ) \\ &= \cos(30^\circ) \\ &= \frac{\sqrt{3}}{2}\end{aligned}$$

Final Answer

$$\Delta = \frac{\sqrt{3}}{2}$$

Question 4 (i)

If $A = \begin{bmatrix} 5 & 6 & -3 \\ -4 & 3 & 2 \\ -4 & -7 & 3 \end{bmatrix}$, then write the cofactor of the element a_{21} of its 2nd row.

TYPE 4 Minors and Cofactors of a Determinant

Given: Matrix $A = \begin{bmatrix} 5 & 6 & -3 \\ -4 & 3 & 2 \\ -4 & -7 & 3 \end{bmatrix}$

Concept / Formula Used: $A_{ij} = (-1)^{i+j} M_{ij}$ M_{ij} a_{ij}

Solution:

The element at a_{21} is -4 . To find its minor M_{21} , delete the 2nd row and 1st column:

$$M_{21} = \begin{vmatrix} 6 & -3 \\ -7 & 3 \end{vmatrix}$$

Evaluating minor M_{21} :

$$\begin{aligned} M_{21} &= (6)(3) - (-3)(-7) \\ &= 18 - 21 \\ &= -3 \end{aligned}$$

Calculating the cofactor A_{21} :

$$\begin{aligned} A_{21} &= (-1)^{2+1} M_{21} \\ &= (-1)^3 (-3) \\ &= (-1)(-3) \\ &= 3 \end{aligned}$$

Final Answer

$$A_{21} = 3$$

Question 4 (ii)

If A_{ij} is the cofactor of the element a_{ij} of the determinant $\begin{vmatrix} 2 & -3 & 5 \\ 6 & 0 & 4 \\ 1 & 5 & -7 \end{vmatrix}$, then write the value of $a_{32}A_{32}$.

TYPE 4 Minors and Cofactors of a Determinant

Given: Determinant $\Delta = \begin{vmatrix} 2 & -3 & 5 \\ 6 & 0 & 4 \\ 1 & 5 & -7 \end{vmatrix}$

Concept / Formula Used: $A_{32} = (-1)^{3+2} M_{32}$

Solution:

Identifying element a_{32} located at row 3, column 2:

$$a_{32} = 5$$

The minor M_{32} is formed by deleting row 3 and column 2:

$$M_{32} = \begin{vmatrix} 2 & 5 \\ 6 & 4 \end{vmatrix}$$

Evaluating M_{32} :

$$\begin{aligned} M_{32} &= (2)(4) - (5)(6) \\ &= 8 - 30 \\ &= -22 \end{aligned}$$

Calculating cofactor A_{32} :

$$\begin{aligned} A_{32} &= (-1)^{3+2} M_{32} \\ &= (-1)^5 (-22) \\ &= (-1)(-22) \\ &= 22 \end{aligned}$$

Evaluating product $a_{32}A_{32}$:

$$\begin{aligned} a_{32}A_{32} &= 5 \times 22 \\ &= 110 \end{aligned}$$

Final Answer

$$a_{32}A_{32} = 110$$

Question 5 (i)

Find the value of x if:

$$\begin{vmatrix} 2x+5 & 3 \\ 5x+2 & 9 \end{vmatrix} = 0$$

TYPE 5 Solving Determinant Equations

Given: $\begin{vmatrix} 2x+5 & 3 \\ 5x+2 & 9 \end{vmatrix} = 0$

Concept / Formula Used: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

Solution:

Expanding the 2×2 determinant:

$$\begin{aligned} (2x+5)(9) - (3)(5x+2) &= 0 \\ 18x + 45 - 15x - 6 &= 0 \end{aligned}$$

Grouping like terms:

$$\begin{aligned} (18x - 15x) + (45 - 6) &= 0 \\ 3x + 39 &= 0 \\ 3x &= -39 \\ x &= \frac{-39}{3} \\ x &= -13 \end{aligned}$$

Final Answer

$$x = -13$$

Question 5 (ii)

Find the value of x if:

$$\begin{vmatrix} x+2 & 3 \\ x+5 & 4 \end{vmatrix} = 3$$

TYPE 5 Solving Determinant Equations

Given: $\begin{vmatrix} x+2 & 3 \\ x+5 & 4 \end{vmatrix} = 3$

Concept / Formula Used: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

Solution:

Expanding the 2×2 determinant:

$$\begin{aligned}(x+2)(4) - (3)(x+5) &= 3 \\ 4x + 8 - 3x - 15 &= 3\end{aligned}$$

Combining like terms on the LHS:

$$\begin{aligned}(4x - 3x) + (8 - 15) &= 3 \\ x - 7 &= 3 \\ x &= 3 + 7 \\ x &= 10\end{aligned}$$

Final Answer

$$x = 10$$

Question 5 (iii)

Find the value of x if:

$$\begin{vmatrix} 3x & 7 \\ -2 & 4 \end{vmatrix} = \begin{vmatrix} 8 & 7 \\ 6 & 4 \end{vmatrix}$$

TYPE 5 Solving Determinant Equations

Given: $\begin{vmatrix} 3x & 7 \\ -2 & 4 \end{vmatrix} = \begin{vmatrix} 8 & 7 \\ 6 & 4 \end{vmatrix}$

Concept / Formula Used: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

Solution:

Evaluating LHS determinant:

$$\begin{aligned}\text{LHS} &= (3x)(4) - (7)(-2) \\ &= 12x + 14\end{aligned}$$

Evaluating RHS determinant:

$$\begin{aligned}\text{RHS} &= (8)(4) - (7)(6) \\ &= 32 - 42 \\ &= -10\end{aligned}$$

Equating LHS and RHS:

$$\begin{aligned}12x + 14 &= -10 \\ 12x &= -10 - 14 \\ 12x &= -24 \\ x &= \frac{-24}{12} \\ x &= -2\end{aligned}$$

Final Answer

$$x = -2$$

Question 6 (i)

If $\begin{vmatrix} x+1 & x-1 \\ x-3 & x+2 \end{vmatrix} = \begin{vmatrix} 4 & -1 \\ 1 & 3 \end{vmatrix}$, then write the value of x .

TYPE 5 Solving Determinant Equations

Given: $\begin{vmatrix} x+1 & x-1 \\ x-3 & x+2 \end{vmatrix} = \begin{vmatrix} 4 & -1 \\ 1 & 3 \end{vmatrix}$

Concept / Formula Used: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

Solution:

Evaluating LHS determinant:

$$\begin{aligned}\text{LHS} &= (x+1)(x+2) - (x-1)(x-3) \\ &= (x^2 + 3x + 2) - (x^2 - 4x + 3) \\ &= x^2 + 3x + 2 - x^2 + 4x - 3 \\ &= 7x - 1\end{aligned}$$

Evaluating RHS determinant:

$$\begin{aligned}\text{RHS} &= (4)(3) - (-1)(1) \\ &= 12 - (-1) \\ &= 12 + 1 \\ &= 13\end{aligned}$$

Equating LHS and RHS:

$$\begin{aligned}7x - 1 &= 13 \\7x &= 13 + 1 \\7x &= 14 \\x &= \frac{14}{7} \\x &= 2\end{aligned}$$

Final Answer

$$x = 2$$

Question 6 (ii)

If $\begin{vmatrix} \sqrt{6} & x \\ \sqrt{20} & \sqrt{24} \end{vmatrix} = \begin{vmatrix} 6 & 2 \\ 5 & 2 \end{vmatrix}$, then write the value of x .

TYPE 5 Solving Determinant Equations

Given: $\begin{vmatrix} \sqrt{6} & x \\ \sqrt{20} & \sqrt{24} \end{vmatrix} = \begin{vmatrix} 6 & 2 \\ 5 & 2 \end{vmatrix}$

Concept / Formula Used: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

Solution:

Evaluating LHS determinant:

$$\begin{aligned}\text{LHS} &= (\sqrt{6})(\sqrt{24}) - (x)(\sqrt{20}) \\&= \sqrt{144} - x\sqrt{20} \\&= 12 - x(2\sqrt{5}) \\&= 12 - 2\sqrt{5}x\end{aligned}$$

Evaluating RHS determinant:

$$\begin{aligned}\text{RHS} &= (6)(2) - (2)(5) \\&= 12 - 10 \\&= 2\end{aligned}$$

Equating LHS and RHS:

$$\begin{aligned}
 12 - 2\sqrt{5}x &= 2 \\
 -2\sqrt{5}x &= 2 - 12 \\
 -2\sqrt{5}x &= -10 \\
 x &= \frac{-10}{-2\sqrt{5}} \\
 x &= \frac{5}{\sqrt{5}} \\
 x &= \sqrt{5}
 \end{aligned}$$

Final Answer

$$x = \sqrt{5}$$

Question 7 (i)

If $x \in \mathbf{N}$ and $\begin{vmatrix} x & 3 \\ 4 & x \end{vmatrix} = \begin{vmatrix} 4 & -3 \\ 0 & 1 \end{vmatrix}$, find the value(s) of x .

TYPE 5 Solving Determinant Equations

Given: $\begin{vmatrix} x & 3 \\ 4 & x \end{vmatrix} = \begin{vmatrix} 4 & -3 \\ 0 & 1 \end{vmatrix}$, $x \in \mathbf{N}$

Concept / Formula Used: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

Solution:

Evaluating LHS determinant:

$$\begin{aligned}
 \text{LHS} &= (x)(x) - (3)(4) \\
 &= x^2 - 12
 \end{aligned}$$

Evaluating RHS determinant:

$$\begin{aligned}
 \text{RHS} &= (4)(1) - (-3)(0) \\
 &= 4 - 0 \\
 &= 4
 \end{aligned}$$

Equating LHS and RHS:

$$\begin{aligned}
 x^2 - 12 &= 4 \\
 x^2 &= 4 + 12 \\
 x^2 &= 16 \\
 x &= \pm 4
 \end{aligned}$$

Since $x \in \mathbf{N}$ (natural numbers), x must be positive:

$$x = 4$$

Final Answer

$$x = 4$$

Question 7 (ii)

If $x \in \mathbf{I}$ and $\begin{vmatrix} 2x & 3 \\ -1 & x \end{vmatrix} = \begin{vmatrix} 3 & 1 \\ x & 3 \end{vmatrix}$, find the value(s) of x .

TYPE 5 Solving Determinant Equations

Given: $\begin{vmatrix} 2x & 3 \\ -1 & x \end{vmatrix} = \begin{vmatrix} 3 & 1 \\ x & 3 \end{vmatrix}, x \in \mathbf{I}$

Concept / Formula Used: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

Solution:

Evaluating LHS determinant:

$$\begin{aligned} \text{LHS} &= (2x)(x) - (3)(-1) \\ &= 2x^2 + 3 \end{aligned}$$

Evaluating RHS determinant:

$$\begin{aligned} \text{RHS} &= (3)(3) - (1)(x) \\ &= 9 - x \end{aligned}$$

Equating LHS and RHS:

$$\begin{aligned} 2x^2 + 3 &= 9 - x \\ 2x^2 + x + 3 - 9 &= 0 \\ 2x^2 + x - 6 &= 0 \end{aligned}$$

Factorizing the quadratic equation $2x^2 + x - 6 = 0$:

$$\begin{aligned} 2x^2 + 4x - 3x - 6 &= 0 \\ 2x(x + 2) - 3(x + 2) &= 0 \\ (2x - 3)(x + 2) &= 0 \end{aligned}$$

Solving for roots:

$$\begin{aligned} 2x - 3 = 0 &\implies x = \frac{3}{2} \\ x + 2 = 0 &\implies x = -2 \end{aligned}$$

Since $x \in \mathbf{I}$ (set of integers), $x = \frac{3}{2} \notin \mathbf{I}$. Therefore, the only integer solution is:

$$x = -2$$

Final Answer

$$x = -2$$

Question 8

If $A = \begin{bmatrix} 1 & 2 \\ 4 & 2 \end{bmatrix}$, then show that $|2A| = 4|A|$.

TYPE 6 Determinant Properties of Scalar Multiplication

★ Likely Exam Question

To Prove: $|2A| = 4|A|$

Given: Matrix $A = \begin{bmatrix} 1 & 2 \\ 4 & 2 \end{bmatrix}$

Concept / Formula Used: $n \times n$ A $|kA| = k^n|A|$ 2×2 $|2A| = 2^2|A| = 4|A|$

Solution:

First, calculate the determinant of A :

$$\begin{aligned} |A| &= \begin{vmatrix} 1 & 2 \\ 4 & 2 \end{vmatrix} \\ &= (1)(2) - (2)(4) \\ &= 2 - 8 \\ &= -6 \end{aligned}$$

Evaluating the RHS:

$$\begin{aligned} \text{RHS} &= 4|A| \\ &= 4 \times (-6) \\ &= -24 \end{aligned}$$

Now, determine the matrix $2A$:

$$\begin{aligned} 2A &= 2 \begin{bmatrix} 1 & 2 \\ 4 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 4 \\ 8 & 4 \end{bmatrix} \end{aligned}$$

Calculate the determinant of $2A$:

$$\begin{aligned}|2A| &= \begin{vmatrix} 2 & 4 \\ 8 & 4 \end{vmatrix} \\ &= (2)(4) - (4)(8) \\ &= 8 - 32 \\ &= -24\end{aligned}$$

Comparing the two sides:

$$|2A| = -24 = 4 \times (-6) = 4|A|$$

Hence, $|2A| = 4|A|$ is verified.

Hence Proved

$|2A| = 4|A|$ – verified since $|2A| = -24$ and $4|A| = 4(-6) = -24$.

Question 9 (i)

Evaluate the determinant:

$$\begin{vmatrix} 0 & -1 & 2 \\ 1 & 0 & 3 \\ -2 & -3 & 0 \end{vmatrix}$$

TYPE 3 Evaluation of 3x3 Determinants

Given: $\Delta = \begin{vmatrix} 0 & -1 & 2 \\ 1 & 0 & 3 \\ -2 & -3 & 0 \end{vmatrix}$

Concept / Formula Used: Expansion along the first row: $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - b_1 \begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1 \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}$

Solution:

Expanding along the first row:

$$\Delta = 0 \begin{vmatrix} 0 & 3 \\ -3 & 0 \end{vmatrix} - (-1) \begin{vmatrix} 1 & 3 \\ -2 & 0 \end{vmatrix} + 2 \begin{vmatrix} 1 & 0 \\ -2 & -3 \end{vmatrix}$$

Evaluating each 2×2 minor:

$$\begin{aligned}\begin{vmatrix} 0 & 3 \\ -3 & 0 \end{vmatrix} &= (0)(0) - (3)(-3) = 0 + 9 = 9 \\ \begin{vmatrix} 1 & 3 \\ -2 & 0 \end{vmatrix} &= (1)(0) - (3)(-2) = 0 + 6 = 6 \\ \begin{vmatrix} 1 & 0 \\ -2 & -3 \end{vmatrix} &= (1)(-3) - (0)(-2) = -3 - 0 = -3\end{aligned}$$

Substituting back:

$$\begin{aligned}\Delta &= 0(9) + 1(6) + 2(-3) \\ &= 0 + 6 - 6 \\ &= 0\end{aligned}$$

Final Answer

$$\Delta = 0$$

Question 9 (ii)

Evaluate the determinant:

$$\begin{vmatrix} 3 & -4 & 5 \\ 1 & 1 & -2 \\ 2 & 3 & 1 \end{vmatrix}$$

TYPE 3 Evaluation of 3x3 Determinants

Given: $\Delta = \begin{vmatrix} 3 & -4 & 5 \\ 1 & 1 & -2 \\ 2 & 3 & 1 \end{vmatrix}$

Concept / Formula Used: Expansion along the first row: $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - b_1 \begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1 \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}$

Solution:

Expanding along the first row:

$$\Delta = 3 \begin{vmatrix} 1 & -2 \\ 3 & 1 \end{vmatrix} - (-4) \begin{vmatrix} 1 & -2 \\ 2 & 1 \end{vmatrix} + 5 \begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix}$$

Evaluating each 2×2 minor:

$$\begin{aligned}\begin{vmatrix} 1 & -2 \\ 3 & 1 \end{vmatrix} &= (1)(1) - (-2)(3) = 1 + 6 = 7 \\ \begin{vmatrix} 1 & -2 \\ 2 & 1 \end{vmatrix} &= (1)(1) - (-2)(2) = 1 + 4 = 5 \\ \begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix} &= (1)(3) - (1)(2) = 3 - 2 = 1\end{aligned}$$

Substituting back:

$$\begin{aligned}\Delta &= 3(7) + 4(5) + 5(1) \\ &= 21 + 20 + 5 \\ &= 46\end{aligned}$$

Final Answer

$$\Delta = 46$$

Question 9 (iii)

Evaluate the determinant:

$$\begin{vmatrix} 2 & -1 & -2 \\ 0 & 2 & -1 \\ 3 & -5 & 0 \end{vmatrix}$$

TYPE 3 Evaluation of 3x3 Determinants

Given: $\Delta = \begin{vmatrix} 2 & -1 & -2 \\ 0 & 2 & -1 \\ 3 & -5 & 0 \end{vmatrix}$

Concept / Formula Used: Expansion along the first row: $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - b_1 \begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1 \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}$

Solution:

Expanding along the first row:

$$\Delta = 2 \begin{vmatrix} 2 & -1 \\ -5 & 0 \end{vmatrix} - (-1) \begin{vmatrix} 0 & -1 \\ 3 & 0 \end{vmatrix} + (-2) \begin{vmatrix} 0 & 2 \\ 3 & -5 \end{vmatrix}$$

Evaluating each 2×2 minor:

$$\begin{aligned} \begin{vmatrix} 2 & -1 \\ -5 & 0 \end{vmatrix} &= (2)(0) - (-1)(-5) = 0 - 5 = -5 \\ \begin{vmatrix} 0 & -1 \\ 3 & 0 \end{vmatrix} &= (0)(0) - (-1)(3) = 0 + 3 = 3 \\ \begin{vmatrix} 0 & 2 \\ 3 & -5 \end{vmatrix} &= (0)(-5) - (2)(3) = 0 - 6 = -6 \end{aligned}$$

Substituting back:

$$\begin{aligned} \Delta &= 2(-5) + 1(3) - 2(-6) \\ &= -10 + 3 + 12 \\ &= 5 \end{aligned}$$

Final Answer

$$\Delta = 5$$

Question 10

Prove that:

$$\begin{vmatrix} 1 & a & b \\ -a & 1 & c \\ -b & -c & 1 \end{vmatrix} = 1 + a^2 + b^2 + c^2$$

TYPE 3 Evaluation of 3x3 Determinants**★ Likely Exam Question**

To Prove: $\begin{vmatrix} 1 & a & b \\ -a & 1 & c \\ -b & -c & 1 \end{vmatrix} = 1 + a^2 + b^2 + c^2$

Given: $\Delta = \begin{vmatrix} 1 & a & b \\ -a & 1 & c \\ -b & -c & 1 \end{vmatrix}$

Concept / Formula Used: 3×3 **Solution:**Expanding LHS determinant along the first row R_1 :

$$\text{LHS} = 1 \cdot \begin{vmatrix} 1 & c \\ -c & 1 \end{vmatrix} - a \cdot \begin{vmatrix} -a & c \\ -b & 1 \end{vmatrix} + b \cdot \begin{vmatrix} -a & 1 \\ -b & -c \end{vmatrix}$$

Evaluating each 2×2 minor separately:

$$\begin{aligned} \begin{vmatrix} 1 & c \\ -c & 1 \end{vmatrix} &= (1)(1) - (c)(-c) = 1 + c^2 \\ \begin{vmatrix} -a & c \\ -b & 1 \end{vmatrix} &= (-a)(1) - (c)(-b) = -a + bc \\ \begin{vmatrix} -a & 1 \\ -b & -c \end{vmatrix} &= (-a)(-c) - (1)(-b) = ac + b \end{aligned}$$

Substituting these minors back into the expansion:

$$\begin{aligned} \text{LHS} &= 1(1 + c^2) - a(-a + bc) + b(ac + b) \\ &= 1 + c^2 + a^2 - abc + abc + b^2 \end{aligned}$$

Cancelling equal and opposite terms $-abc + abc$:

$$\begin{aligned} \text{LHS} &= 1 + c^2 + a^2 + b^2 \\ &= 1 + a^2 + b^2 + c^2 \\ &= \text{RHS} \end{aligned}$$

Hence Proved*LHS = RHS. Hence proved.*

Question 11

Show that the value of the determinant $\begin{vmatrix} 0 & \tan x & 1 \\ 1 & -\sec x & 0 \\ \sec x & 0 & \tan x \end{vmatrix}$ is independent of x .

TYPE 2 Evaluation of Determinants Using Trigonometric Identities

To Prove: $\Delta = \begin{vmatrix} 0 & \tan x & 1 \\ 1 & -\sec x & 0 \\ \sec x & 0 & \tan x \end{vmatrix}$ is independent of x .

Given: $\Delta = \begin{vmatrix} 0 & \tan x & 1 \\ 1 & -\sec x & 0 \\ \sec x & 0 & \tan x \end{vmatrix}$

Concept / Formula Used: $\sec^2 x - \tan^2 x = 1$

Solution:

Expanding the determinant along row 1 (R_1):

$$\Delta = 0 \cdot \begin{vmatrix} -\sec x & 0 \\ 0 & \tan x \end{vmatrix} - \tan x \cdot \begin{vmatrix} 1 & 0 \\ \sec x & \tan x \end{vmatrix} + 1 \cdot \begin{vmatrix} 1 & -\sec x \\ \sec x & 0 \end{vmatrix}$$

Evaluating the individual 2×2 minors:

$$\begin{aligned} \begin{vmatrix} 1 & 0 \\ \sec x & \tan x \end{vmatrix} &= (1)(\tan x) - (0)(\sec x) = \tan x \\ \begin{vmatrix} 1 & -\sec x \\ \sec x & 0 \end{vmatrix} &= (1)(0) - (-\sec x)(\sec x) = 0 + \sec^2 x = \sec^2 x \end{aligned}$$

Substituting these values back into the expanded expression:

$$\begin{aligned} \Delta &= 0 - \tan x(\tan x) + 1(\sec^2 x) \\ &= -\tan^2 x + \sec^2 x \\ &= \sec^2 x - \tan^2 x \end{aligned}$$

Using the standard identity $\sec^2 x - \tan^2 x = 1$:

$$\Delta = 1$$

Since 1 is a constant value containing no variable x , the determinant is independent of x .

Hence Proved

LHS = 1, which is independent of x . Hence proved.

Question 12

Using cofactors of elements of second row, evaluate $\Delta = \begin{vmatrix} 5 & 3 & 8 \\ 2 & 0 & 1 \\ 1 & 2 & 3 \end{vmatrix}$.

TYPE 7 Expansion of Determinant Using Cofactors

★ Likely Exam Question

Given: $\Delta = \begin{vmatrix} 5 & 3 & 8 \\ 2 & 0 & 1 \\ 1 & 2 & 3 \end{vmatrix}$

Concept / Formula Used: $\Delta = a_{21}A_{21} + a_{22}A_{22} + a_{23}A_{23}$ $A_{ij} = (-1)^{i+j}M_{ij}$

Solution:

Elements of the second row are:

$$a_{21} = 2, \quad a_{22} = 0, \quad a_{23} = 1$$

Calculating cofactor A_{21} :

$$\begin{aligned} A_{21} &= (-1)^{2+1} \begin{vmatrix} 3 & 8 \\ 2 & 3 \end{vmatrix} \\ &= (-1)^3 ((3)(3) - (8)(2)) \\ &= (-1)(9 - 16) \\ &= (-1)(-7) \\ &= 7 \end{aligned}$$

Calculating cofactor A_{22} :

$$\begin{aligned} A_{22} &= (-1)^{2+2} \begin{vmatrix} 5 & 8 \\ 1 & 3 \end{vmatrix} \\ &= (-1)^4 ((5)(3) - (8)(1)) \\ &= (1)(15 - 8) \\ &= 7 \end{aligned}$$

Calculating cofactor A_{23} :

$$\begin{aligned} A_{23} &= (-1)^{2+3} \begin{vmatrix} 5 & 3 \\ 1 & 2 \end{vmatrix} \\ &= (-1)^5 ((5)(2) - (3)(1)) \\ &= (-1)(10 - 3) \\ &= (-1)(7) \\ &= -7 \end{aligned}$$

Expanding Δ along the second row using cofactors:

$$\begin{aligned}\Delta &= a_{21}A_{21} + a_{22}A_{22} + a_{23}A_{23} \\ &= (2)(7) + (0)(7) + (1)(-7) \\ &= 14 + 0 - 7 \\ &= 7\end{aligned}$$

Final Answer

$$\Delta = 7$$

4.2 – Properties of Determinants and Proving Identities

Question 1 (i)

Without expanding, find the value of:

$$\begin{vmatrix} 102 & 18 & 36 \\ 1 & 3 & 4 \\ 17 & 3 & 6 \end{vmatrix}$$

TYPE 8 Evaluation of Determinants Using Properties

To Find: The value of $\begin{vmatrix} 102 & 18 & 36 \\ 1 & 3 & 4 \\ 17 & 3 & 6 \end{vmatrix}$

Solution:

Let the given determinant be Δ :

$$\Delta = \begin{vmatrix} 102 & 18 & 36 \\ 1 & 3 & 4 \\ 17 & 3 & 6 \end{vmatrix}$$

Factoring out 6 from the first row R_1 :

$$\Delta = 6 \begin{vmatrix} 17 & 3 & 6 \\ 1 & 3 & 4 \\ 17 & 3 & 6 \end{vmatrix}$$

Observing row R_1 and row R_3 :

$$\begin{aligned}R_1 &= [17 \quad 3 \quad 6] \\ R_3 &= [17 \quad 3 \quad 6]\end{aligned}$$

Since $R_1 = R_3$, by the property of determinants:

$$\Delta = 6 \times 0$$

$$\Delta = 0$$

Final Answer

0

Question 1 (ii)

Without expanding, find the value of:

$$\begin{vmatrix} 2 & 7 & 65 \\ 3 & 8 & 75 \\ 5 & 9 & 86 \end{vmatrix}$$

TYPE 8 Evaluation of Determinants Using Properties

To Find: The value of $\begin{vmatrix} 2 & 7 & 65 \\ 3 & 8 & 75 \\ 5 & 9 & 86 \end{vmatrix}$

Concept / Formula Used: $C_i \rightarrow C_i - kC_j$

Solution:

Let the given determinant be Δ :

$$\Delta = \begin{vmatrix} 2 & 7 & 65 \\ 3 & 8 & 75 \\ 5 & 9 & 86 \end{vmatrix}$$

Applying the column operation $C_3 \rightarrow C_3 - 9C_2$:

$$\text{First row entry of } C_3 = 65 - 9(7) = 65 - 63 = 2$$

$$\text{Second row entry of } C_3 = 75 - 9(8) = 75 - 72 = 3$$

$$\text{Third row entry of } C_3 = 86 - 9(9) = 86 - 81 = 5$$

Substituting these transformed values back into C_3 :

$$\Delta = \begin{vmatrix} 2 & 7 & 2 \\ 3 & 8 & 3 \\ 5 & 9 & 5 \end{vmatrix}$$

Since column C_1 and column C_3 are identical ($C_1 = C_3$):

$$\Delta = 0$$

Final Answer

0

Question 1 (iii)

Without expanding, find the value of:

$$\begin{vmatrix} 1 & 3 & 5 \\ 7 & 9 & 11 \\ 13 & 15 & 17 \end{vmatrix}$$

TYPE 8 Evaluation of Determinants Using Properties

To Find: The value of $\begin{vmatrix} 1 & 3 & 5 \\ 7 & 9 & 11 \\ 13 & 15 & 17 \end{vmatrix}$

Solution:

Let the given determinant be Δ :

$$\Delta = \begin{vmatrix} 1 & 3 & 5 \\ 7 & 9 & 11 \\ 13 & 15 & 17 \end{vmatrix}$$

Applying elementary row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$R_2 - R_1 = [7 - 1 \quad 9 - 3 \quad 11 - 5] = [6 \quad 6 \quad 6]$$

$$R_3 - R_1 = [13 - 1 \quad 15 - 3 \quad 17 - 5] = [12 \quad 12 \quad 12]$$

Substituting these into the determinant:

$$\Delta = \begin{vmatrix} 1 & 3 & 5 \\ 6 & 6 & 6 \\ 12 & 12 & 12 \end{vmatrix}$$

Since row R_2 and row R_3 are identical ($R_2 = R_3$):

$$\Delta = 0$$

Final Answer

0

Question 2 (i)

Without expanding, find the value of:

$$\begin{vmatrix} 5 & 5 & 5 \\ a & b & c \\ b+c & c+a & a+b \end{vmatrix}$$

TYPE 8 Evaluation of Determinants Using Properties

★ Likely Exam Question

To Find: The value of $\begin{vmatrix} 5 & 5 & 5 \\ a & b & c \\ b+c & c+a & a+b \end{vmatrix}$

Concept / Formula Used: $|kA| = k|A|$

Solution:

Let the given determinant be Δ :

$$\Delta = \begin{vmatrix} 5 & 5 & 5 \\ a & b & c \\ b+c & c+a & a+b \end{vmatrix}$$

Factoring out 5 from the first row R_1 :

$$\Delta = 5 \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ b+c & c+a & a+b \end{vmatrix}$$

Applying the row operation $R_3 \rightarrow R_3 + R_2$:

$$\begin{aligned} R_3 &= [(b+c) + a \quad (c+a) + b \quad (a+b) + c] \\ &= [a+b+c \quad a+b+c \quad a+b+c] \end{aligned}$$

Substituting this back into the determinant:

$$\Delta = 5 \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a+b+c & a+b+c & a+b+c \end{vmatrix}$$

Factoring out $(a+b+c)$ from the third row R_3 :

$$\Delta = 5(a+b+c) \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix}$$

Since row R_1 and row R_3 are identical:

$$\begin{aligned} \Delta &= 5(a+b+c) \times 0 \\ \Delta &= 0 \end{aligned}$$

Final Answer

0

Question 2 (ii)

Without expanding, find the value of:

$$\begin{vmatrix} a & b & c \\ a+2x & b+2y & c+2z \\ x & y & z \end{vmatrix}$$

TYPE 8 Evaluation of Determinants Using Properties

★ **Likely Exam Question**

To Find: The value of $\begin{vmatrix} a & b & c \\ a+2x & b+2y & c+2z \\ x & y & z \end{vmatrix}$

Solution:

Let the given determinant be Δ :

$$\Delta = \begin{vmatrix} a & b & c \\ a+2x & b+2y & c+2z \\ x & y & z \end{vmatrix}$$

Applying the row operation $R_2 \rightarrow R_2 - R_1 - 2R_3$:

$$\text{First entry of } R_2 = (a+2x) - a - 2(x) = 0$$

$$\text{Second entry of } R_2 = (b+2y) - b - 2(y) = 0$$

$$\text{Third entry of } R_2 = (c+2z) - c - 2(z) = 0$$

Substituting these entries into R_2 :

$$\Delta = \begin{vmatrix} a & b & c \\ 0 & 0 & 0 \\ x & y & z \end{vmatrix}$$

Since all elements of row R_2 are zero:

$$\Delta = 0$$

Final Answer

0

Question 3 (i)

Without expanding, find the value of:

$$\begin{vmatrix} a-b & b-c & c-a \\ x-y & y-z & z-x \\ p-q & q-r & r-p \end{vmatrix}$$

TYPE 8 Evaluation of Determinants Using Properties

★ Likely Exam Question

To Find: The value of $\begin{vmatrix} a-b & b-c & c-a \\ x-y & y-z & z-x \\ p-q & q-r & r-p \end{vmatrix}$

Concept / Formula Used: $(u-v) + (v-w) + (w-u) = 0$

Solution:

Let the given determinant be Δ :

$$\Delta = \begin{vmatrix} a-b & b-c & c-a \\ x-y & y-z & z-x \\ p-q & q-r & r-p \end{vmatrix}$$

Applying the column operation $C_1 \rightarrow C_1 + C_2 + C_3$:

$$\text{First row entry of } C_1 = (a-b) + (b-c) + (c-a) = 0$$

$$\text{Second row entry of } C_1 = (x-y) + (y-z) + (z-x) = 0$$

$$\text{Third row entry of } C_1 = (p-q) + (q-r) + (r-p) = 0$$

Substituting these entries into C_1 :

$$\Delta = \begin{vmatrix} 0 & b-c & c-a \\ 0 & y-z & z-x \\ 0 & q-r & r-p \end{vmatrix}$$

Since all elements in the first column C_1 are zero:

$$\Delta = 0$$

Final Answer

0

Question 3 (ii)

Without expanding, find the value of:

$$\begin{vmatrix} \beta\gamma & 1 & \alpha(\beta+\gamma) \\ \gamma\alpha & 1 & \beta(\gamma+\alpha) \\ \alpha\beta & 1 & \gamma(\alpha+\beta) \end{vmatrix}$$

TYPE 8 Evaluation of Determinants Using Properties

★ Likely Exam Question

To Find: The value of $\begin{vmatrix} \beta\gamma & 1 & \alpha(\beta + \gamma) \\ \gamma\alpha & 1 & \beta(\gamma + \alpha) \\ \alpha\beta & 1 & \gamma(\alpha + \beta) \end{vmatrix}$

Solution:

Let the given determinant be Δ :

$$\Delta = \begin{vmatrix} \beta\gamma & 1 & \alpha\beta + \alpha\gamma \\ \gamma\alpha & 1 & \beta\gamma + \beta\alpha \\ \alpha\beta & 1 & \gamma\alpha + \gamma\beta \end{vmatrix}$$

Applying the column operation $C_3 \rightarrow C_3 + C_1$:

First row entry of $C_3 = (\alpha\beta + \alpha\gamma) + \beta\gamma = \alpha\beta + \beta\gamma + \gamma\alpha$

Second row entry of $C_3 = (\beta\gamma + \beta\alpha) + \gamma\alpha = \alpha\beta + \beta\gamma + \gamma\alpha$

Third row entry of $C_3 = (\gamma\alpha + \gamma\beta) + \alpha\beta = \alpha\beta + \beta\gamma + \gamma\alpha$

Substituting back into column C_3 :

$$\Delta = \begin{vmatrix} \beta\gamma & 1 & \alpha\beta + \beta\gamma + \gamma\alpha \\ \gamma\alpha & 1 & \alpha\beta + \beta\gamma + \gamma\alpha \\ \alpha\beta & 1 & \alpha\beta + \beta\gamma + \gamma\alpha \end{vmatrix}$$

Factoring out $(\alpha\beta + \beta\gamma + \gamma\alpha)$ from column C_3 :

$$\Delta = (\alpha\beta + \beta\gamma + \gamma\alpha) \begin{vmatrix} \beta\gamma & 1 & 1 \\ \gamma\alpha & 1 & 1 \\ \alpha\beta & 1 & 1 \end{vmatrix}$$

Since column C_2 and column C_3 are identical:

$$\Delta = (\alpha\beta + \beta\gamma + \gamma\alpha) \times 0$$

$$\Delta = 0$$

Final Answer

0

Question 4 (i)

Without expanding, find the value of:

$$\begin{vmatrix} \frac{1}{a} & 1 & bc \\ \frac{1}{b} & 1 & ca \\ \frac{1}{c} & 1 & ab \end{vmatrix}$$

TYPE 8 Evaluation of Determinants Using Properties

To Find: The value of $\begin{vmatrix} \frac{1}{a} & 1 & bc \\ \frac{1}{b} & 1 & ca \\ \frac{1}{c} & 1 & ab \end{vmatrix}$

Concept / Formula Used: R_1, R_2, R_3 a, b, c abc

Solution:

Let the given determinant be Δ :

$$\Delta = \begin{vmatrix} \frac{1}{a} & 1 & bc \\ \frac{1}{b} & 1 & ca \\ \frac{1}{c} & 1 & ab \end{vmatrix}$$

Multiplying rows R_1, R_2, R_3 by a, b, c respectively and dividing the determinant by abc :

$$\begin{aligned} \Delta &= \frac{1}{abc} \begin{vmatrix} a \cdot \frac{1}{a} & a \cdot 1 & a \cdot bc \\ b \cdot \frac{1}{b} & b \cdot 1 & b \cdot ca \\ c \cdot \frac{1}{c} & c \cdot 1 & c \cdot ab \end{vmatrix} \\ &= \frac{1}{abc} \begin{vmatrix} 1 & a & abc \\ 1 & b & abc \\ 1 & c & abc \end{vmatrix} \end{aligned}$$

Factoring out abc from column C_3 :

$$\begin{aligned} \Delta &= \frac{abc}{abc} \begin{vmatrix} 1 & a & 1 \\ 1 & b & 1 \\ 1 & c & 1 \end{vmatrix} \\ &= 1 \cdot \begin{vmatrix} 1 & a & 1 \\ 1 & b & 1 \\ 1 & c & 1 \end{vmatrix} \end{aligned}$$

Since column C_1 and column C_3 are identical:

$$\Delta = 0$$

Final Answer

0

Question 4 (ii)

Without expanding, find the value of:

$$\begin{vmatrix} a+b & 2a+b & 3a+b \\ 2a+b & 3a+b & 4a+b \\ 4a+b & 5a+b & 6a+b \end{vmatrix}$$

TYPE 8 Evaluation of Determinants Using Properties

To Find: The value of $\begin{vmatrix} a+b & 2a+b & 3a+b \\ 2a+b & 3a+b & 4a+b \\ 4a+b & 5a+b & 6a+b \end{vmatrix}$

Solution:

Let the given determinant be Δ :

$$\Delta = \begin{vmatrix} a+b & 2a+b & 3a+b \\ 2a+b & 3a+b & 4a+b \\ 4a+b & 5a+b & 6a+b \end{vmatrix}$$

Applying the column operations $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_2$:

$$C_2 - C_1 = \begin{bmatrix} (2a+b) - (a+b) \\ (3a+b) - (2a+b) \\ (5a+b) - (4a+b) \end{bmatrix} = \begin{bmatrix} a \\ a \\ a \end{bmatrix}$$

$$C_3 - C_2 = \begin{bmatrix} (3a+b) - (2a+b) \\ (4a+b) - (3a+b) \\ (6a+b) - (5a+b) \end{bmatrix} = \begin{bmatrix} a \\ a \\ a \end{bmatrix}$$

Substituting these into the determinant:

$$\Delta = \begin{vmatrix} a+b & a & a \\ 2a+b & a & a \\ 4a+b & a & a \end{vmatrix}$$

Since column C_2 and column C_3 are identical ($C_2 = C_3$):

$$\Delta = 0$$

Final Answer

0

Question 5

If A is a square matrix of order 2 and $|A| = -5$, find the value of $|3A|$.

TYPE 6 Determinant Properties of Scalar Multiplication

Given: Order of matrix $n = 2$, $|A| = -5$

To Find: The value of $|3A|$

Concept / Formula Used: $A \quad n \quad |kA| = k^n|A|$

Solution:

Given that A is a square matrix of order $n = 2$ and scalar $k = 3$: Using the standard result $|kA| = k^n|A|$ with $n = 2$ and $k = 3$:

$$\begin{aligned} |3A| &= 3^2|A| \\ &= 9|A| \end{aligned}$$

Substituting $|A| = -5$:

$$\begin{aligned}|3A| &= 9 \times (-5) \\ &= -45\end{aligned}$$

Final Answer

-45

Question 6

If A is a square matrix of order 3 and $|A| = 4$, then write the value of $|2A|$.

TYPE 6 Determinant Properties of Scalar Multiplication

Given: Order of matrix $n = 3$, $|A| = 4$

To Find: The value of $|2A|$

Concept / Formula Used: $A \quad n \quad |kA| = k^n|A|$

Solution:

Given $n = 3$, $k = 2$, and $|A| = 4$: Using the standard result $|kA| = k^n|A|$:

$$\begin{aligned}|2A| &= 2^3|A| \\ &= 8|A|\end{aligned}$$

Substituting $|A| = 4$:

$$\begin{aligned}|2A| &= 8 \times 4 \\ &= 32\end{aligned}$$

Final Answer

32

Question 7

If A is a square matrix of order 3 and $|A| = 5$, find the value of $|3A|$.

TYPE 6 Determinant Properties of Scalar Multiplication

Given: Order of matrix $n = 3$, $|A| = 5$

To Find: The value of $|3A|$

Concept / Formula Used: $A \quad n \quad |kA| = k^n|A|$

Solution:

Given $n = 3$, $k = 3$, and $|A| = 5$: Using the standard result $|kA| = k^n|A|$:

$$\begin{aligned}|3A| &= 3^3|A| \\ &= 27|A|\end{aligned}$$

Substituting $|A| = 5$:

$$\begin{aligned}|3A| &= 27 \times 5 \\ &= 135\end{aligned}$$

Final Answer

135

Question 8

If A is a square matrix of order 3 and $|A| = -2$, find the value of $|-5A|$.

TYPE 6 Determinant Properties of Scalar Multiplication

Given: Order of matrix $n = 3$, $|A| = -2$

To Find: The value of $|-5A|$

Concept / Formula Used: A n $|kA| = k^n|A|$

Solution:

Given $n = 3$, $k = -5$, and $|A| = -2$: Using the standard result $|kA| = k^n|A|$:

$$\begin{aligned}|-5A| &= (-5)^3|A| \\ &= -125|A|\end{aligned}$$

Substituting $|A| = -2$:

$$\begin{aligned}|-5A| &= -125 \times (-2) \\ &= 250\end{aligned}$$

Final Answer

250

Question 9

If A is a matrix of order 3×3 and its determinant is 4, then find $|3A|$.

TYPE 6 Determinant Properties of Scalar Multiplication

Given: Order of matrix $n = 3$, $|A| = 4$

To Find: The value of $|3A|$

Concept / Formula Used: $A \quad n \quad |kA| = k^n|A|$

Solution:

Given $n = 3$, $k = 3$, and $|A| = 4$: Using the standard result $|kA| = k^n|A|$:

$$\begin{aligned}|3A| &= 3^3|A| \\ &= 27|A|\end{aligned}$$

Substituting $|A| = 4$:

$$\begin{aligned}|3A| &= 27 \times 4 \\ &= 108\end{aligned}$$

Final Answer

108

Question 10

If A is a square matrix of order 3 and $|3A| = k|A|$, then find the value of k .

TYPE 6 Determinant Properties of Scalar Multiplication

Given: Order of matrix $n = 3$, $|3A| = k|A|$

To Find: The value of k

Concept / Formula Used: $A \quad n \quad |cA| = c^n|A|$

Solution:

Using the property $|cA| = c^n|A|$ with $c = 3$ and $n = 3$:

$$\begin{aligned}|3A| &= 3^3|A| \\ |3A| &= 27|A|\end{aligned}$$

Equating this to the given relation $|3A| = k|A|$:

$$27|A| = k|A|$$

Comparing both sides gives:

$$k = 27$$

Final Answer

27

Question 11 (i)

If A is a square matrix such that $|A| = 5$, write the value of $|AA^T|$.

TYPE 9 Determinant Properties of Matrix Operations**Given:** $|A| = 5$ **To Find:** The value of $|AA^T|$ **Concept / Formula Used:** $|A^T| = |A|$ $|AB| = |A||B|$ **Solution:**Using the product rule for determinants $|AB| = |A||B|$:

$$|AA^T| = |A| \cdot |A^T|$$

Using the property that the determinant of a transpose equals the determinant of the matrix, $|A^T| = |A|$:

$$\begin{aligned} |AA^T| &= |A| \cdot |A| \\ &= |A|^2 \end{aligned}$$

Substituting $|A| = 5$:

$$\begin{aligned} |AA^T| &= 5^2 \\ &= 25 \end{aligned}$$

Final Answer

25

Question 11 (ii)If A is a square matrix satisfying $AA' = I$, write the value of $|A|$.**TYPE 9 Determinant Properties of Matrix Operations****Given:** $AA' = I$ **To Find:** The value of $|A|$ **Concept / Formula Used:** $|AB| = |A||B|$ $|A'| = |A|$ $|I| = 1$ **Solution:**Taking the determinant on both sides of $AA' = I$:

$$|AA'| = |I|$$

Applying product property $|AB| = |A||B|$:

$$|A| \cdot |A'| = 1$$

Since $|A'| = |A|$:

$$\begin{aligned}|A| \cdot |A| &= 1 \\ |A|^2 &= 1 \\ |A| &= \pm 1\end{aligned}$$

Final Answer

± 1

Question 12

If A is a square matrix of order 3 such that $A^2 = 2A$, then find the value of $|A|$.

TYPE 9 Determinant Properties of Matrix Operations

Given: Order of matrix $n = 3$, $A^2 = 2A$

To Find: The value of $|A|$

Concept / Formula Used: $|A^2| = |A|^2$ $|kA| = k^n|A|$ n

Solution:

Given equation:

$$A^2 = 2A$$

Taking determinant on both sides:

$$|A^2| = |2A|$$

Using properties $|A^2| = |A|^2$ and $|2A| = 2^3|A|$ for $n = 3$:

$$\begin{aligned}|A|^2 &= 8|A| \\ |A|^2 - 8|A| &= 0 \\ |A|(|A| - 8) &= 0\end{aligned}$$

Solving for $|A|$ gives two possible solutions:

$$|A| = 0 \quad \text{or} \quad |A| = 8$$

Final Answer

0 or 8

Question 13

If $A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -4 \\ 3 & -2 \end{bmatrix}$, find $|AB|$.

TYPE 9 Determinant Properties of Matrix Operations

Given: $A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -4 \\ 3 & -2 \end{bmatrix}$

To Find: The value of $|AB|$

Concept / Formula Used: $2 \times 2 \quad \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad ad - bc \quad |AB| = |A||B|$

Solution:

Calculating $|A|$:

$$\begin{aligned} |A| &= \begin{vmatrix} 1 & 2 \\ 3 & -1 \end{vmatrix} \\ &= (1)(-1) - (2)(3) \\ &= -1 - 6 \\ &= -7 \end{aligned}$$

Calculating $|B|$:

$$\begin{aligned} |B| &= \begin{vmatrix} 1 & -4 \\ 3 & -2 \end{vmatrix} \\ &= (1)(-2) - (-4)(3) \\ &= -2 + 12 \\ &= 10 \end{aligned}$$

Using the determinant product property $|AB| = |A||B|$:

$$\begin{aligned} |AB| &= (-7) \times 10 \\ &= -70 \end{aligned}$$

Final Answer

-70

Question 14

If $A = \begin{bmatrix} x & 2 \\ 2 & x \end{bmatrix}$ and $|A^3| = 125$, find the value(s) of x .

TYPE 9 Determinant Properties of Matrix Operations

Given: $A = \begin{bmatrix} x & 2 \\ 2 & x \end{bmatrix}$, $|A^3| = 125$

To Find: The value(s) of x

Concept / Formula Used: $|A^k| = |A|^k$ $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

Solution:

Using the property $|A^3| = |A|^3$:

$$\begin{aligned}|A|^3 &= 125 \\ |A| &= \sqrt[3]{125} \\ |A| &= 5\end{aligned}$$

Now, evaluating $|A|$ directly from A :

$$\begin{aligned}|A| &= \begin{vmatrix} x & 2 \\ 2 & x \end{vmatrix} \\ &= x \cdot x - 2 \cdot 2 \\ &= x^2 - 4\end{aligned}$$

Equating the two expressions for $|A|$:

$$\begin{aligned}x^2 - 4 &= 5 \\ x^2 &= 5 + 4 \\ x^2 &= 9 \\ x &= \pm 3\end{aligned}$$

Final Answer

$$x = \pm 3$$

Question 15

Using properties of determinants, solve the following equation for x :

$$\begin{vmatrix} x+a & x & x \\ x & x+a & x \\ x & x & x+a \end{vmatrix} = 0$$

TYPE 5 Solving Determinant Equations

Given: $\begin{vmatrix} x+a & x & x \\ x & x+a & x \\ x & x & x+a \end{vmatrix} = 0$

To Find: The value of x

Concept / Formula Used: $C_1 \rightarrow C_1 + C_2 + C_3$

Solution:

Let the given determinant equation be $\Delta = 0$:

$$\begin{vmatrix} x+a & x & x \\ x & x+a & x \\ x & x & x+a \end{vmatrix} = 0$$

Applying column operation $C_1 \rightarrow C_1 + C_2 + C_3$:

$$\begin{vmatrix} (x+a) + x + x & x & x \\ x + (x+a) + x & x+a & x \\ x + x + (x+a) & x & x+a \end{vmatrix} = 0$$

$$\begin{vmatrix} 3x+a & x & x \\ 3x+a & x+a & x \\ 3x+a & x & x+a \end{vmatrix} = 0$$

Factoring out $(3x+a)$ from column C_1 :

$$(3x+a) \begin{vmatrix} 1 & x & x \\ 1 & x+a & x \\ 1 & x & x+a \end{vmatrix} = 0$$

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$(3x+a) \begin{vmatrix} 1 & x & x \\ 0 & a & 0 \\ 0 & 0 & a \end{vmatrix} = 0$$

Expanding along column C_1 :

$$\begin{aligned} (3x+a) \cdot 1 \cdot (a \cdot a - 0 \cdot 0) &= 0 \\ a^2(3x+a) &= 0 \end{aligned}$$

Assuming $a \neq 0$:

$$\begin{aligned} 3x+a &= 0 \\ 3x &= -a \\ x &= -\frac{a}{3} \end{aligned}$$

Final Answer

$$x = -\frac{a}{3}$$

Question 16

Show that one root of the equation

$$\begin{vmatrix} x+a & b & c \\ b & x+c & a \\ c & a & x+b \end{vmatrix} = 0$$

is $-(a+b+c)$.

TYPE 5 Solving Determinant Equations

To Prove: One root of $\begin{vmatrix} x+a & b & c \\ b & x+c & a \\ c & a & x+b \end{vmatrix} = 0$ is $x = -(a+b+c)$

Concept / Formula Used: $(x-r) \quad \Delta(x) = 0 \quad r$

Solution:

Consider the given equation:

$$\begin{vmatrix} x+a & b & c \\ b & x+c & a \\ c & a & x+b \end{vmatrix} = 0$$

Applying column operation $C_1 \rightarrow C_1 + C_2 + C_3$:

$$\text{First row entry of } C_1 = (x+a) + b + c = x + a + b + c$$

$$\text{Second row entry of } C_1 = b + (x+c) + a = x + a + b + c$$

$$\text{Third row entry of } C_1 = c + a + (x+b) = x + a + b + c$$

Substituting these into column C_1 :

$$\begin{vmatrix} x+a+b+c & b & c \\ x+a+b+c & x+c & a \\ x+a+b+c & a & x+b \end{vmatrix} = 0$$

Factoring out $(x+a+b+c)$ from column C_1 :

$$(x+a+b+c) \begin{vmatrix} 1 & b & c \\ 1 & x+c & a \\ 1 & a & x+b \end{vmatrix} = 0$$

Setting the linear factor to zero:

$$x + a + b + c = 0$$

$$x = -(a+b+c)$$

Thus, $x = -(a+b+c)$ is a root of the equation.

Hence Proved

One root of the given equation is $-(a+b+c)$.

Question 17

Without expanding, prove that

$$\begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix} = \begin{vmatrix} x & y & z \\ p & q & r \\ a & b & c \end{vmatrix} = \begin{vmatrix} y & b & q \\ x & a & p \\ z & c & r \end{vmatrix}$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix} = \begin{vmatrix} x & y & z \\ p & q & r \\ a & b & c \end{vmatrix} = \begin{vmatrix} y & b & q \\ x & a & p \\ z & c & r \end{vmatrix}$

Concept / Formula Used: (-1)

Solution:

Let $\Delta_1 = \begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix}$.

Step 1: Transform Δ_1 to Δ_2 : Interchanging $R_1 \leftrightarrow R_2$:

$$\Delta_1 = - \begin{vmatrix} x & y & z \\ a & b & c \\ p & q & r \end{vmatrix}$$

Interchanging $R_2 \leftrightarrow R_3$:

$$\begin{aligned} \Delta_1 &= (-1)(-1) \begin{vmatrix} x & y & z \\ p & q & r \\ a & b & c \end{vmatrix} \\ &= \begin{vmatrix} x & y & z \\ p & q & r \\ a & b & c \end{vmatrix} = \Delta_2 \end{aligned}$$

Step 2: Transform Δ_1 to Δ_3 : Taking transpose of Δ_1 (since $|A^T| = |A|$):

$$\Delta_1 = \begin{vmatrix} a & x & p \\ b & y & q \\ c & z & r \end{vmatrix}$$

Interchanging $R_1 \leftrightarrow R_2$:

$$\Delta_1 = - \begin{vmatrix} b & y & q \\ a & x & p \\ c & z & r \end{vmatrix}$$

Interchanging $C_1 \leftrightarrow C_2$:

$$\begin{aligned} \Delta_1 &= (-1)(-1) \begin{vmatrix} y & b & q \\ x & a & p \\ z & c & r \end{vmatrix} \\ &= \begin{vmatrix} y & b & q \\ x & a & p \\ z & c & r \end{vmatrix} = \Delta_3 \end{aligned}$$

Thus, $\Delta_1 = \Delta_2 = \Delta_3$.

Hence Proved

$\Delta_1 = \Delta_2 = \Delta_3$, hence proved.

Question 18

Without expanding, prove that

$$\begin{vmatrix} a & a^2 & bc \\ b & b^2 & ca \\ c & c^2 & ab \end{vmatrix} = \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} a & a^2 & bc \\ b & b^2 & ca \\ c & c^2 & ab \end{vmatrix} = \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}$

Concept / Formula Used: a, b, c abc

Solution:

Let LHS = Δ :

$$\Delta = \begin{vmatrix} a & a^2 & bc \\ b & b^2 & ca \\ c & c^2 & ab \end{vmatrix}$$

Multiplying rows R_1, R_2, R_3 by a, b, c respectively and dividing by abc :

$$\begin{aligned} \Delta &= \frac{1}{abc} \begin{vmatrix} a \cdot a & a \cdot a^2 & a \cdot bc \\ b \cdot b & b \cdot b^2 & b \cdot ca \\ c \cdot c & c \cdot c^2 & c \cdot ab \end{vmatrix} \\ &= \frac{1}{abc} \begin{vmatrix} a^2 & a^3 & abc \\ b^2 & b^3 & abc \\ c^2 & c^3 & abc \end{vmatrix} \end{aligned}$$

Factoring out abc from column C_3 :

$$\begin{aligned} \Delta &= \frac{abc}{abc} \begin{vmatrix} a^2 & a^3 & 1 \\ b^2 & b^3 & 1 \\ c^2 & c^3 & 1 \end{vmatrix} \\ &= \begin{vmatrix} a^2 & a^3 & 1 \\ b^2 & b^3 & 1 \\ c^2 & c^3 & 1 \end{vmatrix} \end{aligned}$$

Interchanging $C_2 \leftrightarrow C_3$:

$$\Delta = - \begin{vmatrix} a^2 & 1 & a^3 \\ b^2 & 1 & b^3 \\ c^2 & 1 & c^3 \end{vmatrix}$$

Interchanging $C_1 \leftrightarrow C_2$:

$$\begin{aligned}\Delta &= (-1)(-1) \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} \\ &= \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = \text{RHS}\end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 19 (i)

Use properties of determinants to solve for x :

$$\begin{vmatrix} x+a & b & c \\ c & x+b & a \\ a & b & x+c \end{vmatrix} = 0$$

TYPE 5 Solving Determinant Equations

★ Likely Exam Question

Given: $\begin{vmatrix} x+a & b & c \\ c & x+b & a \\ a & b & x+c \end{vmatrix} = 0$

To Find: The values of x

Concept / Formula Used: $C_1 \rightarrow C_1 + C_2 + C_3 \quad (x + a + b + c)$

Solution:

Consider the given equation:

$$\begin{vmatrix} x+a & b & c \\ c & x+b & a \\ a & b & x+c \end{vmatrix} = 0$$

Applying $C_1 \rightarrow C_1 + C_2 + C_3$:

$$\begin{vmatrix} x+a+b+c & b & c \\ x+a+b+c & x+b & a \\ x+a+b+c & b & x+c \end{vmatrix} = 0$$

Factoring out $(x + a + b + c)$ from column C_1 :

$$(x+a+b+c) \begin{vmatrix} 1 & b & c \\ 1 & x+b & a \\ 1 & b & x+c \end{vmatrix} = 0$$

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$(x + a + b + c) \begin{vmatrix} 1 & b & c \\ 0 & x & a - c \\ 0 & 0 & x \end{vmatrix} = 0$$

Expanding along column C_1 :

$$(x + a + b + c) \cdot 1 \cdot (x \cdot x - (a - c) \cdot 0) = 0$$

$$x^2(x + a + b + c) = 0$$

Equating each factor to zero:

$$x^2 = 0 \implies x = 0$$

$$x + a + b + c = 0 \implies x = -(a + b + c)$$

Final Answer

$$x = 0 \text{ or } x = -(a + b + c)$$

Question 19 (ii)

Use properties of determinants to solve for x :

$$\begin{vmatrix} 3-x & -1 & 1 \\ -1 & 5-x & -1 \\ 1 & -1 & 3-x \end{vmatrix} = 0$$

TYPE 5 Solving Determinant Equations

Given: $\begin{vmatrix} 3-x & -1 & 1 \\ -1 & 5-x & -1 \\ 1 & -1 & 3-x \end{vmatrix} = 0$

To Find: The values of x

Solution:

Consider the equation:

$$\begin{vmatrix} 3-x & -1 & 1 \\ -1 & 5-x & -1 \\ 1 & -1 & 3-x \end{vmatrix} = 0$$

Applying $R_1 \rightarrow R_1 + R_3$:

$$R_1 = [(3-x) + 1 \quad -1 + (-1) \quad 1 + (3-x)]$$

$$= [4-x \quad -2 \quad 4-x]$$

Substituting into R_1 :

$$\begin{vmatrix} 4-x & -2 & 4-x \\ -1 & 5-x & -1 \\ 1 & -1 & 3-x \end{vmatrix} = 0$$

Factoring out $(4 - x)$ from row R_1 :

$$(4 - x) \begin{vmatrix} 1 & -2 & 1 \\ -1 & 5 - x & -1 \\ 1 & -1 & 3 - x \end{vmatrix} = 0$$

Applying $C_3 \rightarrow C_3 - C_1$:

$$(4 - x) \begin{vmatrix} 1 & -2 & 0 \\ -1 & 5 - x & 0 \\ 1 & -1 & 2 - x \end{vmatrix} = 0$$

Expanding along column C_3 :

$$\begin{aligned} (4 - x)(2 - x) \begin{vmatrix} 1 & -2 \\ -1 & 5 - x \end{vmatrix} &= 0 \\ (4 - x)(2 - x)[(1)(5 - x) - (-2)(-1)] &= 0 \\ (4 - x)(2 - x)(5 - x - 2) &= 0 \\ (4 - x)(2 - x)(3 - x) &= 0 \end{aligned}$$

Solving for roots:

$$x = 2, 3, 4$$

Final Answer

$$x = 2, 3, 4$$

Question 20

If $s = a + b + c$, then prove that

$$\begin{vmatrix} s + c & a & b \\ c & s + a & b \\ c & a & s + b \end{vmatrix} = 2s^3$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} s + c & a & b \\ c & s + a & b \\ c & a & s + b \end{vmatrix} = 2s^3$ given $s = a + b + c$

Solution:

Let LHS = Δ :

$$\Delta = \begin{vmatrix} s + c & a & b \\ c & s + a & b \\ c & a & s + b \end{vmatrix}$$

Applying row operations $R_1 \rightarrow R_1 - R_3$ and $R_2 \rightarrow R_2 - R_3$:

$$\begin{aligned} R_1 - R_3 &= [(s + c) - c \quad a - a \quad b - (s + b)] = [s \quad 0 \quad -s] \\ R_2 - R_3 &= [c - c \quad (s + a) - a \quad b - (s + b)] = [0 \quad s \quad -s] \end{aligned}$$

Substituting these into the determinant:

$$\Delta = \begin{vmatrix} s & 0 & -s \\ 0 & s & -s \\ c & a & s+b \end{vmatrix}$$

Factoring out s from R_1 and s from R_2 :

$$\Delta = s^2 \begin{vmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ c & a & s+b \end{vmatrix}$$

Applying column operation $C_3 \rightarrow C_3 + C_1 + C_2$:

$$\text{Third entry of } C_3 = (s+b) + c + a = s + (a+b+c)$$

Given $a+b+c = s$, this entry becomes $s + s = 2s$:

$$\Delta = s^2 \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ c & a & 2s \end{vmatrix}$$

Expanding along row R_1 :

$$\begin{aligned} \Delta &= s^2 \cdot 1 \cdot (1 \cdot 2s - 0 \cdot a) \\ &= s^2 \cdot 2s \\ &= 2s^3 = \text{RHS} \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 21

If a, b and c are in A.P., then find the value of

$$\begin{vmatrix} 2y+4 & 5y+7 & 8y+a \\ 3y+5 & 6y+8 & 9y+b \\ 4y+6 & 7y+9 & 10y+c \end{vmatrix}$$

TYPE 8 Evaluation of Determinants Using Properties

★ Likely Exam Question

Given: a, b, c are in A.P.

To Find: The value of $\begin{vmatrix} 2y+4 & 5y+7 & 8y+a \\ 3y+5 & 6y+8 & 9y+b \\ 4y+6 & 7y+9 & 10y+c \end{vmatrix}$

Concept / Formula Used: a, b, c $2b = a + c \implies a - 2b + c = 0$

Solution:

Since a, b, c are in Arithmetic Progression:

$$\begin{aligned} 2b &= a + c \\ a - 2b + c &= 0 \end{aligned}$$

Let the given determinant be Δ :

$$\Delta = \begin{vmatrix} 2y + 4 & 5y + 7 & 8y + a \\ 3y + 5 & 6y + 8 & 9y + b \\ 4y + 6 & 7y + 9 & 10y + c \end{vmatrix}$$

Applying the row operation $R_1 \rightarrow R_1 - 2R_2 + R_3$:

$$\begin{aligned} \text{First entry of } R_1 &= (2y + 4) - 2(3y + 5) + (4y + 6) \\ &= 2y - 6y + 4y + 4 - 10 + 6 = 0 \end{aligned}$$

$$\begin{aligned} \text{Second entry of } R_1 &= (5y + 7) - 2(6y + 8) + (7y + 9) \\ &= 5y - 12y + 7y + 7 - 16 + 9 = 0 \end{aligned}$$

$$\begin{aligned} \text{Third entry of } R_1 &= (8y + a) - 2(9y + b) + (10y + c) \\ &= 8y - 18y + 10y + a - 2b + c \\ &= a - 2b + c = 0 \end{aligned}$$

Substituting these entries into R_1 :

$$\Delta = \begin{vmatrix} 0 & 0 & 0 \\ 3y + 5 & 6y + 8 & 9y + b \\ 4y + 6 & 7y + 9 & 10y + c \end{vmatrix}$$

Since row R_1 is entirely composed of zeros:

$$\Delta = 0$$

Final Answer

0

Question 22

Using the properties of determinant, prove that

$$\begin{vmatrix} x - 3 & x - 4 & x - \alpha \\ x - 2 & x - 3 & x - \beta \\ x - 1 & x - 2 & x - \gamma \end{vmatrix} = 0$$

where α, β and γ are in A.P.

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} x-3 & x-4 & x-\alpha \\ x-2 & x-3 & x-\beta \\ x-1 & x-2 & x-\gamma \end{vmatrix} = 0$ given α, β, γ in A.P.

Concept / Formula Used: $\alpha, \beta, \gamma \implies 2\beta = \alpha + \gamma \implies \alpha - 2\beta + \gamma = 0$

Solution:

Given α, β, γ are in A.P.:

$$\begin{aligned} 2\beta &= \alpha + \gamma \\ \alpha - 2\beta + \gamma &= 0 \end{aligned}$$

Let LHS = Δ :

$$\Delta = \begin{vmatrix} x-3 & x-4 & x-\alpha \\ x-2 & x-3 & x-\beta \\ x-1 & x-2 & x-\gamma \end{vmatrix}$$

Applying the row operation $R_1 \rightarrow R_1 - 2R_2 + R_3$:

$$\begin{aligned} \text{First entry of } R_1 &= (x-3) - 2(x-2) + (x-1) \\ &= x - 2x + x - 3 + 4 - 1 = 0 \end{aligned}$$

$$\begin{aligned} \text{Second entry of } R_1 &= (x-4) - 2(x-3) + (x-2) \\ &= x - 2x + x - 4 + 6 - 2 = 0 \end{aligned}$$

$$\begin{aligned} \text{Third entry of } R_1 &= (x-\alpha) - 2(x-\beta) + (x-\gamma) \\ &= x - 2x + x - \alpha + 2\beta - \gamma \\ &= -(\alpha - 2\beta + \gamma) = 0 \end{aligned}$$

Substituting these entries back into R_1 :

$$\Delta = \begin{vmatrix} 0 & 0 & 0 \\ x-2 & x-3 & x-\beta \\ x-1 & x-2 & x-\gamma \end{vmatrix}$$

Since all elements of row R_1 are zero:

$$\Delta = 0 = \text{RHS}$$

Hence Proved

LHS = RHS, hence proved.

Question 23

Using properties of determinants, prove that:

$$\begin{vmatrix} a+b & b+c & c+a \\ b+c & c+a & a+b \\ c+a & a+b & b+c \end{vmatrix} = 2 \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove:
$$\begin{vmatrix} a+b & b+c & c+a \\ b+c & c+a & a+b \\ c+a & a+b & b+c \end{vmatrix} = 2 \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$$

Solution:

Let LHS = Δ :

$$\Delta = \begin{vmatrix} a+b & b+c & c+a \\ b+c & c+a & a+b \\ c+a & a+b & b+c \end{vmatrix}$$

Applying column operation $C_1 \rightarrow C_1 + C_2 + C_3$:

$$\Delta = \begin{vmatrix} 2(a+b+c) & b+c & c+a \\ 2(a+b+c) & c+a & a+b \\ 2(a+b+c) & a+b & b+c \end{vmatrix}$$

Factoring out 2 from column C_1 :

$$\Delta = 2 \begin{vmatrix} a+b+c & b+c & c+a \\ a+b+c & c+a & a+b \\ a+b+c & a+b & b+c \end{vmatrix}$$

Applying $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_1$:

$$C_2 - C_1 = \begin{bmatrix} (b+c) - (a+b+c) \\ (c+a) - (a+b+c) \\ (a+b) - (a+b+c) \end{bmatrix} = \begin{bmatrix} -a \\ -b \\ -c \end{bmatrix}$$

$$C_3 - C_1 = \begin{bmatrix} (c+a) - (a+b+c) \\ (a+b) - (a+b+c) \\ (b+c) - (a+b+c) \end{bmatrix} = \begin{bmatrix} -b \\ -c \\ -a \end{bmatrix}$$

Substituting these into column C_2 and C_3 :

$$\Delta = 2 \begin{vmatrix} a+b+c & -a & -b \\ a+b+c & -b & -c \\ a+b+c & -c & -a \end{vmatrix}$$

Factoring out (-1) from C_2 and (-1) from C_3 :

$$\Delta = 2(-1)(-1) \begin{vmatrix} a+b+c & a & b \\ a+b+c & b & c \\ a+b+c & c & a \end{vmatrix}$$

$$= 2 \begin{vmatrix} a+b+c & a & b \\ a+b+c & b & c \\ a+b+c & c & a \end{vmatrix}$$

Applying $C_1 \rightarrow C_1 - C_2 - C_3$:

$$C_1 = \begin{bmatrix} (a+b+c) - a - b \\ (a+b+c) - b - c \\ (a+b+c) - c - a \end{bmatrix} = \begin{bmatrix} c \\ a \\ b \end{bmatrix}$$

So we have:

$$\Delta = 2 \begin{vmatrix} c & a & b \\ a & b & c \\ b & c & a \end{vmatrix}$$

Interchanging $C_1 \leftrightarrow C_3$:

$$\Delta = -2 \begin{vmatrix} b & a & c \\ c & b & a \\ a & c & b \end{vmatrix}$$

Interchanging $C_1 \leftrightarrow C_2$:

$$\begin{aligned} \Delta &= (-2)(-1) \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \\ &= 2 \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = \text{RHS} \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 24 (i)

Prove that

$$\begin{vmatrix} 1+a & b & c \\ a & 1+b & c \\ a & b & 1+c \end{vmatrix} = 1+a+b+c$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} 1+a & b & c \\ a & 1+b & c \\ a & b & 1+c \end{vmatrix} = 1+a+b+c$

Concept / Formula Used: $C_1 \rightarrow C_1 + C_2 + C_3$

Solution:

Let $\text{LHS} = \Delta$:

$$\Delta = \begin{vmatrix} 1+a & b & c \\ a & 1+b & c \\ a & b & 1+c \end{vmatrix}$$

Applying column operation $C_1 \rightarrow C_1 + C_2 + C_3$:

$$\Delta = \begin{vmatrix} 1+a+b+c & b & c \\ 1+a+b+c & 1+b & c \\ 1+a+b+c & b & 1+c \end{vmatrix}$$

Factoring out $(1+a+b+c)$ from column C_1 :

$$\Delta = (1+a+b+c) \begin{vmatrix} 1 & b & c \\ 1 & 1+b & c \\ 1 & b & 1+c \end{vmatrix}$$

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$\begin{aligned} R_2 - R_1 &= [1-1 \quad (1+b)-b \quad c-c] = [0 \quad 1 \quad 0] \\ R_3 - R_1 &= [1-1 \quad b-b \quad (1+c)-c] = [0 \quad 0 \quad 1] \end{aligned}$$

Substituting these into the determinant:

$$\Delta = (1+a+b+c) \begin{vmatrix} 1 & b & c \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

Expanding along column C_1 :

$$\begin{aligned} \Delta &= (1+a+b+c) \cdot 1 \cdot (1 \cdot 1 - 0 \cdot 0) \\ &= 1+a+b+c = \text{RHS} \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 24 (ii)

Prove that

$$\begin{vmatrix} a+x & y & z \\ x & a+y & z \\ x & y & a+z \end{vmatrix} = a^2(a+x+y+z)$$

TYPE 10 Proving Determinant Identities Using Properties

★ Likely Exam Question

To Prove: $\begin{vmatrix} a+x & y & z \\ x & a+y & z \\ x & y & a+z \end{vmatrix} = a^2(a+x+y+z)$

Concept / Formula Used: $C_1 \rightarrow C_1 + C_2 + C_3$ $(a+x+y+z)$

Solution:

Let $LHS = \Delta$:

$$\Delta = \begin{vmatrix} a+x & y & z \\ x & a+y & z \\ x & y & a+z \end{vmatrix}$$

Applying column operation $C_1 \rightarrow C_1 + C_2 + C_3$:

$$\Delta = \begin{vmatrix} a+x+y+z & y & z \\ a+x+y+z & a+y & z \\ a+x+y+z & y & a+z \end{vmatrix}$$

Factoring out $(a+x+y+z)$ from column C_1 :

$$\Delta = (a+x+y+z) \begin{vmatrix} 1 & y & z \\ 1 & a+y & z \\ 1 & y & a+z \end{vmatrix}$$

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$\begin{aligned} R_2 - R_1 &= [1-1 \quad (a+y)-y \quad z-z] = [0 \quad a \quad 0] \\ R_3 - R_1 &= [1-1 \quad y-y \quad (a+z)-z] = [0 \quad 0 \quad a] \end{aligned}$$

Substituting these into the determinant:

$$\Delta = (a+x+y+z) \begin{vmatrix} 1 & y & z \\ 0 & a & 0 \\ 0 & 0 & a \end{vmatrix}$$

Expanding along column C_1 :

$$\begin{aligned} \Delta &= (a+x+y+z) \cdot 1 \cdot (a \cdot a - 0 \cdot 0) \\ &= a^2(a+x+y+z) = RHS \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 24 (iii)

Prove that

$$\begin{vmatrix} y+k & y & y \\ y & y+k & y \\ y & y & y+k \end{vmatrix} = k^2(3y+k)$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} y+k & y & y \\ y & y+k & y \\ y & y & y+k \end{vmatrix} = k^2(3y+k)$

Concept / Formula Used: $C_1 \rightarrow C_1 + C_2 + C_3$ $(3y + k)$

Solution:

Let LHS = Δ :

$$\Delta = \begin{vmatrix} y+k & y & y \\ y & y+k & y \\ y & y & y+k \end{vmatrix}$$

Applying column operation $C_1 \rightarrow C_1 + C_2 + C_3$:

$$\Delta = \begin{vmatrix} 3y+k & y & y \\ 3y+k & y+k & y \\ 3y+k & y & y+k \end{vmatrix}$$

Factoring out $(3y + k)$ from column C_1 :

$$\Delta = (3y + k) \begin{vmatrix} 1 & y & y \\ 1 & y+k & y \\ 1 & y & y+k \end{vmatrix}$$

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$R_2 - R_1 = [1 - 1 \quad (y+k) - y \quad y - y] = [0 \quad k \quad 0]$$

$$R_3 - R_1 = [1 - 1 \quad y - y \quad (y+k) - y] = [0 \quad 0 \quad k]$$

Substituting these into the determinant:

$$\Delta = (3y + k) \begin{vmatrix} 1 & y & y \\ 0 & k & 0 \\ 0 & 0 & k \end{vmatrix}$$

Expanding along column C_1 :

$$\begin{aligned} \Delta &= (3y + k) \cdot 1 \cdot (k \cdot k - 0 \cdot 0) \\ &= k^2(3y + k) = \text{RHS} \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 24 (iv)

Prove that

$$\begin{vmatrix} x+4 & 2x & 2x \\ 2x & x+4 & 2x \\ 2x & 2x & x+4 \end{vmatrix} = (5x+4)(4-x)^2$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} x+4 & 2x & 2x \\ 2x & x+4 & 2x \\ 2x & 2x & x+4 \end{vmatrix} = (5x+4)(4-x)^2$

Concept / Formula Used: $C_1 \rightarrow C_1 + C_2 + C_3$ $(5x + 4)$

Solution:

Let LHS = Δ :

$$\Delta = \begin{vmatrix} x+4 & 2x & 2x \\ 2x & x+4 & 2x \\ 2x & 2x & x+4 \end{vmatrix}$$

Applying column operation $C_1 \rightarrow C_1 + C_2 + C_3$:

$$\text{First entry of } C_1 = (x+4) + 2x + 2x = 5x + 4$$

$$\text{Second entry of } C_1 = 2x + (x+4) + 2x = 5x + 4$$

$$\text{Third entry of } C_1 = 2x + 2x + (x+4) = 5x + 4$$

Substituting into C_1 :

$$\Delta = \begin{vmatrix} 5x+4 & 2x & 2x \\ 5x+4 & x+4 & 2x \\ 5x+4 & 2x & x+4 \end{vmatrix}$$

Factoring out $(5x + 4)$ from column C_1 :

$$\Delta = (5x + 4) \begin{vmatrix} 1 & 2x & 2x \\ 1 & x+4 & 2x \\ 1 & 2x & x+4 \end{vmatrix}$$

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$R_2 - R_1 = [1 - 1 \quad (x+4) - 2x \quad 2x - 2x] = [0 \quad 4 - x \quad 0]$$

$$R_3 - R_1 = [1 - 1 \quad 2x - 2x \quad (x+4) - 2x] = [0 \quad 0 \quad 4 - x]$$

Substituting these into the determinant:

$$\Delta = (5x + 4) \begin{vmatrix} 1 & 2x & 2x \\ 0 & 4 - x & 0 \\ 0 & 0 & 4 - x \end{vmatrix}$$

Expanding along column C_1 :

$$\begin{aligned} \Delta &= (5x + 4) \cdot 1 \cdot [(4 - x)(4 - x) - 0 \cdot 0] \\ &= (5x + 4)(4 - x)^2 = \text{RHS} \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 25

Show that:

$$\begin{vmatrix} a+b+nc & na-a & nb-b \\ nc-c & b+c+na & nb-b \\ nc-c & na-a & c+a+nb \end{vmatrix} = n(a+b+c)^3$$

TYPE 10 Proving Determinant Identities Using Properties**★ Likely Exam Question**

To Prove: $\begin{vmatrix} a+b+nc & na-a & nb-b \\ nc-c & b+c+na & nb-b \\ nc-c & na-a & c+a+nb \end{vmatrix} = n(a+b+c)^3$

Concept / Formula Used: $\det(A) = \det(A^T)$

$$\begin{vmatrix} ka_1 & b_1 & c_1 \\ ka_2 & b_2 & c_2 \\ ka_3 & b_3 & c_3 \end{vmatrix} = k \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

Solution:Let the given determinant be Δ :

$$\Delta = \begin{vmatrix} a+b+nc & (n-1)a & (n-1)b \\ (n-1)c & b+c+na & (n-1)b \\ (n-1)c & (n-1)a & c+a+nb \end{vmatrix}$$

Applying the column operation $C_1 \rightarrow C_1 + C_2 + C_3$:Row 1 element of $C_1 = (a+b+nc) + (na-a) + (nb-b) = n(a+b+c)$ Row 2 element of $C_1 = (nc-c) + (b+c+na) + (nb-b) = n(a+b+c)$ Row 3 element of $C_1 = (nc-c) + (na-a) + (c+a+nb) = n(a+b+c)$ Substituting these into C_1 :

$$\Delta = \begin{vmatrix} n(a+b+c) & (n-1)a & (n-1)b \\ n(a+b+c) & b+c+na & (n-1)b \\ n(a+b+c) & (n-1)a & c+a+nb \end{vmatrix}$$

Factoring out $n(a+b+c)$ from C_1 :

$$\Delta = n(a+b+c) \begin{vmatrix} 1 & (n-1)a & (n-1)b \\ 1 & b+c+na & (n-1)b \\ 1 & (n-1)a & c+a+nb \end{vmatrix}$$

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$\begin{aligned} R_2 - R_1 &= (1-1 \quad (b+c+na)-(na-a) \quad (nb-b)-(nb-b)) \\ &= (0 \quad a+b+c \quad 0) \end{aligned}$$

$$\begin{aligned} R_3 - R_1 &= (1-1 \quad (na-a)-(na-a) \quad (c+a+nb)-(nb-b)) \\ &= (0 \quad 0 \quad a+b+c) \end{aligned}$$

The determinant becomes:

$$\Delta = n(a+b+c) \begin{vmatrix} 1 & (n-1)a & (n-1)b \\ 0 & a+b+c & 0 \\ 0 & 0 & a+b+c \end{vmatrix}$$

Expanding along C_1 :

$$\begin{aligned} \Delta &= n(a+b+c) \cdot 1 \cdot [(a+b+c)(a+b+c) - 0] \\ &= n(a+b+c) \cdot (a+b+c)^2 \\ &= n(a+b+c)^3 \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 26 (i)

Prove that:

$$\begin{vmatrix} 1 & x & yz \\ 1 & y & zx \\ 1 & z & xy \end{vmatrix} = \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = (x-y)(y-z)(z-x)$$

TYPE 10 Proving Determinant Identities Using Properties

★ Likely Exam Question

To Prove: $\begin{vmatrix} 1 & x & yz \\ 1 & y & zx \\ 1 & z & xy \end{vmatrix} = \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = (x-y)(y-z)(z-x)$

Concept / Formula Used: $\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$

Solution:

Let $\Delta_1 = \begin{vmatrix} 1 & x & yz \\ 1 & y & zx \\ 1 & z & xy \end{vmatrix}$.

Multiplying R_1 by x , R_2 by y , and R_3 by z :

$$\Delta_1 = \frac{1}{xyz} \begin{vmatrix} x & x^2 & xyz \\ y & y^2 & xyz \\ z & z^2 & xyz \end{vmatrix}$$

Factoring out xyz from C_3 :

$$\Delta_1 = \frac{xyz}{xyz} \begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} = \begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix}$$

Interchanging $C_3 \leftrightarrow C_2$ (introducing a negative sign):

$$\Delta_1 = - \begin{vmatrix} x & 1 & x^2 \\ y & 1 & y^2 \\ z & 1 & z^2 \end{vmatrix}$$

Interchanging $C_1 \leftrightarrow C_2$ (introducing a second negative sign):

$$\Delta_1 = \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = \Delta_2$$

Now evaluating $\Delta_2 = \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$:

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$\Delta_2 = \begin{vmatrix} 1 & x & x^2 \\ 0 & y-x & y^2-x^2 \\ 0 & z-x & z^2-x^2 \end{vmatrix}$$

Factoring out $(y-x)$ from R_2 and $(z-x)$ from R_3 :

$$\Delta_2 = (y-x)(z-x) \begin{vmatrix} 1 & x & x^2 \\ 0 & 1 & y+x \\ 0 & 1 & z+x \end{vmatrix}$$

Expanding along C_1 :

$$\begin{aligned} \Delta_2 &= (y-x)(z-x) [1 \cdot ((z+x) - (y+x))] \\ &= (y-x)(z-x)(z-y) \\ &= [-(x-y)][-(z-x)][-(y-z)] \\ &= (x-y)(y-z)(z-x) \end{aligned}$$

Hence Proved

$LHS = MiddleTerm = RHS$, hence proved.

Question 26 (ii)

Prove that:

$$\begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ x^3 & y^3 & z^3 \end{vmatrix} = xyz(x-y)(y-z)(z-x)$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ x^3 & y^3 & z^3 \end{vmatrix} = xyz(x-y)(y-z)(z-x)$

Concept / Formula Used: $\begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ x^3 & y^3 & z^3 \end{vmatrix} = xyz \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix}$

Solution:

Let $\Delta = \begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ x^3 & y^3 & z^3 \end{vmatrix}$.

Factoring out x from C_1 , y from C_2 , and z from C_3 :

$$\Delta = xyz \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix}$$

Applying column operations $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_1$:

$$\Delta = xyz \begin{vmatrix} 1 & 0 & 0 \\ x & y-x & z-x \\ x^2 & y^2-x^2 & z^2-x^2 \end{vmatrix}$$

Factoring out $(y-x)$ from C_2 and $(z-x)$ from C_3 :

$$\Delta = xyz(y-x)(z-x) \begin{vmatrix} 1 & 0 & 0 \\ x & 1 & 1 \\ x^2 & y+x & z+x \end{vmatrix}$$

Expanding along R_1 :

$$\begin{aligned} \Delta &= xyz(y-x)(z-x) [1 \cdot ((z+x) - (y+x))] \\ &= xyz(y-x)(z-x)(z-y) \\ &= xyz[-(x-y)][-(z-x)][-(y-z)] \\ &= xyz(x-y)(y-z)(z-x) \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 27 (i)

Prove that:

$$\begin{vmatrix} x & y & 1 \\ \alpha & x & 1 \\ \alpha & \beta & 1 \end{vmatrix} = (x-\alpha)(x-\beta)$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} x & y & 1 \\ \alpha & x & 1 \\ \alpha & \beta & 1 \end{vmatrix} = (x-\alpha)(x-\beta)$

Concept / Formula Used: $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$

Solution:

$$\text{Let } \Delta = \begin{vmatrix} x & y & 1 \\ \alpha & x & 1 \\ \alpha & \beta & 1 \end{vmatrix}.$$

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$\Delta = \begin{vmatrix} x & y & 1 \\ \alpha - x & x - y & 0 \\ \alpha - x & \beta - y & 0 \end{vmatrix}$$

Expanding along C_3 :

$$\begin{aligned} \Delta &= 1 \cdot [(\alpha - x)(\beta - y) - (x - y)(\alpha - x)] \\ &= (\alpha - x)[(\beta - y) - (x - y)] \\ &= (\alpha - x)[\beta - y - x + y] \\ &= (\alpha - x)(\beta - x) \\ &= [-(x - \alpha)][-(x - \beta)] \\ &= (x - \alpha)(x - \beta) \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 27 (ii)

Prove that:

$$\begin{vmatrix} 1 & x & y \\ 1 & x + y & y \\ 1 & x & x + y \end{vmatrix} = xy$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} 1 & x & y \\ 1 & x + y & y \\ 1 & x & x + y \end{vmatrix} = xy$

Concept / Formula Used: $\det(\text{Upper Triangular Matrix}) = \text{Product of diagonal elements}$

Solution:

$$\text{Let } \Delta = \begin{vmatrix} 1 & x & y \\ 1 & x + y & y \\ 1 & x & x + y \end{vmatrix}.$$

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$\begin{aligned} R_2 - R_1 &= (1-1 \quad (x+y)-x \quad y-y) = (0 \quad y \quad 0) \\ R_3 - R_1 &= (1-1 \quad x-x \quad (x+y)-y) = (0 \quad 0 \quad x) \end{aligned}$$

Substituting these rows back into the determinant:

$$\Delta = \begin{vmatrix} 1 & x & y \\ 0 & y & 0 \\ 0 & 0 & x \end{vmatrix}$$

Since this is an upper triangular matrix, the determinant is the product of its diagonal elements:

$$\begin{aligned} \Delta &= 1 \cdot y \cdot x \\ &= xy \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 27 (iii)

Prove that:

$$\begin{vmatrix} y+z & x & y \\ z+x & z & x \\ x+y & y & z \end{vmatrix} = (x+y+z)(x-z)^2$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} y+z & x & y \\ z+x & z & x \\ x+y & y & z \end{vmatrix} = (x+y+z)(x-z)^2$

Concept / Formula Used: $R_1 \rightarrow R_1 + R_2 + R_3 \implies$ factor out $(x+y+z)$

Solution:

Let $\Delta = \begin{vmatrix} y+z & x & y \\ z+x & z & x \\ x+y & y & z \end{vmatrix}$.

Applying $R_1 \rightarrow R_1 + R_2 + R_3$:

$$\text{Column 1 element} = (y+z) + (z+x) + (x+y) = 2(x+y+z)$$

$$\text{Column 2 element} = x + z + y = x + y + z$$

$$\text{Column 3 element} = y + x + z = x + y + z$$

Substituting into R_1 :

$$\Delta = \begin{vmatrix} 2(x+y+z) & x+y+z & x+y+z \\ z+x & z & x \\ x+y & y & z \end{vmatrix}$$

Factoring out $(x + y + z)$ from R_1 :

$$\Delta = (x + y + z) \begin{vmatrix} 2 & 1 & 1 \\ z + x & z & x \\ x + y & y & z \end{vmatrix}$$

Applying column operations $C_1 \rightarrow C_1 - 2C_2$ and $C_3 \rightarrow C_3 - C_2$:

$$C_1 - 2C_2 = \begin{pmatrix} 2 - 2(1) \\ (z + x) - 2z \\ (x + y) - 2y \end{pmatrix} = \begin{pmatrix} 0 \\ x - z \\ x - y \end{pmatrix}$$

$$C_3 - C_2 = \begin{pmatrix} 1 - 1 \\ x - z \\ z - y \end{pmatrix} = \begin{pmatrix} 0 \\ x - z \\ z - y \end{pmatrix}$$

Substituting these back into the determinant:

$$\Delta = (x + y + z) \begin{vmatrix} 0 & 1 & 0 \\ x - z & z & x - z \\ x - y & y & z - y \end{vmatrix}$$

Expanding along R_1 :

$$\begin{aligned} \Delta &= (x + y + z) \cdot (-1) \cdot [(x - z)(z - y) - (x - z)(x - y)] \\ &= (x + y + z) \cdot (-1)(x - z) \cdot [(z - y) - (x - y)] \\ &= -(x + y + z)(x - z)(z - x) \\ &= -(x + y + z)(x - z)[-(x - z)] \\ &= (x + y + z)(x - z)^2 \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 28

Prove that:

$$\begin{vmatrix} -a^2 & ab & ac \\ ab & -b^2 & bc \\ ca & bc & -c^2 \end{vmatrix} = 4a^2b^2c^2$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} -a^2 & ab & ac \\ ab & -b^2 & bc \\ ca & bc & -c^2 \end{vmatrix} = 4a^2b^2c^2$

Concept / Formula Used: $\det(kA) = k^n \det(A)$ $n \times n$ A

Solution:

$$\text{Let } \Delta = \begin{vmatrix} -a^2 & ab & ac \\ ab & -b^2 & bc \\ ca & bc & -c^2 \end{vmatrix}.$$

Factoring out a from R_1 , b from R_2 , and c from R_3 :

$$\Delta = abc \begin{vmatrix} -a & b & c \\ a & -b & c \\ a & b & -c \end{vmatrix}$$

Now factoring out a from C_1 , b from C_2 , and c from C_3 :

$$\Delta = a^2b^2c^2 \begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix}$$

Applying row operations $R_2 \rightarrow R_2 + R_1$ and $R_3 \rightarrow R_3 + R_1$:

$$R_2 + R_1 = (1 + (-1) \quad -1 + 1 \quad 1 + 1) = (0 \quad 0 \quad 2)$$

$$R_3 + R_1 = (1 + (-1) \quad 1 + 1 \quad -1 + 1) = (0 \quad 2 \quad 0)$$

Substituting into the determinant:

$$\Delta = a^2b^2c^2 \begin{vmatrix} -1 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & 2 & 0 \end{vmatrix}$$

Expanding along C_1 :

$$\begin{aligned} \Delta &= a^2b^2c^2 \cdot (-1) \cdot [(0)(0) - (2)(2)] \\ &= a^2b^2c^2 \cdot (-1) \cdot (-4) \\ &= 4a^2b^2c^2 \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 29

Prove that:

$$\begin{vmatrix} 1 & \alpha & \alpha^2 + \beta\gamma \\ 1 & \beta & \beta^2 + \gamma\alpha \\ 1 & \gamma & \gamma^2 + \alpha\beta \end{vmatrix} = 2(\alpha - \beta)(\beta - \gamma)(\gamma - \alpha)$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} 1 & \alpha & \alpha^2 + \beta\gamma \\ 1 & \beta & \beta^2 + \gamma\alpha \\ 1 & \gamma & \gamma^2 + \alpha\beta \end{vmatrix} = 2(\alpha - \beta)(\beta - \gamma)(\gamma - \alpha)$

Concept / Formula Used: $\begin{vmatrix} a & b & c+d \\ e & f & g+h \\ i & j & k+l \end{vmatrix} = \begin{vmatrix} a & b & c \\ e & f & g \\ i & j & k \end{vmatrix} + \begin{vmatrix} a & b & d \\ e & f & h \\ i & j & l \end{vmatrix}$

Solution:

Let $\Delta = \begin{vmatrix} 1 & \alpha & \alpha^2 + \beta\gamma \\ 1 & \beta & \beta^2 + \gamma\alpha \\ 1 & \gamma & \gamma^2 + \alpha\beta \end{vmatrix}$.

Splitting C_3 into two determinants:

$$\Delta = \begin{vmatrix} 1 & \alpha & \alpha^2 \\ 1 & \beta & \beta^2 \\ 1 & \gamma & \gamma^2 \end{vmatrix} + \begin{vmatrix} 1 & \alpha & \beta\gamma \\ 1 & \beta & \gamma\alpha \\ 1 & \gamma & \alpha\beta \end{vmatrix} = \Delta_1 + \Delta_2$$

Using the standard result for Δ_1 :

$$\Delta_1 = (\alpha - \beta)(\beta - \gamma)(\gamma - \alpha)$$

For $\Delta_2 = \begin{vmatrix} 1 & \alpha & \beta\gamma \\ 1 & \beta & \gamma\alpha \\ 1 & \gamma & \alpha\beta \end{vmatrix}$, multiply R_1 by α , R_2 by β , and R_3 by γ :

$$\Delta_2 = \frac{1}{\alpha\beta\gamma} \begin{vmatrix} \alpha & \alpha^2 & \alpha\beta\gamma \\ \beta & \beta^2 & \alpha\beta\gamma \\ \gamma & \gamma^2 & \alpha\beta\gamma \end{vmatrix}$$

Factoring out $\alpha\beta\gamma$ from C_3 :

$$\Delta_2 = \frac{\alpha\beta\gamma}{\alpha\beta\gamma} \begin{vmatrix} \alpha & \alpha^2 & 1 \\ \beta & \beta^2 & 1 \\ \gamma & \gamma^2 & 1 \end{vmatrix} = \begin{vmatrix} \alpha & \alpha^2 & 1 \\ \beta & \beta^2 & 1 \\ \gamma & \gamma^2 & 1 \end{vmatrix}$$

Interchanging $C_3 \leftrightarrow C_2$, then $C_2 \leftrightarrow C_1$:

$$\Delta_2 = - \begin{vmatrix} \alpha & 1 & \alpha^2 \\ \beta & 1 & \beta^2 \\ \gamma & 1 & \gamma^2 \end{vmatrix} = \begin{vmatrix} 1 & \alpha & \alpha^2 \\ 1 & \beta & \beta^2 \\ 1 & \gamma & \gamma^2 \end{vmatrix} = (\alpha - \beta)(\beta - \gamma)(\gamma - \alpha)$$

Adding Δ_1 and Δ_2 :

$$\begin{aligned} \Delta &= \Delta_1 + \Delta_2 \\ &= (\alpha - \beta)(\beta - \gamma)(\gamma - \alpha) + (\alpha - \beta)(\beta - \gamma)(\gamma - \alpha) \\ &= 2(\alpha - \beta)(\beta - \gamma)(\gamma - \alpha) \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 30

Prove that:

$$\begin{vmatrix} a^2 + 1 & ab & ac \\ ab & b^2 + 1 & bc \\ ac & bc & c^2 + 1 \end{vmatrix} = \begin{vmatrix} a^2 + 1 & b^2 & c^2 \\ a^2 & b^2 + 1 & c^2 \\ a^2 & b^2 & c^2 + 1 \end{vmatrix} = 1 + a^2 + b^2 + c^2$$

TYPE 10 Proving Determinant Identities Using Properties

★ Likely Exam Question

To Prove: $\begin{vmatrix} a^2 + 1 & ab & ac \\ ab & b^2 + 1 & bc \\ ac & bc & c^2 + 1 \end{vmatrix} = \begin{vmatrix} a^2 + 1 & b^2 & c^2 \\ a^2 & b^2 + 1 & c^2 \\ a^2 & b^2 & c^2 + 1 \end{vmatrix} = 1 + a^2 + b^2 + c^2$

Concept / Formula Used: $C_1 \rightarrow C_1 + C_2 + C_3 \implies$ factor out $(1 + a^2 + b^2 + c^2)$ **Solution:**

Let $\Delta_1 = \begin{vmatrix} a^2 + 1 & ab & ac \\ ab & b^2 + 1 & bc \\ ac & bc & c^2 + 1 \end{vmatrix}$.

Multiplying R_1 by a , R_2 by b , and R_3 by c :

$$\Delta_1 = \frac{1}{abc} \begin{vmatrix} a(a^2 + 1) & a^2b & a^2c \\ ab^2 & b(b^2 + 1) & b^2c \\ ac^2 & bc^2 & c(c^2 + 1) \end{vmatrix}$$

Factoring out a from C_1 , b from C_2 , and c from C_3 :

$$\Delta_1 = \frac{abc}{abc} \begin{vmatrix} a^2 + 1 & b^2 & c^2 \\ a^2 & b^2 + 1 & c^2 \\ a^2 & b^2 & c^2 + 1 \end{vmatrix} = \begin{vmatrix} a^2 + 1 & b^2 & c^2 \\ a^2 & b^2 + 1 & c^2 \\ a^2 & b^2 & c^2 + 1 \end{vmatrix} = \Delta_2$$

Now evaluating Δ_2 :Applying column operation $C_1 \rightarrow C_1 + C_2 + C_3$:

$$\Delta_2 = \begin{vmatrix} 1 + a^2 + b^2 + c^2 & b^2 & c^2 \\ 1 + a^2 + b^2 + c^2 & b^2 + 1 & c^2 \\ 1 + a^2 + b^2 + c^2 & b^2 & c^2 + 1 \end{vmatrix}$$

Factoring out $(1 + a^2 + b^2 + c^2)$ from C_1 :

$$\Delta_2 = (1 + a^2 + b^2 + c^2) \begin{vmatrix} 1 & b^2 & c^2 \\ 1 & b^2 + 1 & c^2 \\ 1 & b^2 & c^2 + 1 \end{vmatrix}$$

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$\Delta_2 = (1 + a^2 + b^2 + c^2) \begin{vmatrix} 1 & b^2 & c^2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

Expanding along C_1 :

$$\begin{aligned}\Delta_2 &= (1 + a^2 + b^2 + c^2) \cdot 1 \cdot (1 - 0) \\ &= 1 + a^2 + b^2 + c^2\end{aligned}$$

Hence Proved

$LHS = MiddleTerm = RHS$, hence proved.

Question 31

Prove that:

$$\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ bc & ca & ab \end{vmatrix} = \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(ab+bc+ca)$$

TYPE 10 Proving Determinant Identities Using Properties

★ Likely Exam Question

To Prove: $\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ bc & ca & ab \end{vmatrix} = \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(ab+bc+ca)$

Concept / Formula Used: $\begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(ab+bc+ca)$

Solution:

$$\text{Let } \Delta_1 = \begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ bc & ca & ab \end{vmatrix}.$$

Multiplying C_1 by a , C_2 by b , and C_3 by c :

$$\Delta_1 = \frac{1}{abc} \begin{vmatrix} a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \\ abc & abc & abc \end{vmatrix}$$

Factoring out abc from R_3 :

$$\Delta_1 = \frac{abc}{abc} \begin{vmatrix} a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \\ 1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \\ 1 & 1 & 1 \end{vmatrix}$$

Interchanging $R_3 \leftrightarrow R_2$, then $R_2 \leftrightarrow R_1$:

$$\Delta_1 = - \begin{vmatrix} a^2 & b^2 & c^2 \\ 1 & 1 & 1 \\ a^3 & b^3 & c^3 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix}$$

Taking transpose ($\det(A) = \det(A^T)$):

$$\Delta_1 = \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = \Delta_2$$

Now evaluating Δ_2 :

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$\Delta_2 = \begin{vmatrix} 1 & a^2 & a^3 \\ 0 & b^2 - a^2 & b^3 - a^3 \\ 0 & c^2 - a^2 & c^3 - a^3 \end{vmatrix}$$

Factoring out $(b - a)$ from R_2 and $(c - a)$ from R_3 :

$$\Delta_2 = (b - a)(c - a) \begin{vmatrix} 1 & a^2 & a^3 \\ 0 & b + a & b^2 + ab + a^2 \\ 0 & c + a & c^2 + ac + a^2 \end{vmatrix}$$

Expanding along C_1 :

$$\Delta_2 = (b - a)(c - a) [(b + a)(c^2 + ac + a^2) - (c + a)(b^2 + ab + a^2)]$$

Simplifying the terms inside the square bracket:

$$\text{First product} = (b + a)(c^2 + ac + a^2) = bc^2 + abc + ba^2 + ac^2 + a^2c + a^3$$

$$\text{Second product} = (c + a)(b^2 + ab + a^2) = cb^2 + abc + ca^2 + ab^2 + a^2b + a^3$$

$$\text{Difference} = (bc^2 - cb^2) + (ac^2 - ab^2)$$

$$= bc(c - b) + a(c^2 - b^2)$$

$$= (c - b) [bc + a(c + b)]$$

$$= (c - b)(ab + bc + ca)$$

Substituting back into Δ_2 :

$$\begin{aligned} \Delta_2 &= (b - a)(c - a)(c - b)(ab + bc + ca) \\ &= [-(a - b)][-(c - a)][-(b - c)](ab + bc + ca) \\ &= (a - b)(b - c)(c - a)(ab + bc + ca) \end{aligned}$$

Hence Proved

LHS = MiddleTerm = RHS, hence proved.

Question 32

Prove that:

$$\begin{vmatrix} a^2 & a^2 - (b - c)^2 & bc \\ b^2 & b^2 - (c - a)^2 & ca \\ c^2 & c^2 - (a - b)^2 & ab \end{vmatrix} = (a - b)(b - c)(c - a)(a + b + c)(a^2 + b^2 + c^2)$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove:
$$\begin{vmatrix} a^2 & a^2 - (b-c)^2 & bc \\ b^2 & b^2 - (c-a)^2 & ca \\ c^2 & c^2 - (a-b)^2 & ab \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c)(a^2+b^2+c^2)$$

Concept / Formula Used: $(b-c)^2 + 2bc = b^2 + c^2$

$c)(c-a)(a+b+c)$

$$\begin{vmatrix} 1 & a^2 & bc \\ 1 & b^2 & ca \\ 1 & c^2 & ab \end{vmatrix} = (a-b)(b-c)(c-a)$$

Solution:

Let $\Delta = \begin{vmatrix} a^2 & a^2 - (b-c)^2 & bc \\ b^2 & b^2 - (c-a)^2 & ca \\ c^2 & c^2 - (a-b)^2 & ab \end{vmatrix}$.

Applying $C_2 \rightarrow C_1 - C_2 + 2C_3$:

$$\text{Row 1} = a^2 - [a^2 - (b-c)^2] + 2bc = (b-c)^2 + 2bc = b^2 + c^2$$

$$\text{Row 2} = b^2 - [b^2 - (c-a)^2] + 2ca = (c-a)^2 + 2ca = c^2 + a^2$$

$$\text{Row 3} = c^2 - [c^2 - (a-b)^2] + 2ab = (a-b)^2 + 2ab = a^2 + b^2$$

The determinant becomes:

$$\Delta = \begin{vmatrix} a^2 & b^2 + c^2 & bc \\ b^2 & c^2 + a^2 & ca \\ c^2 & a^2 + b^2 & ab \end{vmatrix}$$

Now applying $C_2 \rightarrow C_2 + C_1$:

$$\Delta = \begin{vmatrix} a^2 & a^2 + b^2 + c^2 & bc \\ b^2 & a^2 + b^2 + c^2 & ca \\ c^2 & a^2 + b^2 + c^2 & ab \end{vmatrix}$$

Factoring out $(a^2 + b^2 + c^2)$ from C_2 :

$$\Delta = (a^2 + b^2 + c^2) \begin{vmatrix} a^2 & 1 & bc \\ b^2 & 1 & ca \\ c^2 & 1 & ab \end{vmatrix}$$

Interchanging $C_1 \leftrightarrow C_2$:

$$\Delta = -(a^2 + b^2 + c^2) \begin{vmatrix} 1 & a^2 & bc \\ 1 & b^2 & ca \\ 1 & c^2 & ab \end{vmatrix}$$

Evaluating $\begin{vmatrix} 1 & a^2 & bc \\ 1 & b^2 & ca \\ 1 & c^2 & ab \end{vmatrix}$: Applying $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$\begin{vmatrix} 1 & a^2 & bc \\ 0 & b^2 - a^2 & ca - bc \\ 0 & c^2 - a^2 & ab - bc \end{vmatrix} = \begin{vmatrix} 1 & a^2 & bc \\ 0 & (b-a)(b+a) & -c(b-a) \\ 0 & (c-a)(c+a) & -b(c-a) \end{vmatrix}$$

Factoring out $(b - a)$ from R_2 and $(c - a)$ from R_3 :

$$= (b - a)(c - a) \begin{vmatrix} 1 & a^2 & bc \\ 0 & b + a & -c \\ 0 & c + a & -b \end{vmatrix}$$

Expanding along C_1 :

$$\begin{aligned} &= (b - a)(c - a) [-b(b + a) - (-c)(c + a)] \\ &= (b - a)(c - a) [-b^2 - ab + c^2 + ac] \\ &= (b - a)(c - a) [(c^2 - b^2) + a(c - b)] \\ &= (b - a)(c - a)(c - b)(a + b + c) \\ &= -(a - b)(b - c)(c - a)(a + b + c) \end{aligned}$$

Substituting this back into Δ :

$$\begin{aligned} \Delta &= -(a^2 + b^2 + c^2) [-(a - b)(b - c)(c - a)(a + b + c)] \\ &= (a - b)(b - c)(c - a)(a + b + c)(a^2 + b^2 + c^2) \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 33

Prove that:

$$\begin{vmatrix} (b + c)^2 & a^2 & bc \\ (c + a)^2 & b^2 & ca \\ (a + b)^2 & c^2 & ab \end{vmatrix} = (a - b)(b - c)(c - a)(a + b + c)(a^2 + b^2 + c^2)$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} (b + c)^2 & a^2 & bc \\ (c + a)^2 & b^2 & ca \\ (a + b)^2 & c^2 & ab \end{vmatrix} = (a - b)(b - c)(c - a)(a + b + c)(a^2 + b^2 + c^2)$

Concept / Formula Used: $(b + c)^2 - 2bc = b^2 + c^2$

Solution:

Let $\Delta = \begin{vmatrix} (b + c)^2 & a^2 & bc \\ (c + a)^2 & b^2 & ca \\ (a + b)^2 & c^2 & ab \end{vmatrix}$.

Applying $C_1 \rightarrow C_1 - C_2 - 2C_3$:

$$\text{Row 1} = (b + c)^2 - a^2 - 2bc = b^2 + c^2 + 2bc - a^2 - 2bc = b^2 + c^2 - a^2$$

$$\text{Row 2} = (c + a)^2 - b^2 - 2ca = c^2 + a^2 + 2ca - b^2 - 2ca = c^2 + a^2 - b^2$$

$$\text{Row 3} = (a + b)^2 - c^2 - 2ab = a^2 + b^2 + 2ab - c^2 - 2ab = a^2 + b^2 - c^2$$

Substituting back into C_1 :

$$\Delta = \begin{vmatrix} b^2 + c^2 - a^2 & a^2 & bc \\ c^2 + a^2 - b^2 & b^2 & ca \\ a^2 + b^2 - c^2 & c^2 & ab \end{vmatrix}$$

Applying $C_1 \rightarrow C_1 + C_2$:

$$\Delta = \begin{vmatrix} b^2 + c^2 & a^2 & bc \\ c^2 + a^2 & b^2 & ca \\ a^2 + b^2 & c^2 & ab \end{vmatrix}$$

Applying $C_1 \rightarrow C_1 + C_2$:

$$\Delta = \begin{vmatrix} a^2 + b^2 + c^2 & a^2 & bc \\ a^2 + b^2 + c^2 & b^2 & ca \\ a^2 + b^2 + c^2 & c^2 & ab \end{vmatrix}$$

Factoring out $(a^2 + b^2 + c^2)$ from C_1 :

$$\Delta = (a^2 + b^2 + c^2) \begin{vmatrix} 1 & a^2 & bc \\ 1 & b^2 & ca \\ 1 & c^2 & ab \end{vmatrix}$$

Using the standard result derived in Question 32 for $\begin{vmatrix} 1 & a^2 & bc \\ 1 & b^2 & ca \\ 1 & c^2 & ab \end{vmatrix}$:

$$\begin{vmatrix} 1 & a^2 & bc \\ 1 & b^2 & ca \\ 1 & c^2 & ab \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c)$$

Thus:

$$\Delta = (a-b)(b-c)(c-a)(a+b+c)(a^2 + b^2 + c^2)$$

Hence Proved

LHS = RHS, hence proved.

Question 34 (i)

Prove that:

$$\begin{vmatrix} a+b+c & -c & -b \\ -c & a+b+c & -a \\ -b & -a & a+b+c \end{vmatrix} = 2(a+b)(b+c)(c+a)$$

TYPE 10 Proving Determinant Identities Using Properties

★ Likely Exam Question

To Prove: $\begin{vmatrix} a+b+c & -c & -b \\ -c & a+b+c & -a \\ -b & -a & a+b+c \end{vmatrix} = 2(a+b)(b+c)(c+a)$

Concept / Formula Used: $C_1 \rightarrow C_1 + C_2 + C_3 \Rightarrow$
 first column elements become a, b, c

Solution:

$$\text{Let } \Delta = \begin{vmatrix} a+b+c & -c & -b \\ -c & a+b+c & -a \\ -b & -a & a+b+c \end{vmatrix}.$$

Applying $C_1 \rightarrow C_1 + C_2 + C_3$:

$$\text{Row 1} = (a+b+c) - c - b = a$$

$$\text{Row 2} = -c + (a+b+c) - a = b$$

$$\text{Row 3} = -b - a + (a+b+c) = c$$

The determinant becomes:

$$\Delta = \begin{vmatrix} a & -c & -b \\ b & a+b+c & -a \\ c & -a & a+b+c \end{vmatrix}$$

Applying $R_1 \rightarrow R_1 + R_2$:

$$\Delta = \begin{vmatrix} a+b & a+b & -(a+b) \\ b & a+b+c & -a \\ c & -a & a+b+c \end{vmatrix}$$

Factoring out $(a+b)$ from R_1 :

$$\Delta = (a+b) \begin{vmatrix} 1 & 1 & -1 \\ b & a+b+c & -a \\ c & -a & a+b+c \end{vmatrix}$$

Applying column operations $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 + C_1$:

$$C_2 - C_1 = \begin{pmatrix} 1-1 \\ (a+b+c)-b \\ -a-c \end{pmatrix} = \begin{pmatrix} 0 \\ a+c \\ -(a+c) \end{pmatrix}$$

$$C_3 + C_1 = \begin{pmatrix} -1+1 \\ -a+b \\ (a+b+c)+c \end{pmatrix} = \begin{pmatrix} 0 \\ b-a \\ a+b+2c \end{pmatrix}$$

Substituting these into the determinant:

$$\Delta = (a+b) \begin{vmatrix} 1 & 0 & 0 \\ b & a+c & b-a \\ c & -(a+c) & a+b+2c \end{vmatrix}$$

Expanding along R_1 :

$$\begin{aligned} \Delta &= (a+b) [(a+c)(a+b+2c) - (b-a)(-(a+c))] \\ &= (a+b) [(a+c)(a+b+2c) + (b-a)(a+c)] \end{aligned}$$

Factoring out $(a + c)$ from the square bracket:

$$\begin{aligned}\Delta &= (a + b)(a + c)[(a + b + 2c) + (b - a)] \\ &= (a + b)(a + c)[2b + 2c] \\ &= 2(a + b)(b + c)(c + a)\end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 34 (ii)

Prove that:

$$\begin{vmatrix} a+x & y & z \\ x & a+y & z \\ x & y & a+z \end{vmatrix} = a^2(a+x+y+z)$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} a+x & y & z \\ x & a+y & z \\ x & y & a+z \end{vmatrix} = a^2(a+x+y+z)$

Concept / Formula Used: $C_1 \rightarrow C_1 + C_2 + C_3 \implies$ factor out $(a + x + y + z)$

Solution:

Let $\Delta = \begin{vmatrix} a+x & y & z \\ x & a+y & z \\ x & y & a+z \end{vmatrix}$.

Applying $C_1 \rightarrow C_1 + C_2 + C_3$:

$$\Delta = \begin{vmatrix} a+x+y+z & y & z \\ a+x+y+z & a+y & z \\ a+x+y+z & y & a+z \end{vmatrix}$$

Factoring out $(a + x + y + z)$ from C_1 :

$$\Delta = (a + x + y + z) \begin{vmatrix} 1 & y & z \\ 1 & a+y & z \\ 1 & y & a+z \end{vmatrix}$$

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$\begin{aligned}R_2 - R_1 &= (1 - 1 \quad (a + y) - y \quad z - z) = (0 \quad a \quad 0) \\ R_3 - R_1 &= (1 - 1 \quad y - y \quad (a + z) - z) = (0 \quad 0 \quad a)\end{aligned}$$

Substituting these rows into the determinant:

$$\Delta = (a + x + y + z) \begin{vmatrix} 1 & y & z \\ 0 & a & 0 \\ 0 & 0 & a \end{vmatrix}$$

Expanding along C_1 :

$$\begin{aligned}\Delta &= (a + x + y + z) \cdot 1 \cdot (a \cdot a - 0) \\ &= a^2(a + x + y + z)\end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 35

Prove that:

$$\begin{vmatrix} y+z & x+y & x \\ z+x & y+z & y \\ x+y & z+x & z \end{vmatrix} = x^3 + y^3 + z^3 - 3xyz$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} y+z & x+y & x \\ z+x & y+z & y \\ x+y & z+x & z \end{vmatrix} = x^3 + y^3 + z^3 - 3xyz$

Concept / Formula Used: $x^3 + y^3 + z^3 - 3xyz = (x+y+z)(x^2 + y^2 + z^2 - xy - yz - zx)$

Solution:

Let $\Delta = \begin{vmatrix} y+z & x+y & x \\ z+x & y+z & y \\ x+y & z+x & z \end{vmatrix}$.

Applying $C_2 \rightarrow C_2 - C_3$:

$$\Delta = \begin{vmatrix} y+z & y & x \\ z+x & z & y \\ x+y & x & z \end{vmatrix}$$

Applying $C_1 \rightarrow C_1 - C_2$:

$$\Delta = \begin{vmatrix} z & y & x \\ x & z & y \\ y & x & z \end{vmatrix}$$

Applying $R_1 \rightarrow R_1 + R_2 + R_3$:

$$\Delta = \begin{vmatrix} x+y+z & x+y+z & x+y+z \\ x & z & y \\ y & x & z \end{vmatrix}$$

Factoring out $(x + y + z)$ from R_1 :

$$\Delta = (x + y + z) \begin{vmatrix} 1 & 1 & 1 \\ x & z & y \\ y & x & z \end{vmatrix}$$

Applying $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_1$:

$$\Delta = (x + y + z) \begin{vmatrix} 1 & 0 & 0 \\ x & z - x & y - x \\ y & x - y & z - y \end{vmatrix}$$

Expanding along R_1 :

$$\begin{aligned} \Delta &= (x + y + z) [(z - x)(z - y) - (y - x)(x - y)] \\ &= (x + y + z) [(z^2 - zy - zx + xy) + (x - y)^2] \\ &= (x + y + z) [z^2 - yz - zx + xy + x^2 - 2xy + y^2] \\ &= (x + y + z) [x^2 + y^2 + z^2 - xy - yz - zx] \\ &= x^3 + y^3 + z^3 - 3xyz \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 36

Prove that:

$$\begin{vmatrix} a & b - c & c - b \\ a - c & b & c - a \\ a - b & b - a & c \end{vmatrix} = (a + b - c)(b + c - a)(c + a - b)$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} a & b - c & c - b \\ a - c & b & c - a \\ a - b & b - a & c \end{vmatrix} = (a + b - c)(b + c - a)(c + a - b)$

Concept / Formula Used: $C_1 \rightarrow C_1 + C_2 \implies$ factor out $(a + b - c)$

Solution:

Let $\Delta = \begin{vmatrix} a & b - c & c - b \\ a - c & b & c - a \\ a - b & b - a & c \end{vmatrix}$.

Applying $C_1 \rightarrow C_1 + C_2$:

$$\text{Row 1} = a + (b - c) = a + b - c$$

$$\text{Row 2} = (a - c) + b = a + b - c$$

$$\text{Row 3} = (a - b) + (b - a) = 0$$

Substituting into C_1 :

$$\Delta = \begin{vmatrix} a + b - c & b - c & c - b \\ a + b - c & b & c - a \\ 0 & b - a & c \end{vmatrix}$$

Factoring out $(a + b - c)$ from C_1 :

$$\Delta = (a + b - c) \begin{vmatrix} 1 & b - c & c - b \\ 1 & b & c - a \\ 0 & b - a & c \end{vmatrix}$$

Applying $R_2 \rightarrow R_2 - R_1$:

$$\begin{aligned} R_2 - R_1 &= (1 - 1 \quad b - (b - c) \quad (c - a) - (c - b)) \\ &= (0 \quad c \quad b - a) \end{aligned}$$

Substituting back into R_2 :

$$\Delta = (a + b - c) \begin{vmatrix} 1 & b - c & c - b \\ 0 & c & b - a \\ 0 & b - a & c \end{vmatrix}$$

Expanding along C_1 :

$$\begin{aligned} \Delta &= (a + b - c) \cdot 1 \cdot [c^2 - (b - a)^2] \\ &= (a + b - c) [c - (b - a)] [c + (b - a)] \\ &= (a + b - c)(c - b + a)(c + b - a) \\ &= (a + b - c)(b + c - a)(c + a - b) \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 37

Prove that:

$$\begin{vmatrix} 1 & bc + ad & b^2c^2 + a^2d^2 \\ 1 & ca + bd & c^2a^2 + b^2d^2 \\ 1 & ab + cd & a^2b^2 + c^2d^2 \end{vmatrix} = (a - b)(a - c)(a - d)(b - c)(b - d)(c - d)$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} 1 & bc + ad & b^2c^2 + a^2d^2 \\ 1 & ca + bd & c^2a^2 + b^2d^2 \\ 1 & ab + cd & a^2b^2 + c^2d^2 \end{vmatrix} = (a - b)(a - c)(a - d)(b - c)(b - d)(c - d)$

Concept / Formula Used: $(bc + ad)^2 - 2abcd = b^2c^2 + a^2d^2$

Solution:

Let $x = bc + ad$, $y = ca + bd$, and $z = ab + cd$.

Note that:

$$\begin{aligned} x^2 &= (bc + ad)^2 = b^2c^2 + a^2d^2 + 2abcd \implies b^2c^2 + a^2d^2 = x^2 - 2abcd \\ y^2 &= (ca + bd)^2 = c^2a^2 + b^2d^2 + 2abcd \implies c^2a^2 + b^2d^2 = y^2 - 2abcd \\ z^2 &= (ab + cd)^2 = a^2b^2 + c^2d^2 + 2abcd \implies a^2b^2 + c^2d^2 = z^2 - 2abcd \end{aligned}$$

Substituting these into the determinant Δ :

$$\Delta = \begin{vmatrix} 1 & x & x^2 - 2abcd \\ 1 & y & y^2 - 2abcd \\ 1 & z & z^2 - 2abcd \end{vmatrix}$$

Applying $C_3 \rightarrow C_3 + 2abcdC_1$:

$$\Delta = \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = (x - y)(y - z)(z - x)$$

Now computing the factors $x - y$, $y - z$, and $z - x$:

$$x - y = (bc + ad) - (ca + bd) = c(b - a) - d(b - a) = (b - a)(c - d) = -(a - b)(c - d)$$

$$y - z = (ca + bd) - (ab + cd) = a(c - b) - d(c - b) = (c - b)(a - d) = -(b - c)(a - d)$$

$$z - x = (ab + cd) - (bc + ad) = b(a - c) - d(a - c) = (a - c)(b - d)$$

Multiplying these factors together:

$$\begin{aligned} \Delta &= [-(a - b)(c - d)] \cdot [-(b - c)(a - d)] \cdot [(a - c)(b - d)] \\ &= (a - b)(b - c)(a - d)(c - d)(a - c)(b - d) \\ &= (a - b)(a - c)(a - d)(b - c)(b - d)(c - d) \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Question 38

If a, b, c are all non-zero and:

$$\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = 0$$

then prove that:

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + 1 = 0$$

TYPE 10 Proving Determinant Identities Using Properties

★ Likely Exam Question

Given: $a, b, c \neq 0$ and $\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = 0$

To Prove: $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + 1 = 0$

Concept / Formula Used: $R_1 \rightarrow R_1 + R_2 + R_3 \implies$ factor out $\left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$

Solution:

$$\text{Let } \Delta = \begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix}.$$

Factoring out a from R_1 , b from R_2 , and c from R_3 :

$$\Delta = abc \begin{vmatrix} \frac{1}{a} + 1 & \frac{1}{b} & \frac{1}{c} \\ \frac{1}{b} & 1 & \frac{1}{c} \\ \frac{1}{c} & \frac{1}{c} & 1 \end{vmatrix}$$

Applying $R_1 \rightarrow R_1 + R_2 + R_3$:

$$\Delta = abc \begin{vmatrix} 1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} & 1 + \frac{1}{b} + \frac{1}{c} & 1 + \frac{1}{c} \\ \frac{1}{b} & 1 & \frac{1}{c} \\ \frac{1}{c} & \frac{1}{c} & 1 \end{vmatrix}$$

Factoring out $\left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$ from R_1 :

$$\Delta = abc \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \begin{vmatrix} 1 & 1 & 1 \\ \frac{1}{b} & 1 & \frac{1}{c} \\ \frac{1}{c} & \frac{1}{c} & 1 \end{vmatrix}$$

Applying $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_1$:

$$\Delta = abc \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \begin{vmatrix} 1 & 0 & 0 \\ \frac{1}{b} & 1 & 0 \\ \frac{1}{c} & 0 & 1 \end{vmatrix}$$

Expanding along R_1 :

$$\Delta = abc \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \cdot (1 - 0) = abc \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$$

We are given that $\Delta = 0$. Since a, b, c are non-zero, $abc \neq 0$:

$$abc \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) = 0 \implies 1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 0$$

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + 1 = 0$$

Hence Proved

Hence proved.

Question 39

Without expanding any of the determinants given below, prove that:

$$\begin{vmatrix} a^2 & b^2 & c^2 \\ (a+1)^2 & (b+1)^2 & (c+1)^2 \\ (a-1)^2 & (b-1)^2 & (c-1)^2 \end{vmatrix} = 4 \begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix}$$

TYPE 10 Proving Determinant Identities Using Properties

To Prove: $\begin{vmatrix} a^2 & b^2 & c^2 \\ (a+1)^2 & (b+1)^2 & (c+1)^2 \\ (a-1)^2 & (b-1)^2 & (c-1)^2 \end{vmatrix} = 4 \begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix}$

Concept / Formula Used: $(a+1)^2 - a^2 = 2a + 1$ and $(a-1)^2 - a^2 = -2a + 1$

Solution:

Let $\Delta = \begin{vmatrix} a^2 & b^2 & c^2 \\ (a+1)^2 & (b+1)^2 & (c+1)^2 \\ (a-1)^2 & (b-1)^2 & (c-1)^2 \end{vmatrix}$.

Expanding the algebraic squares in R_2 and R_3 :

$$\Delta = \begin{vmatrix} a^2 & b^2 & c^2 \\ a^2 + 2a + 1 & b^2 + 2b + 1 & c^2 + 2c + 1 \\ a^2 - 2a + 1 & b^2 - 2b + 1 & c^2 - 2c + 1 \end{vmatrix}$$

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$\Delta = \begin{vmatrix} a^2 & b^2 & c^2 \\ 2a + 1 & 2b + 1 & 2c + 1 \\ -2a + 1 & -2b + 1 & -2c + 1 \end{vmatrix}$$

Applying $R_2 \rightarrow R_2 + R_3$:

$$\Delta = \begin{vmatrix} a^2 & b^2 & c^2 \\ 2 & 2 & 2 \\ -2a + 1 & -2b + 1 & -2c + 1 \end{vmatrix}$$

Factoring out 2 from R_2 :

$$\Delta = 2 \begin{vmatrix} a^2 & b^2 & c^2 \\ 1 & 1 & 1 \\ -2a + 1 & -2b + 1 & -2c + 1 \end{vmatrix}$$

Applying $R_3 \rightarrow R_3 - R_2$:

$$\Delta = 2 \begin{vmatrix} a^2 & b^2 & c^2 \\ 1 & 1 & 1 \\ -2a & -2b & -2c \end{vmatrix}$$

Factoring out -2 from R_3 :

$$\Delta = 2 \cdot (-2) \begin{vmatrix} a^2 & b^2 & c^2 \\ 1 & 1 & 1 \\ a & b & c \end{vmatrix} = -4 \begin{vmatrix} a^2 & b^2 & c^2 \\ 1 & 1 & 1 \\ a & b & c \end{vmatrix}$$

Interchanging $R_2 \leftrightarrow R_3$ (multiplying by -1):

$$\Delta = 4 \begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix}$$

Hence Proved

LHS = RHS, hence proved.

Question 40 (i)

Using properties of determinants, solve the following equation for x :

$$\begin{vmatrix} 3x-8 & 3 & 3 \\ 3 & 3x-8 & 3 \\ 3 & 3 & 3x-8 \end{vmatrix} = 0$$

TYPE 5 Solving Determinant Equations

To Find: The values of x satisfying the determinant equation.

Concept / Formula Used: $C_1 \rightarrow C_1 + C_2 + C_3 \implies$ factor out $(3x - 2)$

Solution:

Given equation:

$$\begin{vmatrix} 3x-8 & 3 & 3 \\ 3 & 3x-8 & 3 \\ 3 & 3 & 3x-8 \end{vmatrix} = 0$$

Applying column operation $C_1 \rightarrow C_1 + C_2 + C_3$:

$$\text{Row 1} = (3x - 8) + 3 + 3 = 3x - 2$$

$$\text{Row 2} = 3 + (3x - 8) + 3 = 3x - 2$$

$$\text{Row 3} = 3 + 3 + (3x - 8) = 3x - 2$$

Substituting into C_1 :

$$\begin{vmatrix} 3x-2 & 3 & 3 \\ 3x-2 & 3x-8 & 3 \\ 3x-2 & 3 & 3x-8 \end{vmatrix} = 0$$

Factoring out $(3x - 2)$ from C_1 :

$$(3x-2) \begin{vmatrix} 1 & 3 & 3 \\ 1 & 3x-8 & 3 \\ 1 & 3 & 3x-8 \end{vmatrix} = 0$$

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$(3x - 2) \begin{vmatrix} 1 & 3 & 3 \\ 0 & 3x - 11 & 0 \\ 0 & 0 & 3x - 11 \end{vmatrix} = 0$$

Expanding along C_1 :

$$(3x - 2) \cdot 1 \cdot [(3x - 11)(3x - 11) - 0] = 0$$

$$(3x - 2)(3x - 11)^2 = 0$$

Equating each factor to zero:

$$3x - 2 = 0 \implies x = \frac{2}{3}$$

$$(3x - 11)^2 = 0 \implies x = \frac{11}{3}$$

Final Answer

$$x = \frac{2}{3} \quad \text{or} \quad x = \frac{11}{3}$$

Question 40 (ii)

Using properties of determinants, solve the following equation for x :

$$\begin{vmatrix} x+1 & 3 & 5 \\ 2 & x+2 & 5 \\ 2 & 3 & x+4 \end{vmatrix} = 0$$

TYPE 5 Solving Determinant Equations

To Find: The values of x satisfying the determinant equation.

Concept / Formula Used: $C_1 \rightarrow C_1 + C_2 + C_3 \implies$ factor out $(x + 9)$

Solution:

Given equation:

$$\begin{vmatrix} x+1 & 3 & 5 \\ 2 & x+2 & 5 \\ 2 & 3 & x+4 \end{vmatrix} = 0$$

Applying column operation $C_1 \rightarrow C_1 + C_2 + C_3$:

$$\text{Row 1} = (x + 1) + 3 + 5 = x + 9$$

$$\text{Row 2} = 2 + (x + 2) + 5 = x + 9$$

$$\text{Row 3} = 2 + 3 + (x + 4) = x + 9$$

Substituting into C_1 :

$$\begin{vmatrix} x+9 & 3 & 5 \\ x+9 & x+2 & 5 \\ x+9 & 3 & x+4 \end{vmatrix} = 0$$

Factoring out $(x + 9)$ from C_1 :

$$(x + 9) \begin{vmatrix} 1 & 3 & 5 \\ 1 & x + 2 & 5 \\ 1 & 3 & x + 4 \end{vmatrix} = 0$$

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$(x + 9) \begin{vmatrix} 1 & 3 & 5 \\ 0 & x - 1 & 0 \\ 0 & 0 & x - 1 \end{vmatrix} = 0$$

Expanding along C_1 :

$$(x + 9) \cdot 1 \cdot [(x - 1)(x - 1) - 0] = 0$$

$$(x + 9)(x - 1)^2 = 0$$

Equating each factor to zero:

$$x + 9 = 0 \implies x = -9$$

$$(x - 1)^2 = 0 \implies x = 1$$

Final Answer

$$x = -9 \quad \text{or} \quad x = 1$$

Question 40 (iii)

Using properties of determinants, solve the following equation for x :

$$\begin{vmatrix} x & -6 & -1 \\ 2 & -3x & x - 3 \\ -3 & 2x & x + 2 \end{vmatrix} = 0$$

TYPE 5 Solving Determinant Equations

To Find: The values of x satisfying the determinant equation.

Concept / Formula Used: $\det(A) = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$

Solution:

Given equation:

$$\begin{vmatrix} x & -6 & -1 \\ 2 & -3x & x - 3 \\ -3 & 2x & x + 2 \end{vmatrix} = 0$$

Expanding along R_1 :

$$x \begin{vmatrix} -3x & x - 3 \\ 2x & x + 2 \end{vmatrix} - (-6) \begin{vmatrix} 2 & x - 3 \\ -3 & x + 2 \end{vmatrix} + (-1) \begin{vmatrix} 2 & -3x \\ -3 & 2x \end{vmatrix} = 0$$

Evaluating each 2×2 determinant separately:

$$\begin{aligned}\text{First term} &= x [(-3x)(x+2) - (2x)(x-3)] \\ &= x [-3x^2 - 6x - 2x^2 + 6x] \\ &= x [-5x^2] = -5x^3\end{aligned}$$

$$\begin{aligned}\text{Second term} &= +6 [2(x+2) - (-3)(x-3)] \\ &= 6 [2x + 4 + 3x - 9] \\ &= 6 [5x - 5] = 30x - 30\end{aligned}$$

$$\begin{aligned}\text{Third term} &= -1 [2(2x) - (-3)(-3x)] \\ &= -1 [4x - 9x] \\ &= -1 [-5x] = 5x\end{aligned}$$

Summing all terms together:

$$\begin{aligned}-5x^3 + (30x - 30) + 5x &= 0 \\ -5x^3 + 35x - 30 &= 0\end{aligned}$$

Dividing both sides by -5 :

$$x^3 - 7x + 6 = 0$$

Testing $x = 1$: $1^3 - 7(1) + 6 = 0$, so $(x - 1)$ is a factor. Factorizing $x^3 - 7x + 6$ using synthetic division / grouping:

$$\begin{aligned}x^3 - x^2 + x^2 - x - 6x + 6 &= 0 \\ x^2(x-1) + x(x-1) - 6(x-1) &= 0 \\ (x-1)(x^2 + x - 6) &= 0 \\ (x-1)(x+3)(x-2) &= 0\end{aligned}$$

Setting each factor to zero:

$$x = 1, \quad x = 2, \quad x = -3$$

Final Answer

$$x = 1, \quad x = 2, \quad \text{or} \quad x = -3$$

Question 40 (iv)

Using properties of determinants, solve the following equation for x :

$$\begin{vmatrix} 15 - 2x & 11 & 10 \\ 11 - 3x & 17 & 16 \\ 7 - x & 14 & 13 \end{vmatrix} = 0$$

TYPE 5 Solving Determinant Equations

To Find: The value of x satisfying the determinant equation.

Concept / Formula Used: $C_2 \rightarrow C_2 - C_3 \implies$ column 2 becomes $[1, 1, 1]^T$

Solution:

Given equation:

$$\begin{vmatrix} 15-2x & 11 & 10 \\ 11-3x & 17 & 16 \\ 7-x & 14 & 13 \end{vmatrix} = 0$$

Applying column operation $C_2 \rightarrow C_2 - C_3$:

$$\begin{vmatrix} 15-2x & 1 & 10 \\ 11-3x & 1 & 16 \\ 7-x & 1 & 13 \end{vmatrix} = 0$$

Applying row operations $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$R_2 - R_1 = ((11-3x) - (15-2x) \quad 1-1 \quad 16-10) = (-4-x \quad 0 \quad 6)$$

$$R_3 - R_1 = ((7-x) - (15-2x) \quad 1-1 \quad 13-10) = (x-8 \quad 0 \quad 3)$$

Substituting these into the determinant:

$$\begin{vmatrix} 15-2x & 1 & 10 \\ -4-x & 0 & 6 \\ x-8 & 0 & 3 \end{vmatrix} = 0$$

Expanding along C_2 :

$$\begin{aligned} -1 \cdot [(-4-x)(3) - 6(x-8)] &= 0 \\ -3(-4-x) + 6(x-8) &= 0 \\ 12 + 3x + 6x - 48 &= 0 \\ 9x - 36 &= 0 \\ 9x &= 36 \\ x &= 4 \end{aligned}$$

Final Answer

$$x = 4$$

Question 41

If a, b, c are all different and:

$$\begin{vmatrix} a & a^3 & a^4-1 \\ b & b^3 & b^4-1 \\ c & c^3 & c^4-1 \end{vmatrix} = 0$$

prove that $abc(ab + bc + ca) = a + b + c$.

TYPE 10 Proving Determinant Identities Using Properties

★ Likely Exam Question

Given: a, b, c are distinct and $\begin{vmatrix} a & a^3 & a^4 - 1 \\ b & b^3 & b^4 - 1 \\ c & c^3 & c^4 - 1 \end{vmatrix} = 0$

To Prove: $abc(ab + bc + ca) = a + b + c$

Concept / Formula Used: $\begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix} = (a - b)(b - c)(c - a)(a + b + c)$

Solution:

Let $\Delta = \begin{vmatrix} a & a^3 & a^4 - 1 \\ b & b^3 & b^4 - 1 \\ c & c^3 & c^4 - 1 \end{vmatrix} = 0$.

Splitting C_3 into two determinants:

$$\Delta = \begin{vmatrix} a & a^3 & a^4 \\ b & b^3 & b^4 \\ c & c^3 & c^4 \end{vmatrix} - \begin{vmatrix} a & a^3 & 1 \\ b & b^3 & 1 \\ c & c^3 & 1 \end{vmatrix} = 0$$

For the first determinant, factoring out a from R_1 , b from R_2 , and c from R_3 :

$$\begin{vmatrix} a & a^3 & a^4 \\ b & b^3 & b^4 \\ c & c^3 & c^4 \end{vmatrix} = abc \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}$$

For the second determinant, performing column interchanges $C_3 \leftrightarrow C_2$ then $C_2 \leftrightarrow C_1$:

$$\begin{vmatrix} a & a^3 & 1 \\ b & b^3 & 1 \\ c & c^3 & 1 \end{vmatrix} = - \begin{vmatrix} a & 1 & a^3 \\ b & 1 & b^3 \\ c & 1 & c^3 \end{vmatrix} = \begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix}$$

Substituting both back into the equation:

$$abc \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} - \begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix} = 0$$

Using standard cyclic determinant identities:

$$\begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = (a - b)(b - c)(c - a)(ab + bc + ca)$$

$$\begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix} = (a - b)(b - c)(c - a)(a + b + c)$$

Substituting these into the equation:

$$abc(a - b)(b - c)(c - a)(ab + bc + ca) - (a - b)(b - c)(c - a)(a + b + c) = 0$$

Factoring out $(a - b)(b - c)(c - a)$:

$$(a - b)(b - c)(c - a) [abc(ab + bc + ca) - (a + b + c)] = 0$$

Since a, b, c are distinct, $(a - b) \neq 0$, $(b - c) \neq 0$, and $(c - a) \neq 0$:

$$abc(ab + bc + ca) - (a + b + c) = 0$$

$$abc(ab + bc + ca) = a + b + c$$

Hence Proved

Hence proved.

Question 42

In a triangle ABC , if:

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 + \cos A & 1 + \cos B & 1 + \cos C \\ \cos A + \cos^2 A & \cos B + \cos^2 B & \cos C + \cos^2 C \end{vmatrix} = 0$$

then prove that ABC is an isosceles triangle.

TYPE 10 Proving Determinant Identities Using Properties

★ Likely Exam Question

Given: In $\triangle ABC$, $\begin{vmatrix} 1 & 1 & 1 \\ 1 + \cos A & 1 + \cos B & 1 + \cos C \\ \cos A + \cos^2 A & \cos B + \cos^2 B & \cos C + \cos^2 C \end{vmatrix} = 0$

To Prove: $\triangle ABC$ is an isosceles triangle.

Concept / Formula Used: $\begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix} = (x - y)(y - z)(z - x)$

Solution:

Let $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 + \cos A & 1 + \cos B & 1 + \cos C \\ \cos A + \cos^2 A & \cos B + \cos^2 B & \cos C + \cos^2 C \end{vmatrix} = 0$.

Applying row operation $R_2 \rightarrow R_2 - R_1$:

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ \cos A & \cos B & \cos C \\ \cos A(1 + \cos A) & \cos B(1 + \cos B) & \cos C(1 + \cos C) \end{vmatrix} = 0$$

Applying row operation $R_3 \rightarrow R_3 - R_2$:

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ \cos A & \cos B & \cos C \\ \cos^2 A & \cos^2 B & \cos^2 C \end{vmatrix} = 0$$

This is the standard Vandermonde determinant in terms of $\cos A, \cos B, \cos C$:

$$(\cos A - \cos B)(\cos B - \cos C)(\cos C - \cos A) = 0$$

Equating each factor to zero:

$$\begin{aligned}\cos A - \cos B = 0 &\implies \cos A = \cos B \implies A = B \\ \text{or } \cos B - \cos C = 0 &\implies \cos B = \cos C \implies B = C \\ \text{or } \cos C - \cos A = 0 &\implies \cos C = \cos A \implies C = A\end{aligned}$$

Since at least two angles of $\triangle ABC$ are equal ($A = B$, $B = C$, or $C = A$), triangle ABC is isosceles.

Hence Proved

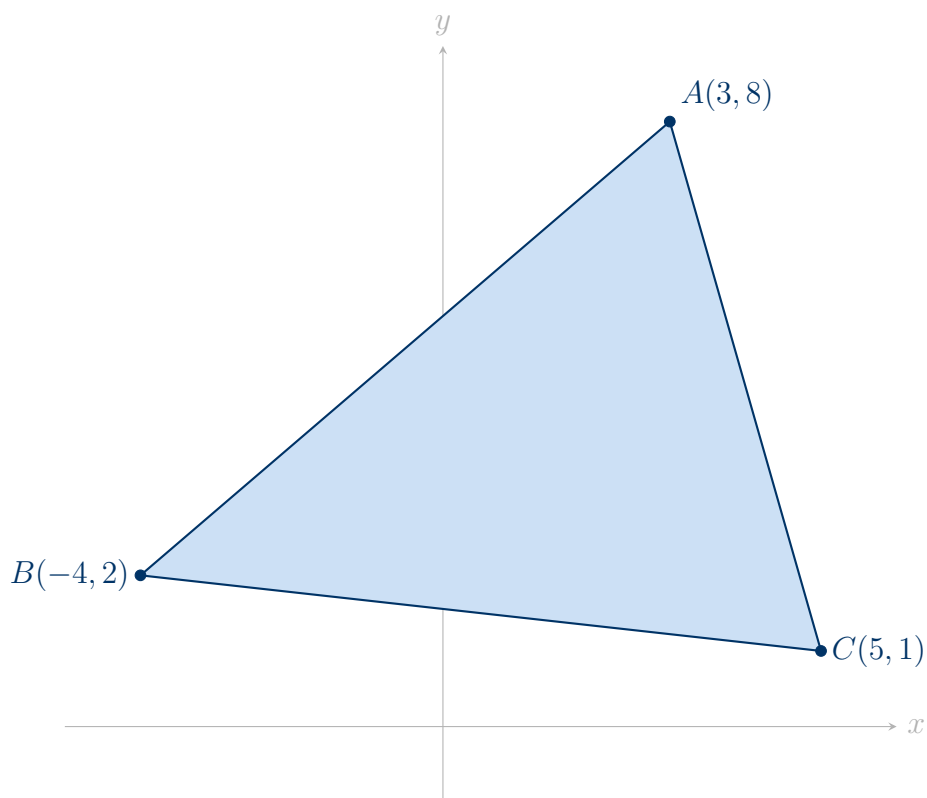
Hence proved.

4.3 – Applications of Determinants: Area, Collinearity and Equation of a Line

Question 1 (i)

Using determinants, find the area of the triangle whose vertices are $(3, 8)$, $(-4, 2)$, and $(5, 1)$.

TYPE 11 Area of Triangle Using Determinants



Triangle with vertices $A(3, 8)$, $B(-4, 2)$, $C(5, 1)$

Given: Vertices of the triangle are $A(3, 8)$, $B(-4, 2)$, and $C(5, 1)$.

Concept / Formula Used: Area of triangle $\Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$

Solution:

Let $(x_1, y_1) = (3, 8)$, $(x_2, y_2) = (-4, 2)$, and $(x_3, y_3) = (5, 1)$.

The area of triangle ABC is given by:

$$\Delta = \frac{1}{2} \begin{vmatrix} 3 & 8 & 1 \\ -4 & 2 & 1 \\ 5 & 1 & 1 \end{vmatrix}$$

Evaluating the determinant by expanding along the first row:

$$\begin{aligned} \begin{vmatrix} 3 & 8 & 1 \\ -4 & 2 & 1 \\ 5 & 1 & 1 \end{vmatrix} &= 3 \cdot \begin{vmatrix} 2 & 1 \\ 1 & 1 \end{vmatrix} - 8 \cdot \begin{vmatrix} -4 & 1 \\ 5 & 1 \end{vmatrix} + 1 \cdot \begin{vmatrix} -4 & 2 \\ 5 & 1 \end{vmatrix} \\ &= 3(2 \cdot 1 - 1 \cdot 1) - 8((-4) \cdot 1 - 1 \cdot 5) + 1((-4) \cdot 1 - 2 \cdot 5) \\ &= 3(2 - 1) - 8(-4 - 5) + 1(-4 - 10) \\ &= 3(1) - 8(-9) + 1(-14) \\ &= 3 + 72 - 14 \\ &= 61 \end{aligned}$$

Substituting this determinant value back into the area expression:

$$\begin{aligned}\Delta &= \frac{1}{2} \times 61 \\ &= \frac{61}{2} \\ &= 30.5 \text{ sq. units}\end{aligned}$$

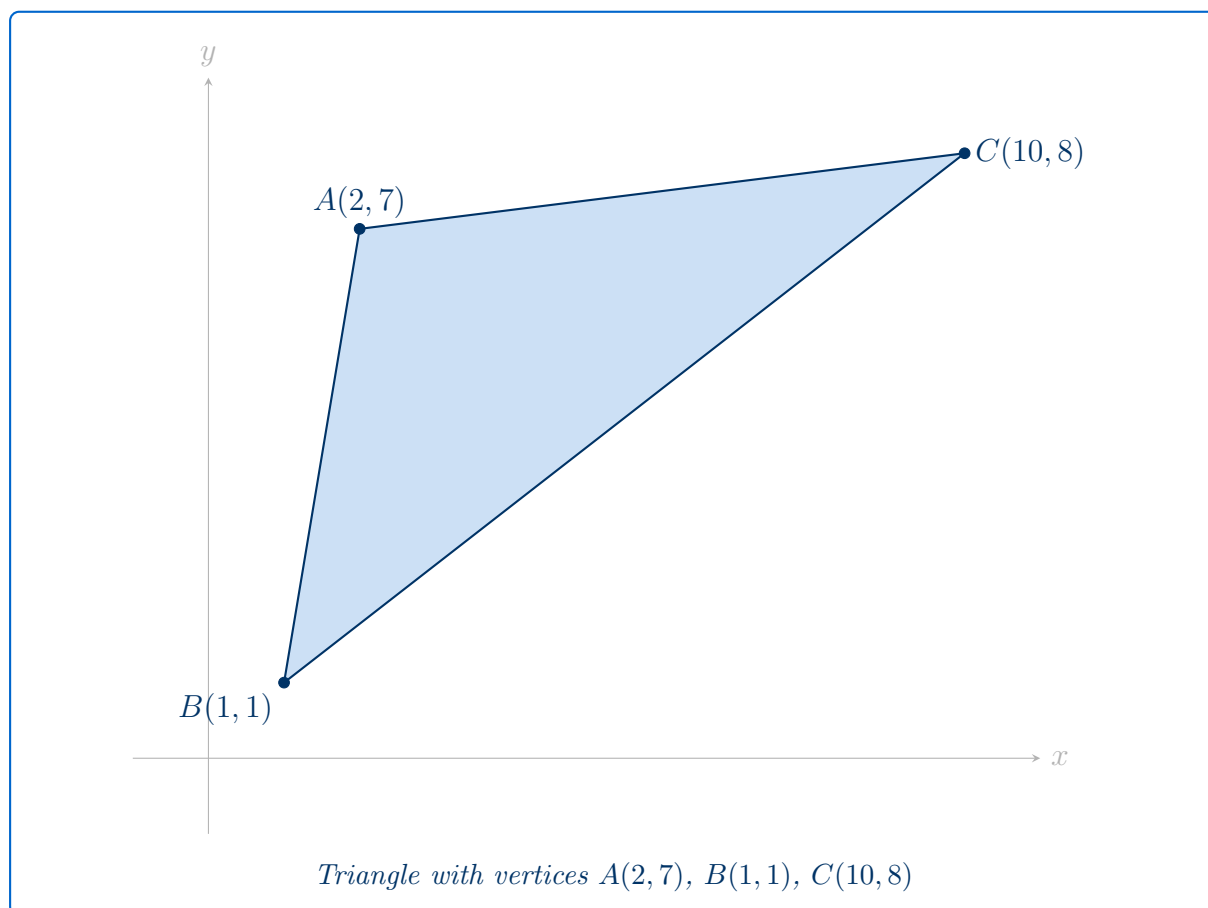
Final Answer

Area = 30.5 sq. units

Question 1 (ii)

Using determinants, find the area of the triangle whose vertices are $(2, 7)$, $(1, 1)$, and $(10, 8)$.

TYPE 11 Area of Triangle Using Determinants



Given: Vertices of the triangle are $A(2, 7)$, $B(1, 1)$, and $C(10, 8)$.

Concept / Formula Used: Area of triangle $\Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$

Solution:

Let $(x_1, y_1) = (2, 7)$, $(x_2, y_2) = (1, 1)$, and $(x_3, y_3) = (10, 8)$.

The area of triangle ABC is given by:

$$\Delta = \frac{1}{2} \begin{vmatrix} 2 & 7 & 1 \\ 1 & 1 & 1 \\ 10 & 8 & 1 \end{vmatrix}$$

Evaluating the determinant by expanding along the first row:

$$\begin{aligned} \begin{vmatrix} 2 & 7 & 1 \\ 1 & 1 & 1 \\ 10 & 8 & 1 \end{vmatrix} &= 2 \cdot \begin{vmatrix} 1 & 1 \\ 8 & 1 \end{vmatrix} - 7 \cdot \begin{vmatrix} 1 & 1 \\ 10 & 1 \end{vmatrix} + 1 \cdot \begin{vmatrix} 1 & 1 \\ 10 & 8 \end{vmatrix} \\ &= 2(1 \cdot 1 - 8 \cdot 1) - 7(1 \cdot 1 - 10 \cdot 1) + 1(1 \cdot 8 - 10 \cdot 1) \\ &= 2(1 - 8) - 7(1 - 10) + 1(8 - 10) \\ &= 2(-7) - 7(-9) + 1(-2) \\ &= -14 + 63 - 2 \\ &= 47 \end{aligned}$$

Substituting this value back into the area expression:

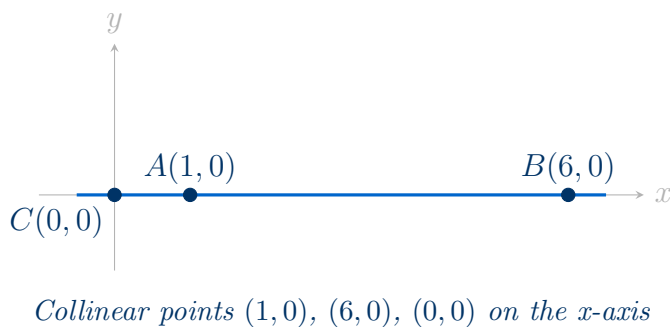
$$\begin{aligned} \Delta &= \frac{1}{2} \times 47 \\ &= \frac{47}{2} \\ &= 23.5 \text{ sq. units} \end{aligned}$$

Final Answer

Area = 23.5 sq. units

Question 2

Show that the points $(1, 0)$, $(6, 0)$, $(0, 0)$ are collinear.

TYPE 12 Collinearity of Points Using Determinants

To Prove: The points $A(1, 0)$, $B(6, 0)$, and $C(0, 0)$ are collinear.

Concept / Formula Used: $(x_1, y_1) \quad (x_2, y_2) \quad (x_3, y_3) \quad \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$

Solution:

Let $(x_1, y_1) = (1, 0)$, $(x_2, y_2) = (6, 0)$, and $(x_3, y_3) = (0, 0)$.

We set up the determinant D for the area of the triangle formed by these three points:

$$D = \begin{vmatrix} 1 & 0 & 1 \\ 6 & 0 & 1 \\ 0 & 0 & 1 \end{vmatrix}$$

Expanding along the second column C_2 :

$$\begin{aligned} D &= -0 \cdot \begin{vmatrix} 6 & 1 \\ 0 & 1 \end{vmatrix} + 0 \cdot \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} - 0 \cdot \begin{vmatrix} 1 & 1 \\ 6 & 1 \end{vmatrix} \\ &= 0 + 0 - 0 \\ &= 0 \end{aligned}$$

Since the value of the determinant is equal to 0, the area of the triangle formed by these points is zero.

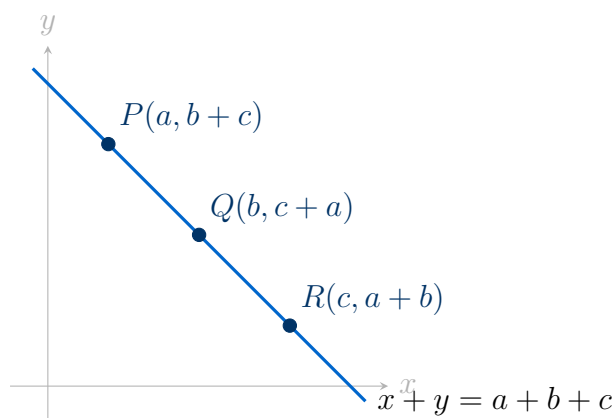
Hence Proved

Since the determinant vanishes, the points $(1, 0)$, $(6, 0)$, and $(0, 0)$ are collinear.

Question 3

Using determinants, determine whether the points $P(a, b+c)$, $Q(b, c+a)$ and $R(c, a+b)$ form a triangle or not.

TYPE 12 Collinearity of Points Using Determinants



Points $P(a, b+c)$, $Q(b, c+a)$, $R(c, a+b)$ on the line $x + y = a + b + c$

Given: Points $P(a, b+c)$, $Q(b, c+a)$, and $R(c, a+b)$.

Concept / Formula Used: $\Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} \neq 0 \quad \Delta = 0$

Solution:

Consider the determinant expression for the area Δ :

$$\Delta = \frac{1}{2} \begin{vmatrix} a & b+c & 1 \\ b & c+a & 1 \\ c & a+b & 1 \end{vmatrix}$$

Applying the column operation $C_2 \rightarrow C_2 + C_1$:

$$\Delta = \frac{1}{2} \begin{vmatrix} a & a+b+c & 1 \\ b & b+c+a & 1 \\ c & c+a+b & 1 \end{vmatrix}$$

Taking out the common factor $(a+b+c)$ from column C_2 :

$$\Delta = \frac{1}{2}(a+b+c) \begin{vmatrix} a & 1 & 1 \\ b & 1 & 1 \\ c & 1 & 1 \end{vmatrix}$$

Using the property that if any two columns of a determinant are identical, the value of the determinant is zero (here $C_2 = C_3$):

$$\begin{aligned} \Delta &= \frac{1}{2}(a+b+c) \times 0 \\ &= 0 \end{aligned}$$

Since the area of the triangle is 0, the points P, Q, R are collinear.

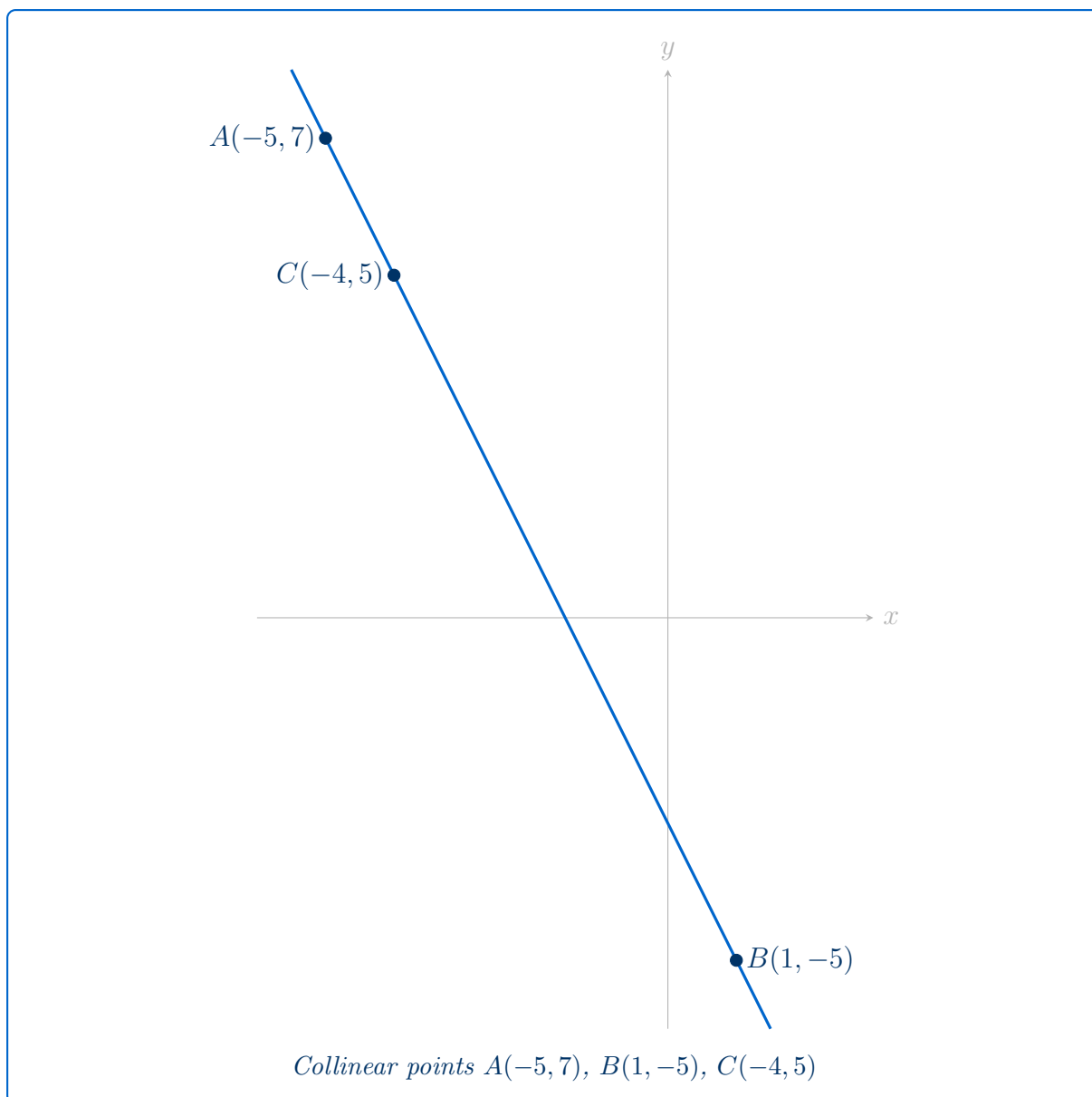
Final Answer

The given points P, Q, R are collinear and do not form a triangle.

Question 4

Using determinants, find the value of λ so that the points $(\lambda, 7)$, $(1, -5)$ and $(-4, 5)$ are collinear.

TYPE 12 Collinearity of Points Using Determinants



Given: Points $A(\lambda, 7)$, $B(1, -5)$, and $C(-4, 5)$ are collinear.

Concept / Formula Used: $\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$

Solution:

Since $A(\lambda, 7)$, $B(1, -5)$, and $C(-4, 5)$ are collinear, the determinant of their coordinates must equal zero:

$$\begin{vmatrix} \lambda & 7 & 1 \\ 1 & -5 & 1 \\ -4 & 5 & 1 \end{vmatrix} = 0$$

Expanding the determinant along the first row:

$$\begin{aligned} \lambda \cdot \begin{vmatrix} -5 & 1 \\ 5 & 1 \end{vmatrix} - 7 \cdot \begin{vmatrix} 1 & 1 \\ -4 & 1 \end{vmatrix} + 1 \cdot \begin{vmatrix} 1 & -5 \\ -4 & 5 \end{vmatrix} &= 0 \\ \lambda((-5) \cdot 1 - 5 \cdot 1) - 7(1 \cdot 1 - 1 \cdot (-4)) + 1(1 \cdot 5 - (-5) \cdot (-4)) &= 0 \\ \lambda(-5 - 5) - 7(1 + 4) + 1(5 - 20) &= 0 \\ \lambda(-10) - 7(5) + 1(-15) &= 0 \\ -10\lambda - 35 - 15 &= 0 \\ -10\lambda - 50 &= 0 \\ -10\lambda &= 50 \\ \lambda &= \frac{50}{-10} \\ \lambda &= -5 \end{aligned}$$

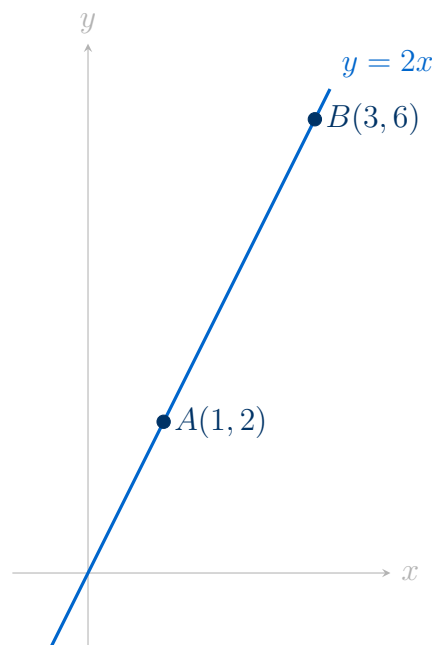
Final Answer

$$\lambda = -5$$

Question 5

Using determinants, find the equation of the line passing through the points (1, 2) and (3, 6).

TYPE 13 Equation of a Line Using Determinants



Line $y = 2x$ passing through (1, 2) and (3, 6)

Given: Points A(1, 2) and B(3, 6).

Concept / Formula Used: $(x_1, y_1) \quad (x_2, y_2) \quad \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$

Solution:

Let $P(x, y)$ be any arbitrary point on the line joining $A(1, 2)$ and $B(3, 6)$.

Since $P(x, y)$, $A(1, 2)$, and $B(3, 6)$ are collinear, the determinant of their coordinates is zero:

$$\begin{vmatrix} x & y & 1 \\ 1 & 2 & 1 \\ 3 & 6 & 1 \end{vmatrix} = 0$$

Expanding along the first row:

$$\begin{aligned} x \cdot \begin{vmatrix} 2 & 1 \\ 6 & 1 \end{vmatrix} - y \cdot \begin{vmatrix} 1 & 1 \\ 3 & 1 \end{vmatrix} + 1 \cdot \begin{vmatrix} 1 & 2 \\ 3 & 6 \end{vmatrix} &= 0 \\ x(2 \cdot 1 - 6 \cdot 1) - y(1 \cdot 1 - 3 \cdot 1) + 1(1 \cdot 6 - 3 \cdot 2) &= 0 \\ x(2 - 6) - y(1 - 3) + 1(6 - 6) &= 0 \\ x(-4) - y(-2) + 1(0) &= 0 \\ -4x + 2y &= 0 \end{aligned}$$

Dividing both sides by -2 :

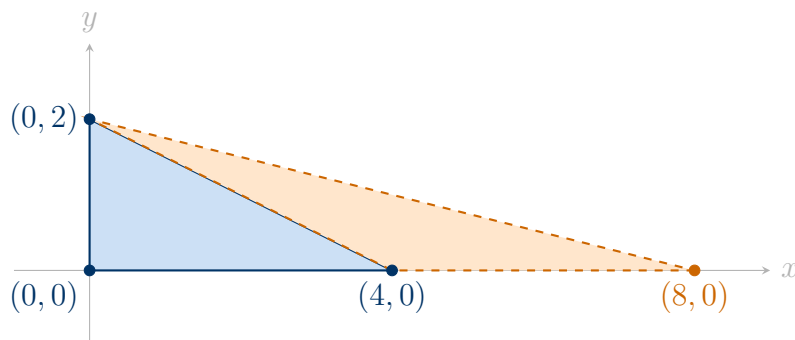
$$\begin{aligned} 2x - y &= 0 \\ y &= 2x \end{aligned}$$

Final Answer

$$y = 2x$$

Question 6

Find the value(s) of k if the area of the triangle with vertices $(k, 0)$, $(4, 0)$ and $(0, 2)$ is 4 sq. units.

TYPE 11 Area of Triangle Using Determinants

Triangles with vertices $(k, 0)$, $(4, 0)$, $(0, 2)$ for $k = 0$ and $k = 8$

Given: Vertices $A(k, 0)$, $B(4, 0)$, $C(0, 2)$, and Area = 4 sq. units.

Concept / Formula Used: $\frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \pm \text{given area}$

Solution:

Setting up the determinant equation with ± 4 since area is absolute:

$$\frac{1}{2} \begin{vmatrix} k & 0 & 1 \\ 4 & 0 & 1 \\ 0 & 2 & 1 \end{vmatrix} = \pm 4$$

Expanding the determinant along the first row:

$$\begin{aligned} \begin{vmatrix} k & 0 & 1 \\ 4 & 0 & 1 \\ 0 & 2 & 1 \end{vmatrix} &= k \cdot \begin{vmatrix} 0 & 1 \\ 2 & 1 \end{vmatrix} - 0 + 1 \cdot \begin{vmatrix} 4 & 0 \\ 0 & 2 \end{vmatrix} \\ &= k(0 \cdot 1 - 2 \cdot 1) + 1(4 \cdot 2 - 0 \cdot 0) \\ &= k(-2) + 8 \\ &= -2k + 8 \end{aligned}$$

Substituting back into the area equation:

$$\begin{aligned} \frac{1}{2}(-2k + 8) &= \pm 4 \\ -k + 4 &= \pm 4 \end{aligned}$$

Solving the two cases separately:

$$\begin{aligned} \text{Case 1: } -k + 4 &= 4 \\ -k &= 0 \\ k &= 0 \end{aligned}$$

$$\begin{aligned} \text{Case 2: } -k + 4 &= -4 \\ -k &= -8 \\ k &= 8 \end{aligned}$$

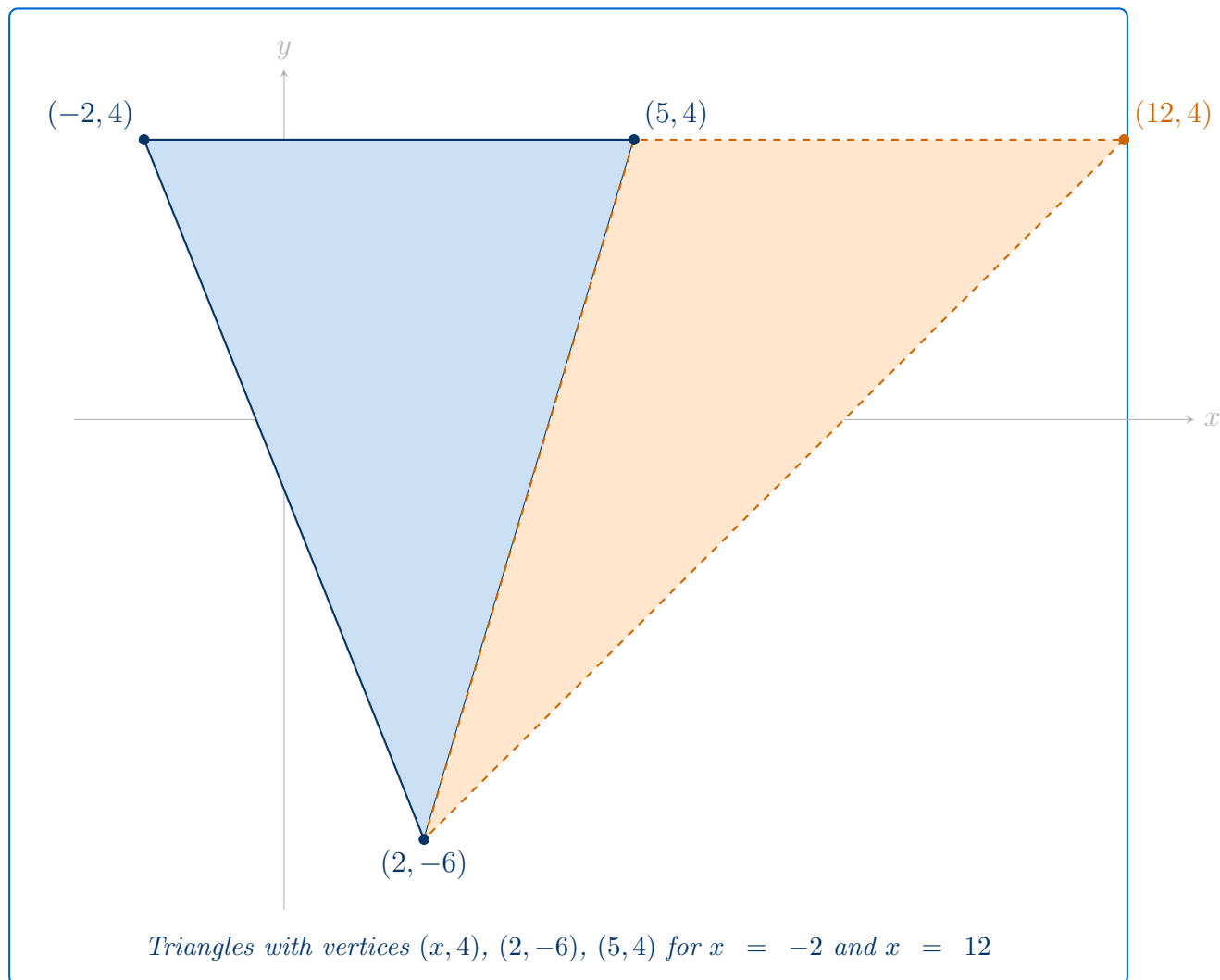
Final Answer

$$k = 0, 8$$

Common Mistake: Taking only +4 and neglecting -4 when setting up the determinant equation will result in missing the second value of k .

Question 7

Find the value(s) of x if the area of the triangle with vertices $(x, 4)$, $(2, -6)$ and $(5, 4)$ is 35 sq. units.

TYPE 11 Area of Triangle Using Determinants

Given: Vertices $A(x, 4)$, $B(2, -6)$, $C(5, 4)$, and Area = 35 sq. units.

Concept / Formula Used: $\frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \pm \text{given area}$

Solution:

Setting up the determinant equation with ± 35 :

$$\frac{1}{2} \begin{vmatrix} x & 4 & 1 \\ 2 & -6 & 1 \\ 5 & 4 & 1 \end{vmatrix} = \pm 35$$

Multiplying both sides by 2:

$$\begin{vmatrix} x & 4 & 1 \\ 2 & -6 & 1 \\ 5 & 4 & 1 \end{vmatrix} = \pm 70$$

Expanding the determinant along the first row:

$$\begin{aligned} x \cdot \begin{vmatrix} -6 & 1 \\ 4 & 1 \end{vmatrix} - 4 \cdot \begin{vmatrix} 2 & 1 \\ 5 & 1 \end{vmatrix} + 1 \cdot \begin{vmatrix} 2 & -6 \\ 5 & 4 \end{vmatrix} &= \pm 70 \\ x((-6) \cdot 1 - 4 \cdot 1) - 4(2 \cdot 1 - 5 \cdot 1) + 1(2 \cdot 4 - (-6) \cdot 5) &= \pm 70 \\ x(-6 - 4) - 4(2 - 5) + 1(8 + 30) &= \pm 70 \\ x(-10) - 4(-3) + 1(38) &= \pm 70 \\ -10x + 12 + 38 &= \pm 70 \\ -10x + 50 &= \pm 70 \end{aligned}$$

Solving the two cases separately:

$$\text{Case 1: } -10x + 50 = 70$$

$$-10x = 20$$

$$x = -2$$

$$\text{Case 2: } -10x + 50 = -70$$

$$-10x = -120$$

$$x = 12$$

Final Answer

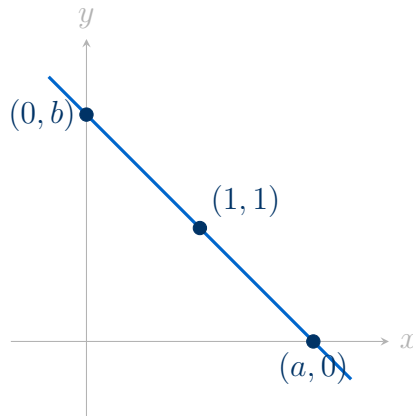
$$x = -2, 12$$

Common Mistake: Forgetting that the value of the determinant can be negative leads to missing $x = 12$.

Question 8

If the points $(a, 0)$, $(0, b)$ and $(1, 1)$ are collinear, prove that $a + b = ab$.

TYPE 12 Collinearity of Points Using Determinants



Collinear points $(a, 0)$, $(0, b)$, $(1, 1)$

To Prove: $a + b = ab$

Concept / Formula Used: (x_1, y_1) (x_2, y_2) (x_3, y_3) $\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$

Solution:

Given that the points $A(a, 0)$, $B(0, b)$, and $C(1, 1)$ are collinear, the determinant formed by their coordinates must equal zero:

$$\begin{vmatrix} a & 0 & 1 \\ 0 & b & 1 \\ 1 & 1 & 1 \end{vmatrix} = 0$$

Expanding along the first row:

$$\begin{aligned} a \cdot \begin{vmatrix} b & 1 \\ 1 & 1 \end{vmatrix} - 0 \cdot \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} + 1 \cdot \begin{vmatrix} 0 & b \\ 1 & 1 \end{vmatrix} &= 0 \\ a(b \cdot 1 - 1 \cdot 1) - 0 + 1(0 \cdot 1 - b \cdot 1) &= 0 \\ a(b - 1) + 1(0 - b) &= 0 \\ ab - a - b &= 0 \end{aligned}$$

Adding $a + b$ to both sides of the equation:

$$\begin{aligned} ab &= a + b \\ a + b &= ab \end{aligned}$$

Hence Proved

LHS = RHS, hence proved that $a + b = ab$.

4.4 – Adjoint and Inverse of a Matrix

Question 1 (i)

For what value of x , is the following matrix singular?

$$\begin{bmatrix} 3-2x & x+1 \\ 2 & 4 \end{bmatrix}$$

TYPE 14 Singular Matrix Condition

Given: The matrix $A = \begin{bmatrix} 3-2x & x+1 \\ 2 & 4 \end{bmatrix}$ is singular.

Concept / Formula Used: A $|A| = 0$

Solution:

Let the given matrix be denoted by A :

$$A = \begin{bmatrix} 3-2x & x+1 \\ 2 & 4 \end{bmatrix}$$

Since the matrix A is singular, its determinant must be equal to zero:

$$|A| = 0$$

We calculate the determinant of the 2×2 matrix:

$$|A| = (3-2x)(4) - (x+1)(2)$$

Setting the determinant to zero:

$$\begin{aligned} (3-2x)(4) - 2(x+1) &= 0 \\ 12 - 8x - 2x - 2 &= 0 \\ 10 - 10x &= 0 \\ -10x &= -10 \\ x &= 1 \end{aligned}$$

Final Answer

$$x = 1$$

Question 1 (ii)

If $0 < x < \pi$ and the matrix $\begin{bmatrix} \cos x & 1 \\ 3 & 4 \cos x \end{bmatrix}$ is singular, find the values of x .

TYPE 14 Singular Matrix Condition

Given: The matrix $A = \begin{bmatrix} \cos x & 1 \\ 3 & 4 \cos x \end{bmatrix}$ is singular, where $0 < x < \pi$.

Concept / Formula Used: A is singular $\Rightarrow |A| = 0$

Solution:

Let the given matrix be denoted by A :

$$A = \begin{bmatrix} \cos x & 1 \\ 3 & 4 \cos x \end{bmatrix}$$

Since the matrix A is singular, its determinant must be equal to zero:

$$|A| = 0$$

We calculate the determinant of the 2×2 matrix:

$$|A| = (\cos x)(4 \cos x) - (1)(3)$$

Setting the determinant to zero:

$$\begin{aligned} 4 \cos^2 x - 3 &= 0 \\ 4 \cos^2 x &= 3 \\ \cos^2 x &= \frac{3}{4} \\ \cos x &= \pm \frac{\sqrt{3}}{2} \end{aligned}$$

We are given the interval $0 < x < \pi$. Let us solve for x in this interval:

$$\begin{aligned} \text{For } \cos x &= \frac{\sqrt{3}}{2} \Rightarrow x = \frac{\pi}{6} \\ \text{For } \cos x &= -\frac{\sqrt{3}}{2} \Rightarrow x = \pi - \frac{\pi}{6} = \frac{5\pi}{6} \end{aligned}$$

Both values lie within the interval $(0, \pi)$.

Final Answer

$$x = \frac{\pi}{6} \text{ or } x = \frac{5\pi}{6}$$

Question 2 (i)

Determine if the following matrix is non-singular:

$$\begin{bmatrix} 6 & -3 & 2 \\ 2 & -1 & 2 \\ -10 & 5 & 2 \end{bmatrix}$$

TYPE 14 Singular Matrix Condition

Given: The matrix $A = \begin{bmatrix} 6 & -3 & 2 \\ 2 & -1 & 2 \\ -10 & 5 & 2 \end{bmatrix}$.

Concept / Formula Used: A $|A| \neq 0$

Solution:

Let us evaluate the determinant of the matrix A :

$$|A| = \begin{vmatrix} 6 & -3 & 2 \\ 2 & -1 & 2 \\ -10 & 5 & 2 \end{vmatrix}$$

Expanding along the first row:

$$|A| = 6 \begin{vmatrix} -1 & 2 \\ 5 & 2 \end{vmatrix} - (-3) \begin{vmatrix} 2 & 2 \\ -10 & 2 \end{vmatrix} + 2 \begin{vmatrix} 2 & -1 \\ -10 & 5 \end{vmatrix}$$

We evaluate each of the 2×2 determinants:

$$\begin{vmatrix} -1 & 2 \\ 5 & 2 \end{vmatrix} = (-1)(2) - (2)(5) = -2 - 10 = -12$$

$$\begin{vmatrix} 2 & 2 \\ -10 & 2 \end{vmatrix} = (2)(2) - (2)(-10) = 4 + 20 = 24$$

$$\begin{vmatrix} 2 & -1 \\ -10 & 5 \end{vmatrix} = (2)(5) - (-1)(-10) = 10 - 10 = 0$$

Substituting these values back into the determinant expression:

$$|A| = 6(-12) + 3(24) + 2(0)$$

$$|A| = -72 + 72 + 0$$

$$|A| = 0$$

Since $|A| = 0$, the matrix is singular.

Final Answer

The matrix is singular.

Question 2 (ii)

Determine if the following matrix is non-singular:

$$\begin{bmatrix} 1 & -3 & 2 \\ 4 & -1 & 2 \\ 3 & 5 & 2 \end{bmatrix}$$

TYPE 14 Singular Matrix Condition

Given: The matrix $B = \begin{bmatrix} 1 & -3 & 2 \\ 4 & -1 & 2 \\ 3 & 5 & 2 \end{bmatrix}$.

Concept / Formula Used: $B \quad |B| \neq 0$

Solution:

Let us evaluate the determinant of the matrix B :

$$|B| = \begin{vmatrix} 1 & -3 & 2 \\ 4 & -1 & 2 \\ 3 & 5 & 2 \end{vmatrix}$$

Expanding along the first row:

$$|B| = 1 \begin{vmatrix} -1 & 2 \\ 5 & 2 \end{vmatrix} - (-3) \begin{vmatrix} 4 & 2 \\ 3 & 2 \end{vmatrix} + 2 \begin{vmatrix} 4 & -1 \\ 3 & 5 \end{vmatrix}$$

We evaluate each of the 2×2 determinants:

$$\begin{vmatrix} -1 & 2 \\ 5 & 2 \end{vmatrix} = (-1)(2) - (2)(5) = -2 - 10 = -12$$

$$\begin{vmatrix} 4 & 2 \\ 3 & 2 \end{vmatrix} = (4)(2) - (2)(3) = 8 - 6 = 2$$

$$\begin{vmatrix} 4 & -1 \\ 3 & 5 \end{vmatrix} = (4)(5) - (-1)(3) = 20 + 3 = 23$$

Substituting these values back into the determinant expression:

$$|B| = 1(-12) + 3(2) + 2(23)$$

$$|B| = -12 + 6 + 46$$

$$|B| = 40$$

Since $|B| = 40 \neq 0$, the matrix is non-singular.

Final Answer

The matrix is non-singular.

Question 3 (i)

Write the adjoint of the matrix:

$$\begin{bmatrix} 2 & -1 \\ 4 & 3 \end{bmatrix}$$

TYPE 15 Adjoint of a Matrix

Given: The matrix $A = \begin{bmatrix} 2 & -1 \\ 4 & 3 \end{bmatrix}$.

Concept / Formula Used: $2 \times 2 \quad A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \text{adj } A = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Solution:

Let the given matrix be:

$$A = \begin{bmatrix} 2 & -1 \\ 4 & 3 \end{bmatrix}$$

Let C_{ij} denote the cofactor of the element a_{ij} of matrix A :

$$C_{11} = (-1)^{1+1}(3) = 3$$

$$C_{12} = (-1)^{1+2}(4) = -4$$

$$C_{21} = (-1)^{2+1}(-1) = 1$$

$$C_{22} = (-1)^{2+2}(2) = 2$$

The cofactor matrix C is:

$$C = \begin{bmatrix} 3 & -4 \\ 1 & 2 \end{bmatrix}$$

The adjoint of A is the transpose of the cofactor matrix C :

$$\begin{aligned} \text{adj } A &= C^T \\ &= \begin{bmatrix} 3 & 1 \\ -4 & 2 \end{bmatrix} \end{aligned}$$

Final Answer

$$\text{adj } A = \begin{bmatrix} 3 & 1 \\ -4 & 2 \end{bmatrix}$$

Question 3 (ii)

Write the adjoint of the matrix:

$$\begin{bmatrix} 3 & 0 \\ -2 & -5 \end{bmatrix}$$

TYPE 15 Adjoint of a Matrix

Given: The matrix $B = \begin{bmatrix} 3 & 0 \\ -2 & -5 \end{bmatrix}$.

Concept / Formula Used: 2×2 $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $\text{adj } B = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Solution:

Let the given matrix be:

$$B = \begin{bmatrix} 3 & 0 \\ -2 & -5 \end{bmatrix}$$

Let C_{ij} denote the cofactor of the element b_{ij} of matrix B :

$$C_{11} = (-1)^{1+1}(-5) = -5$$

$$C_{12} = (-1)^{1+2}(-2) = 2$$

$$C_{21} = (-1)^{2+1}(0) = 0$$

$$C_{22} = (-1)^{2+2}(3) = 3$$

The cofactor matrix C is:

$$C = \begin{bmatrix} -5 & 2 \\ 0 & 3 \end{bmatrix}$$

The adjoint of B is the transpose of the cofactor matrix C :

$$\begin{aligned} \text{adj } B &= C^T \\ &= \begin{bmatrix} -5 & 0 \\ 2 & 3 \end{bmatrix} \end{aligned}$$

Final Answer

$$\text{adj } B = \begin{bmatrix} -5 & 0 \\ 2 & 3 \end{bmatrix}$$

Question 4 (i)

For what value of k , the matrix $\begin{bmatrix} 2 & k \\ 3 & 5 \end{bmatrix}$ has no inverse?

TYPE 14 Singular Matrix Condition

Given: The matrix $A = \begin{bmatrix} 2 & k \\ 3 & 5 \end{bmatrix}$ has no inverse.

Concept / Formula Used: $A \quad |A| = 0$

Solution:

Let the given matrix be:

$$A = \begin{bmatrix} 2 & k \\ 3 & 5 \end{bmatrix}$$

For A to have no inverse, its determinant must be equal to zero:

$$|A| = 0$$

We calculate the determinant of the 2×2 matrix:

$$|A| = (2)(5) - (k)(3)$$

Setting the determinant to zero:

$$\begin{aligned} 10 - 3k &= 0 \\ -3k &= -10 \\ k &= \frac{10}{3} \end{aligned}$$

Final Answer

$$k = \frac{10}{3}$$

Question 4 (ii)

If $A = \begin{bmatrix} 3 & 10 \\ 2 & 7 \end{bmatrix}$, then write A^{-1} .

TYPE 16 Inverse of a Matrix Using Adjoint

Given: The matrix $A = \begin{bmatrix} 3 & 10 \\ 2 & 7 \end{bmatrix}$.

Concept / Formula Used: $A^{-1} = \frac{1}{|A|} \text{adj } A$

Solution:

First, we find the determinant of A :

$$\begin{aligned}|A| &= (3)(7) - (10)(2) \\ &= 21 - 20 \\ &= 1\end{aligned}$$

Since $|A| = 1 \neq 0$, the inverse A^{-1} exists. Next, we find the cofactors of A :

$$\begin{aligned}C_{11} &= 7 \\ C_{12} &= -2 \\ C_{21} &= -10 \\ C_{22} &= 3\end{aligned}$$

The cofactor matrix C is:

$$C = \begin{bmatrix} 7 & -2 \\ -10 & 3 \end{bmatrix}$$

The adjoint of A is the transpose of the cofactor matrix C :

$$\text{adj } A = C^T = \begin{bmatrix} 7 & -10 \\ -2 & 3 \end{bmatrix}$$

Using the formula for the inverse:

$$\begin{aligned}A^{-1} &= \frac{1}{|A|} \text{adj } A \\ &= \frac{1}{1} \begin{bmatrix} 7 & -10 \\ -2 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 7 & -10 \\ -2 & 3 \end{bmatrix}\end{aligned}$$

Final Answer

$$A^{-1} = \begin{bmatrix} 7 & -10 \\ -2 & 3 \end{bmatrix}$$

Question 4 (iii)

Write the inverse of the matrix $\begin{bmatrix} 3 & -1 \\ 1 & 2 \end{bmatrix}$.

TYPE 16 Inverse of a Matrix Using Adjoint

Given: The matrix $A = \begin{bmatrix} 3 & -1 \\ 1 & 2 \end{bmatrix}$.

Concept / Formula Used: $A^{-1} = \frac{1}{|A|} \text{adj } A$

Solution:

Let the given matrix be:

$$A = \begin{bmatrix} 3 & -1 \\ 1 & 2 \end{bmatrix}$$

First, we find the determinant of A :

$$\begin{aligned} |A| &= (3)(2) - (-1)(1) \\ &= 6 + 1 \\ &= 7 \end{aligned}$$

Since $|A| = 7 \neq 0$, the inverse A^{-1} exists. Next, we find the cofactors of A :

$$\begin{aligned} C_{11} &= 2 \\ C_{12} &= -1 \\ C_{21} &= 1 \\ C_{22} &= 3 \end{aligned}$$

The cofactor matrix C is:

$$C = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$$

The adjoint of A is the transpose of the cofactor matrix C :

$$\text{adj } A = C^T = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$$

Using the formula for the inverse:

$$\begin{aligned} A^{-1} &= \frac{1}{|A|} \text{adj } A \\ &= \frac{1}{7} \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} \\ &= \begin{bmatrix} \frac{2}{7} & \frac{1}{7} \\ -\frac{1}{7} & \frac{3}{7} \end{bmatrix} \end{aligned}$$

Final Answer

$$A^{-1} = \begin{bmatrix} \frac{2}{7} & \frac{1}{7} \\ -\frac{1}{7} & \frac{3}{7} \end{bmatrix}$$

Question 5 (i)

If $A = \begin{bmatrix} 8 & 2 \\ 3 & 2 \end{bmatrix}$, then find $|\text{adj } A|$.

TYPE 17 Properties of Adjoint and Inverse Determinants

Given: The matrix $A = \begin{bmatrix} 8 & 2 \\ 3 & 2 \end{bmatrix}$.

Concept / Formula Used: $A \quad n \quad |\text{adj } A| = |A|^{n-1}$

Solution:

The given matrix A is of order $n = 2$. First, we calculate the determinant of A :

$$\begin{aligned} |A| &= (8)(2) - (2)(3) \\ &= 16 - 6 \\ &= 10 \end{aligned}$$

Using the standard result $|\text{adj } A| = |A|^{n-1}$ with $n = 2$:

$$\begin{aligned} |\text{adj } A| &= |A|^{2-1} \\ &= |A|^1 \\ &= 10 \end{aligned}$$

Final Answer

$$|\text{adj } A| = 10$$

Question 5 (ii)

If A is a square matrix of order 3 and $|A| = 5$, then find $|\text{adj } A|$.

TYPE 17 Properties of Adjoint and Inverse Determinants

Given: A square matrix A of order $n = 3$ with $|A| = 5$.

Concept / Formula Used: $A \quad n \quad |\text{adj } A| = |A|^{n-1}$

Solution:

Using the standard result $|\text{adj } A| = |A|^{n-1}$ with $n = 3$:

$$|\text{adj } A| = |A|^{3-1} = |A|^2$$

Substituting the given value $|A| = 5$:

$$\begin{aligned} |\text{adj } A| &= 5^2 \\ &= 25 \end{aligned}$$

Final Answer

$$|\text{adj } A| = 25$$

Question 5 (iii)

If A is a non-singular square matrix of order 3 such that $|\text{adj } A| = 64$, find $|A|$.

TYPE 17 Properties of Adjoint and Inverse Determinants

Given: A square matrix A of order $n = 3$ with $|\text{adj } A| = 64$.

Concept / Formula Used: $A \quad n \quad |\text{adj } A| = |A|^{n-1}$

Solution:

Using the standard result $|\text{adj } A| = |A|^{n-1}$ with $n = 3$:

$$|\text{adj } A| = |A|^{3-1} = |A|^2$$

We are given that $|\text{adj } A| = 64$. Therefore:

$$|A|^2 = 64$$

$$|A| = \pm\sqrt{64}$$

$$|A| = \pm 8$$

Final Answer

$$|A| = \pm 8$$

Question 5 (iv)

If A is a square matrix of order 3 such that $|A| = -2$, then find $|3\text{adj } A|$.

TYPE 17 Properties of Adjoint and Inverse Determinants

Given: A square matrix A of order $n = 3$ with $|A| = -2$.

Concept / Formula Used: $M \quad n \quad k \quad |kM| = k^n|M| \quad |\text{adj } A| = |A|^{n-1}$

Solution:

We need to find $|3\text{adj } A|$. Using the property $|kM| = k^n|M|$ with $k = 3$, $n = 3$, and $M = \text{adj } A$:

$$|3\text{adj } A| = 3^3|\text{adj } A| = 27|\text{adj } A|$$

Using the standard result $|\text{adj } A| = |A|^{n-1}$ with $n = 3$:

$$|\text{adj } A| = |A|^{3-1} = |A|^2$$

Substituting $|A| = -2$:

$$|\text{adj } A| = (-2)^2 = 4$$

Now, substituting this back into the expression:

$$\begin{aligned} |3\text{adj } A| &= 27 \times 4 \\ &= 108 \end{aligned}$$

Final Answer

$$|3\text{adj } A| = 108$$

Question 5 (v)

If $|A^{-1}| = 5$, then find the value of $|A|$.

TYPE 17 Properties of Adjoint and Inverse Determinants

Given: An invertible matrix A with $|A^{-1}| = 5$.

Concept / Formula Used: $A \quad |A^{-1}| = \frac{1}{|A|}$

Solution:

Using the standard result $|A^{-1}| = \frac{1}{|A|}$:

$$\begin{aligned} 5 &= \frac{1}{|A|} \\ |A| &= \frac{1}{5} \end{aligned}$$

Final Answer

$$|A| = \frac{1}{5}$$

Question 5 (vi)

If $|A| = -4$ and $A^{-1} = \begin{bmatrix} -2 & 1 \\ \frac{5}{4} & \frac{1}{2} \end{bmatrix}$, then write $\text{adj } A$.

TYPE 17 Properties of Adjoint and Inverse Determinants

Given: $|A| = -4$ and $A^{-1} = \begin{bmatrix} -2 & 1 \\ \frac{5}{4} & \frac{1}{2} \end{bmatrix}$.

Concept / Formula Used: $A \quad A^{-1} = \frac{1}{|A|} \text{adj } A \quad \text{adj } A = |A|A^{-1}$

Solution:

Using the standard result $\text{adj } A = |A|A^{-1}$:

$$\text{adj } A = -4 \begin{bmatrix} -2 & 1 \\ \frac{5}{4} & \frac{1}{2} \end{bmatrix}$$

Multiplying each element of the matrix by -4 :

$$\begin{aligned}\text{adj } A &= \begin{bmatrix} (-4)(-2) & (-4)(1) \\ (-4)\left(\frac{5}{4}\right) & (-4)\left(\frac{1}{2}\right) \end{bmatrix} \\ &= \begin{bmatrix} 8 & -4 \\ -5 & -2 \end{bmatrix}\end{aligned}$$

Final Answer

$$\text{adj } A = \begin{bmatrix} 8 & -4 \\ -5 & -2 \end{bmatrix}$$

Question 5 (vii)

If $A = \begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix}$ be such that $A^{-1} = kA$, then find the value of k .

TYPE 16 Inverse of a Matrix Using Adjoint

Given: $A = \begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix}$ and $A^{-1} = kA$.

Concept / Formula Used: $A^{-1} = \frac{1}{|A|} \text{adj } A$

Solution:

First, we find the determinant of A :

$$\begin{aligned}|A| &= (2)(-2) - (3)(5) \\ &= -4 - 15 \\ &= -19\end{aligned}$$

Next, we find the cofactors of A :

$$\begin{aligned}C_{11} &= -2 \\ C_{12} &= -5 \\ C_{21} &= -3 \\ C_{22} &= 2\end{aligned}$$

The cofactor matrix C is:

$$C = \begin{bmatrix} -2 & -5 \\ -3 & 2 \end{bmatrix}$$

The adjoint of A is the transpose of the cofactor matrix C :

$$\text{adj } A = C^T = \begin{bmatrix} -2 & -3 \\ -5 & 2 \end{bmatrix}$$

Using the formula for the inverse:

$$\begin{aligned} A^{-1} &= \frac{1}{|A|} \text{adj } A \\ &= \frac{1}{-19} \begin{bmatrix} -2 & -3 \\ -5 & 2 \end{bmatrix} \\ &= \frac{1}{19} \begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix} \end{aligned}$$

Since $A = \begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix}$, we can write:

$$A^{-1} = \frac{1}{19}A$$

Comparing this with the given equation $A^{-1} = kA$:

$$k = \frac{1}{19}$$

Final Answer

$$k = \frac{1}{19}$$

Question 5 (viii)

If $A(\text{adj } A) = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, then find the value of $|\text{adj } A|$.

TYPE 17 Properties of Adjoint and Inverse Determinants

Given: $A(\text{adj } A) = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$.

Concept / Formula Used: $A \quad n \quad A(\text{adj } A) = |A|I_n \quad |\text{adj } A| = |A|^{n-1}$

Solution:

We are given:

$$A(\text{adj } A) = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} = 3 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = 3I_3$$

Using the standard result $A(\text{adj } A) = |A|I_3$:

$$|A|I_3 = 3I_3$$

Comparing the coefficients:

$$|A| = 3$$

Since the order of the matrix is $n = 3$, we use the standard result $|\text{adj } A| = |A|^{n-1}$:

$$\begin{aligned} |\text{adj } A| &= |A|^{3-1} \\ &= |A|^2 \\ &= 3^2 \\ &= 9 \end{aligned}$$

Final Answer

$$|\text{adj } A| = 9$$

Question 5 (ix)

If the value of a determinant of third order is 12, then what is the value of the determinant formed by replacing each element by its co-factor?

TYPE 17 Properties of Adjoint and Inverse Determinants

Given: A square matrix A of order $n = 3$ with $|A| = 12$.

Concept / Formula Used: $A \quad n \quad C \quad |C| = |A|^{n-1}$

Solution:

Let C be the cofactor matrix of A . Using the standard result $|C| = |A|^{n-1}$ with $n = 3$:

$$|C| = |A|^{3-1} = |A|^2$$

Substituting the given value $|A| = 12$:

$$\begin{aligned} |C| &= 12^2 \\ &= 144 \end{aligned}$$

Final Answer

$$144$$

Question 6 (i)

Without computing $\text{adj } A$, find $|\text{adj } A|$ if:

$$A = \begin{bmatrix} 1 & 3 \\ -1 & 4 \end{bmatrix}$$

TYPE 17 Properties of Adjoint and Inverse Determinants

Given: The matrix $A = \begin{bmatrix} 1 & 3 \\ -1 & 4 \end{bmatrix}$.

Concept / Formula Used: $A \quad n \quad |\text{adj } A| = |A|^{n-1}$

Solution:

The given matrix A is of order $n = 2$. First, we find the determinant of A :

$$\begin{aligned}|A| &= (1)(4) - (3)(-1) \\ &= 4 + 3 \\ &= 7\end{aligned}$$

Using the standard result $|\text{adj } A| = |A|^{n-1}$ with $n = 2$:

$$\begin{aligned}|\text{adj } A| &= |A|^{2-1} \\ &= |A|^1 \\ &= 7\end{aligned}$$

Final Answer

$$|\text{adj } A| = 7$$

Question 6 (ii)

Without computing $\text{adj } A$, find $|\text{adj } A|$ if:

$$A = \begin{bmatrix} -2 & 0 & 0 \\ 3 & 4 & 0 \\ 10 & -7 & 3 \end{bmatrix}$$

TYPE 17 Properties of Adjoint and Inverse Determinants

Given: The matrix $A = \begin{bmatrix} -2 & 0 & 0 \\ 3 & 4 & 0 \\ 10 & -7 & 3 \end{bmatrix}$.

Concept / Formula Used: A n $|\text{adj } A| = |A|^{n-1}$

Solution:

The given matrix A is of order $n = 3$. Since A is a lower triangular matrix, its determinant is the product of its diagonal elements:

$$\begin{aligned}|A| &= (-2)(4)(3) \\ &= -24\end{aligned}$$

Using the standard result $|\text{adj } A| = |A|^{n-1}$ with $n = 3$:

$$\begin{aligned}|\text{adj } A| &= |A|^{3-1} \\ &= |A|^2 \\ &= (-24)^2 \\ &= 576\end{aligned}$$

Final Answer

$$|\text{adj } A| = 576$$

Question 6 (iii)

Without computing $\text{adj } A$, find $|\text{adj } A|$ if:

$$A = \begin{bmatrix} 2 & -1 & -2 \\ 0 & 2 & -1 \\ 3 & -5 & 0 \end{bmatrix}$$

TYPE 17 Properties of Adjoint and Inverse Determinants

Given: The matrix $A = \begin{bmatrix} 2 & -1 & -2 \\ 0 & 2 & -1 \\ 3 & -5 & 0 \end{bmatrix}$.

Concept / Formula Used: A is of order n $|\text{adj } A| = |A|^{n-1}$

Solution:

The given matrix A is of order $n = 3$. Let us find the determinant of A by expanding along the first row:

$$|A| = 2 \begin{vmatrix} 2 & -1 \\ -5 & 0 \end{vmatrix} - (-1) \begin{vmatrix} 0 & -1 \\ 3 & 0 \end{vmatrix} + (-2) \begin{vmatrix} 0 & 2 \\ 3 & -5 \end{vmatrix}$$

We evaluate each of the 2×2 determinants:

$$\begin{vmatrix} 2 & -1 \\ -5 & 0 \end{vmatrix} = (2)(0) - (-1)(-5) = 0 - 5 = -5$$

$$\begin{vmatrix} 0 & -1 \\ 3 & 0 \end{vmatrix} = (0)(0) - (-1)(3) = 0 + 3 = 3$$

$$\begin{vmatrix} 0 & 2 \\ 3 & -5 \end{vmatrix} = (0)(-5) - (2)(3) = 0 - 6 = -6$$

Substituting these values back into the determinant expression:

$$|A| = 2(-5) + 1(3) - 2(-6)$$

$$|A| = -10 + 3 + 12$$

$$|A| = 5$$

Using the standard result $|\text{adj } A| = |A|^{n-1}$ with $n = 3$:

$$\begin{aligned} |\text{adj } A| &= |A|^{3-1} \\ &= |A|^2 \\ &= 5^2 \\ &= 25 \end{aligned}$$

Final Answer

$$|\text{adj } A| = 25$$

Question 7 (i)

If $A = \begin{bmatrix} 2 & 1 \\ 7 & 5 \end{bmatrix}$, find $|A \text{adj } A|$.

TYPE 17 Properties of Adjoint and Inverse Determinants

Given: The matrix $A = \begin{bmatrix} 2 & 1 \\ 7 & 5 \end{bmatrix}$.

Concept / Formula Used: $A \quad n \quad A \text{adj } A = |A|I_n \quad |kI_n| = k^n$

Solution:

The given matrix A is of order $n = 2$. First, we find the determinant of A :

$$\begin{aligned} |A| &= (2)(5) - (1)(7) \\ &= 10 - 7 \\ &= 3 \end{aligned}$$

Using the standard result $A \text{adj } A = |A|I_2$:

$$A \text{adj } A = 3I_2$$

Taking the determinant on both sides:

$$|A \text{adj } A| = |3I_2|$$

Using the property $|kI_n| = k^n$ with $k = 3$ and $n = 2$:

$$\begin{aligned} |3I_2| &= 3^2 \\ &= 9 \end{aligned}$$

Final Answer

$$|A \text{adj } A| = 9$$

Question 7 (ii)

If $A = \begin{bmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{bmatrix}$, $a \neq 0$, find $|A \text{adj } A|$.

TYPE 17 Properties of Adjoint and Inverse Determinants

Given: The matrix $A = \begin{bmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{bmatrix}$ with $a \neq 0$.

Concept / Formula Used: A is of order n , $A \text{adj } A = |A|I_n$ and $|kI_n| = k^n$

Solution:

The given matrix A is of order $n = 3$. Since A is a diagonal matrix, its determinant is the product of its diagonal elements:

$$\begin{aligned}|A| &= a \cdot a \cdot a \\ &= a^3\end{aligned}$$

Using the standard result $A \text{adj } A = |A|I_3$:

$$A \text{adj } A = a^3 I_3$$

Taking the determinant on both sides:

$$|A \text{adj } A| = |a^3 I_3|$$

Using the property $|kI_n| = k^n$ with $k = a^3$ and $n = 3$:

$$\begin{aligned}|a^3 I_3| &= (a^3)^3 \\ &= a^9\end{aligned}$$

Final Answer

$$|A \text{adj } A| = a^9$$

Question 7 (iii)

If $A = \begin{bmatrix} 3 & -1 \\ 4 & 5 \end{bmatrix}$, find $\text{adj}(\text{adj } A)$.

TYPE 17 Properties of Adjoint and Inverse Determinants

Given: The matrix $A = \begin{bmatrix} 3 & -1 \\ 4 & 5 \end{bmatrix}$.

Concept / Formula Used: A is of order n , $\text{adj}(\text{adj } A) = |A|^{n-2}A$

Solution:

The given matrix A is of order $n = 2$. First, we find the determinant of A :

$$\begin{aligned}|A| &= (3)(5) - (-1)(4) \\ &= 15 + 4 \\ &= 19\end{aligned}$$

Using the standard result $\text{adj}(\text{adj } A) = |A|^{n-2}A$ with $n = 2$:

$$\begin{aligned}\text{adj}(\text{adj } A) &= |A|^{2-2}A \\ &= |A|^0 A \\ &= 1 \cdot A \\ &= \begin{bmatrix} 3 & -1 \\ 4 & 5 \end{bmatrix}\end{aligned}$$

Final Answer

$$\text{adj}(\text{adj } A) = \begin{bmatrix} 3 & -1 \\ 4 & 5 \end{bmatrix}$$

Question 8 (i)

If A is a non-singular matrix, then show that $|A^{-1}| = \frac{1}{|A|}$.

TYPE 18 Matrix Inverse Properties and Proofs

To Prove: $|A^{-1}| = \frac{1}{|A|}$

Concept / Formula Used: $A \quad AA^{-1} = I \quad |AB| = |A||B|$

Solution:

Since A is non-singular, its inverse A^{-1} exists and satisfies the relation:

$$AA^{-1} = I$$

Taking the determinant on both sides:

$$|AA^{-1}| = |I|$$

Using the multiplicative property of determinants $|AB| = |A||B|$:

$$|A||A^{-1}| = |I|$$

Since the determinant of the identity matrix I is 1:

$$|A||A^{-1}| = 1$$

Since A is non-singular, $|A| \neq 0$. Dividing both sides by $|A|$:

$$|A^{-1}| = \frac{1}{|A|}$$

LHS = RHS, hence proved.

Hence Proved

The relation $|A^{-1}| = \frac{1}{|A|}$ is proved.

Question 8 (ii)

If A is a matrix such that $A^3 = I$, then show that A is invertible.

TYPE 18 Matrix Inverse Properties and Proofs

To Prove: A is invertible.

Concept / Formula Used: $A \quad B \quad AB = BA = I$

Solution:

We are given the matrix equation:

$$A^3 = I$$

We can rewrite A^3 as:

$$A(A^2) = I \quad \text{and} \quad (A^2)A = I$$

Let us define a matrix B such that:

$$B = A^2$$

Substituting B into the equations:

$$AB = I \quad \text{and} \quad BA = I$$

By the definition of the inverse of a matrix, since there exists a matrix $B = A^2$ such that $AB = BA = I$, the matrix A is invertible, and its inverse is:

$$A^{-1} = A^2$$

LHS = RHS, hence proved.

Hence Proved

The matrix A is invertible.

Question 8 (iii)

If A is a matrix such that $A^4 = I$, then show that $A^{-1} = A^3$.

TYPE 18 Matrix Inverse Properties and Proofs

To Prove: $A^{-1} = A^3$

Concept / Formula Used: $B \quad A \quad AB = BA = I$

Solution:

We are given the matrix equation:

$$A^4 = I$$

We can rewrite A^4 as:

$$A(A^3) = I \quad \text{and} \quad (A^3)A = I$$

By the definition of the inverse of a matrix, if there exists a matrix B such that $AB = BA = I$, then $B = A^{-1}$. Here, the matrix $B = A^3$ satisfies this condition. Therefore:

$$A^{-1} = A^3$$

LHS = RHS, hence proved.

Hence Proved

The relation $A^{-1} = A^3$ is proved.

Question 9

If $A = \begin{bmatrix} 3 & -5 \\ 4 & 2 \end{bmatrix}$, find $A(\text{adj } A)$.

TYPE 19 Verification of Matrix Adjoint Properties

Given: The matrix $A = \begin{bmatrix} 3 & -5 \\ 4 & 2 \end{bmatrix}$.

Concept / Formula Used: $A \quad n \quad A(\text{adj } A) = |A|I_n$

Solution:

The given matrix A is of order $n = 2$. First, we find the determinant of A :

$$\begin{aligned} |A| &= (3)(2) - (-5)(4) \\ &= 6 + 20 \\ &= 26 \end{aligned}$$

Using the standard result $A(\text{adj } A) = |A|I_2$:

$$\begin{aligned} A(\text{adj } A) &= 26I_2 \\ &= 26 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 26 & 0 \\ 0 & 26 \end{bmatrix} \end{aligned}$$

Final Answer

$$A(\text{adj } A) = \begin{bmatrix} 26 & 0 \\ 0 & 26 \end{bmatrix}$$

Question 10

If the matrix $\begin{bmatrix} x+4 & x & x \\ x & x+4 & x \\ x & x & x+4 \end{bmatrix}$ is singular, find x .

TYPE 14 Singular Matrix Condition

Given: The matrix $A = \begin{bmatrix} x+4 & x & x \\ x & x+4 & x \\ x & x & x+4 \end{bmatrix}$ is singular.

Concept / Formula Used: $A \quad |A| = 0$

Solution:

Let the given matrix be denoted by A :

$$A = \begin{bmatrix} x+4 & x & x \\ x & x+4 & x \\ x & x & x+4 \end{bmatrix}$$

Since A is singular, its determinant must be equal to zero:

$$|A| = \begin{vmatrix} x+4 & x & x \\ x & x+4 & x \\ x & x & x+4 \end{vmatrix} = 0$$

Let us apply row operations to simplify the determinant. Perform $R_1 \rightarrow R_1 + R_2 + R_3$:

$$|A| = \begin{vmatrix} (x+4) + x + x & x + (x+4) + x & x + x + (x+4) \\ x & x+4 & x \\ x & x & x+4 \end{vmatrix}$$

$$|A| = \begin{vmatrix} 3x+4 & 3x+4 & 3x+4 \\ x & x+4 & x \\ x & x & x+4 \end{vmatrix}$$

Taking out the common factor $(3x+4)$ from the first row:

$$|A| = (3x+4) \begin{vmatrix} 1 & 1 & 1 \\ x & x+4 & x \\ x & x & x+4 \end{vmatrix}$$

Now, perform column operations $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_1$:

$$|A| = (3x+4) \begin{vmatrix} 1 & 1-1 & 1-1 \\ x & (x+4)-x & x-x \\ x & x-x & (x+4)-x \end{vmatrix}$$

$$|A| = (3x+4) \begin{vmatrix} 1 & 0 & 0 \\ x & 4 & 0 \\ x & 0 & 4 \end{vmatrix}$$

Expanding along the first row:

$$|A| = (3x+4) \cdot 1 \cdot \begin{vmatrix} 4 & 0 \\ 0 & 4 \end{vmatrix}$$

$$= (3x+4) \cdot (16 - 0)$$

$$= 16(3x+4)$$

Setting the determinant to zero:

$$16(3x+4) = 0$$

$$3x+4 = 0$$

$$3x = -4$$

$$x = -\frac{4}{3}$$

Final Answer

$$x = -\frac{4}{3}$$

Question 11 (i)

Find the adjoint of the matrix:

$$A = \begin{bmatrix} -2 & 1 \\ 3 & 4 \end{bmatrix}$$

Also verify that $A(\text{adj } A) = |A|I_2 = (\text{adj } A)A$.

TYPE 19 Verification of Matrix Adjoint Properties

Given: The matrix $A = \begin{bmatrix} -2 & 1 \\ 3 & 4 \end{bmatrix}$.

Concept / Formula Used: $A(\text{adj } A) = |A|I_n = (\text{adj } A)A$

Solution:

First, we find the determinant of A :

$$\begin{aligned} |A| &= (-2)(4) - (1)(3) \\ &= -8 - 3 \\ &= -11 \end{aligned}$$

Next, we find the cofactors of A :

$$\begin{aligned} C_{11} &= (-1)^{1+1}(4) = 4 \\ C_{12} &= (-1)^{1+2}(3) = -3 \\ C_{21} &= (-1)^{2+1}(1) = -1 \\ C_{22} &= (-1)^{2+2}(-2) = -2 \end{aligned}$$

The cofactor matrix C is:

$$C = \begin{bmatrix} 4 & -3 \\ -1 & -2 \end{bmatrix}$$

The adjoint of A is the transpose of the cofactor matrix C :

$$\text{adj } A = C^T = \begin{bmatrix} 4 & -1 \\ -3 & -2 \end{bmatrix}$$

Now, let us verify the relation $A(\text{adj } A) = |A|I_2 = (\text{adj } A)A$:

****Step 1: Compute $A(\text{adj } A)$ ****

$$\begin{aligned} A(\text{adj } A) &= \begin{bmatrix} -2 & 1 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 4 & -1 \\ -3 & -2 \end{bmatrix} \\ &= \begin{bmatrix} (-2)(4) + (1)(-3) & (-2)(-1) + (1)(-2) \\ (3)(4) + (4)(-3) & (3)(-1) + (4)(-2) \end{bmatrix} \\ &= \begin{bmatrix} -8 - 3 & 2 - 2 \\ 12 - 12 & -3 - 8 \end{bmatrix} \\ &= \begin{bmatrix} -11 & 0 \\ 0 & -11 \end{bmatrix} \end{aligned}$$

****Step 2: Compute $|A|I_2$ ****

$$\begin{aligned}|A|I_2 &= -11 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} -11 & 0 \\ 0 & -11 \end{bmatrix}\end{aligned}$$

****Step 3: Compute $(\text{adj } A)A$ ****

$$\begin{aligned}(\text{adj } A)A &= \begin{bmatrix} 4 & -1 \\ -3 & -2 \end{bmatrix} \begin{bmatrix} -2 & 1 \\ 3 & 4 \end{bmatrix} \\ &= \begin{bmatrix} (4)(-2) + (-1)(3) & (4)(1) + (-1)(4) \\ (-3)(-2) + (-2)(3) & (-3)(1) + (-2)(4) \end{bmatrix} \\ &= \begin{bmatrix} -8 - 3 & 4 - 4 \\ 6 - 6 & -3 - 8 \end{bmatrix} \\ &= \begin{bmatrix} -11 & 0 \\ 0 & -11 \end{bmatrix}\end{aligned}$$

Since all three expressions yield the same matrix:

$$A(\text{adj } A) = |A|I_2 = (\text{adj } A)A = \begin{bmatrix} -11 & 0 \\ 0 & -11 \end{bmatrix}$$

The relation is verified.

Final Answer

$$\text{adj } A = \begin{bmatrix} 4 & -1 \\ -3 & -2 \end{bmatrix}$$

Question 11 (ii)

Find the adjoint of the matrix:

$$A = \begin{bmatrix} 2 & 3 \\ -4 & -6 \end{bmatrix}$$

Also verify that $A(\text{adj } A) = |A|I_2 = (\text{adj } A)A$.

TYPE 19 Verification of Matrix Adjoint Properties

Given: The matrix $A = \begin{bmatrix} 2 & 3 \\ -4 & -6 \end{bmatrix}$.

Concept / Formula Used: $A(\text{adj } A) = |A|I_n = (\text{adj } A)A$

Solution:

First, we find the determinant of A :

$$\begin{aligned}|A| &= (2)(-6) - (3)(-4) \\ &= -12 + 12 \\ &= 0\end{aligned}$$

Next, we find the cofactors of A :

$$\begin{aligned}C_{11} &= (-1)^{1+1}(-6) = -6 \\ C_{12} &= (-1)^{1+2}(-4) = 4 \\ C_{21} &= (-1)^{2+1}(3) = -3 \\ C_{22} &= (-1)^{2+2}(2) = 2\end{aligned}$$

The cofactor matrix C is:

$$C = \begin{bmatrix} -6 & 4 \\ -3 & 2 \end{bmatrix}$$

The adjoint of A is the transpose of the cofactor matrix C :

$$\text{adj } A = C^T = \begin{bmatrix} -6 & -3 \\ 4 & 2 \end{bmatrix}$$

Now, let us verify the relation $A(\text{adj } A) = |A|I_2 = (\text{adj } A)A$:

****Step 1: Compute $A(\text{adj } A)$ ****

$$\begin{aligned}A(\text{adj } A) &= \begin{bmatrix} 2 & 3 \\ -4 & -6 \end{bmatrix} \begin{bmatrix} -6 & -3 \\ 4 & 2 \end{bmatrix} \\ &= \begin{bmatrix} (2)(-6) + (3)(4) & (2)(-3) + (3)(2) \\ (-4)(-6) + (-6)(4) & (-4)(-3) + (-6)(2) \end{bmatrix} \\ &= \begin{bmatrix} -12 + 12 & -6 + 6 \\ 24 - 24 & 12 - 12 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}\end{aligned}$$

****Step 2: Compute $|A|I_2$ ****

$$\begin{aligned}|A|I_2 &= 0 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}\end{aligned}$$

****Step 3: Compute $(\text{adj } A)A$ ****

$$\begin{aligned}(\text{adj } A)A &= \begin{bmatrix} -6 & -3 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ -4 & -6 \end{bmatrix} \\ &= \begin{bmatrix} (-6)(2) + (-3)(-4) & (-6)(3) + (-3)(-6) \\ (4)(2) + (2)(-4) & (4)(3) + (2)(-6) \end{bmatrix} \\ &= \begin{bmatrix} -12 + 12 & -18 + 18 \\ 8 - 8 & 12 - 12 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}\end{aligned}$$

Since all three expressions yield the same matrix:

$$A(\text{adj } A) = |A|I_2 = (\text{adj } A)A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

The relation is verified.

Final Answer

$$\text{adj } A = \begin{bmatrix} -6 & -3 \\ 4 & 2 \end{bmatrix}$$

Question 12

If $A = \begin{bmatrix} 3 & 1 & 2 \\ 2 & -3 & -1 \\ 1 & 2 & 1 \end{bmatrix}$, find $A(\text{adj } A)$ without calculating $\text{adj } A$.

TYPE 19 Verification of Matrix Adjoint Properties

Given: The matrix $A = \begin{bmatrix} 3 & 1 & 2 \\ 2 & -3 & -1 \\ 1 & 2 & 1 \end{bmatrix}$

To Find: $A(\text{adj } A)$ without calculating $\text{adj } A$

Concept / Formula Used: $A \quad n \quad A(\text{adj } A) = |A|I_n$

Solution:

First, we calculate the determinant of A , denoted by $|A|$:

$$|A| = \begin{vmatrix} 3 & 1 & 2 \\ 2 & -3 & -1 \\ 1 & 2 & 1 \end{vmatrix}$$

Expanding along the first row:

$$|A| = 3 \begin{vmatrix} -3 & -1 \\ 2 & 1 \end{vmatrix} - 1 \begin{vmatrix} 2 & -1 \\ 1 & 1 \end{vmatrix} + 2 \begin{vmatrix} 2 & -3 \\ 1 & 2 \end{vmatrix}$$

Evaluating the 2×2 determinants:

$$\begin{aligned} \begin{vmatrix} -3 & -1 \\ 2 & 1 \end{vmatrix} &= (-3)(1) - (-1)(2) = -3 + 2 = -1 \\ \begin{vmatrix} 2 & -1 \\ 1 & 1 \end{vmatrix} &= (2)(1) - (-1)(1) = 2 + 1 = 3 \\ \begin{vmatrix} 2 & -3 \\ 1 & 2 \end{vmatrix} &= (2)(2) - (-3)(1) = 4 + 3 = 7 \end{aligned}$$

Substituting these values back into the determinant expression:

$$\begin{aligned}|A| &= 3(-1) - 1(3) + 2(7) \\ &= -3 - 3 + 14 \\ &= 8\end{aligned}$$

Using the standard result $A(\text{adj } A) = |A|I_3$:

$$\begin{aligned}A(\text{adj } A) &= 8 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 8 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 8 \end{bmatrix}\end{aligned}$$

Final Answer

$$A(\text{adj } A) = \begin{bmatrix} 8 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 8 \end{bmatrix}$$

Question 13

Find the adjoint of the following matrices:

$$(i) \ A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 3 & 5 \\ -2 & 0 & 1 \end{bmatrix}$$

$$(ii) \ A = \begin{bmatrix} 1 & -1 & 2 \\ 3 & 0 & -2 \\ 1 & 0 & 3 \end{bmatrix}$$

Also verify that $A(\text{adj } A) = |A|I_3 = (\text{adj } A)A$.

TYPE 19 Verification of Matrix Adjoint Properties

Given: The matrices (i) $A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 3 & 5 \\ -2 & 0 & 1 \end{bmatrix}$ and (ii) $A = \begin{bmatrix} 1 & -1 & 2 \\ 3 & 0 & -2 \\ 1 & 0 & 3 \end{bmatrix}$

To Find: The adjoint of each matrix and verification of the identity $A(\text{adj } A) = |A|I_3 = (\text{adj } A)A$

Concept / Formula Used: $A \quad \text{adj } A = C' = [C_{ji}] \quad C_{ij} = (-1)^{i+j}M_{ij}$

Solution:

Part (i): Let $A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 3 & 5 \\ -2 & 0 & 1 \end{bmatrix}$. We first find the cofactors C_{ij} of each element a_{ij} :

$$C_{11} = (-1)^{1+1} \begin{vmatrix} 3 & 5 \\ 0 & 1 \end{vmatrix} = 1(3 - 0) = 3$$

$$C_{12} = (-1)^{1+2} \begin{vmatrix} 2 & 5 \\ -2 & 1 \end{vmatrix} = -1(2 - (-10)) = -12$$

$$C_{13} = (-1)^{1+3} \begin{vmatrix} 2 & 3 \\ -2 & 0 \end{vmatrix} = 1(0 - (-6)) = 6$$

$$C_{21} = (-1)^{2+1} \begin{vmatrix} -1 & 2 \\ 0 & 1 \end{vmatrix} = -1(-1 - 0) = 1$$

$$C_{22} = (-1)^{2+2} \begin{vmatrix} 1 & 2 \\ -2 & 1 \end{vmatrix} = 1(1 - (-4)) = 5$$

$$C_{23} = (-1)^{2+3} \begin{vmatrix} 1 & -1 \\ -2 & 0 \end{vmatrix} = -1(0 - 2) = 2$$

$$C_{31} = (-1)^{3+1} \begin{vmatrix} -1 & 2 \\ 3 & 5 \end{vmatrix} = 1(-5 - 6) = -11$$

$$C_{32} = (-1)^{3+2} \begin{vmatrix} 1 & 2 \\ 2 & 5 \end{vmatrix} = -1(5 - 4) = -1$$

$$C_{33} = (-1)^{3+3} \begin{vmatrix} 1 & -1 \\ 2 & 3 \end{vmatrix} = 1(3 - (-2)) = 5$$

The cofactor matrix C is:

$$C = \begin{bmatrix} 3 & -12 & 6 \\ 1 & 5 & 2 \\ -11 & -1 & 5 \end{bmatrix}$$

Taking the transpose of C to find $\text{adj } A$:

$$\text{adj } A = C' = \begin{bmatrix} 3 & 1 & -11 \\ -12 & 5 & -1 \\ 6 & 2 & 5 \end{bmatrix}$$

Now, we calculate the determinant $|A|$:

$$\begin{aligned} |A| &= a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} \\ &= 1(3) + (-1)(-12) + 2(6) \\ &= 3 + 12 + 12 \\ &= 27 \end{aligned}$$

Next, we verify $A(\text{adj } A)$:

$$\begin{aligned} A(\text{adj } A) &= \begin{bmatrix} 1 & -1 & 2 \\ 2 & 3 & 5 \\ -2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 & -11 \\ -12 & 5 & -1 \\ 6 & 2 & 5 \end{bmatrix} \\ &= \begin{bmatrix} 1(3) + (-1)(-12) + 2(6) & 1(1) + (-1)(5) + 2(2) & 1(-11) + (-1)(-1) + 2(5) \\ 2(3) + 3(-12) + 5(6) & 2(1) + 3(5) + 5(2) & 2(-11) + 3(-1) + 5(5) \\ -2(3) + 0(-12) + 1(6) & -2(1) + 0(5) + 1(2) & -2(-11) + 0(-1) + 1(5) \end{bmatrix} \\ &= \begin{bmatrix} 3 + 12 + 12 & 1 - 5 + 4 & -11 + 1 + 10 \\ 6 - 36 + 30 & 2 + 15 + 10 & -22 - 3 + 25 \\ -6 + 0 + 6 & -2 + 0 + 2 & 22 + 0 + 5 \end{bmatrix} \\ &= \begin{bmatrix} 27 & 0 & 0 \\ 0 & 27 & 0 \\ 0 & 0 & 27 \end{bmatrix} = 27I_3 \end{aligned}$$

Now, we verify $(\text{adj } A)A$:

$$\begin{aligned} (\text{adj } A)A &= \begin{bmatrix} 3 & 1 & -11 \\ -12 & 5 & -1 \\ 6 & 2 & 5 \end{bmatrix} \begin{bmatrix} 1 & -1 & 2 \\ 2 & 3 & 5 \\ -2 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 3(1) + 1(2) + (-11)(-2) & 3(-1) + 1(3) + (-11)(0) & 3(2) + 1(5) + (-11)(1) \\ -12(1) + 5(2) + (-1)(-2) & -12(-1) + 5(3) + (-1)(0) & -12(2) + 5(5) + (-1)(1) \\ 6(1) + 2(2) + 5(-2) & 6(-1) + 2(3) + 5(0) & 6(2) + 2(5) + 5(1) \end{bmatrix} \\ &= \begin{bmatrix} 3 + 2 + 22 & -3 + 3 + 0 & 6 + 5 - 11 \\ -12 + 10 + 2 & 12 + 15 + 0 & -24 + 25 - 1 \\ 6 + 4 - 10 & -6 + 6 + 0 & 12 + 10 + 5 \end{bmatrix} \\ &= \begin{bmatrix} 27 & 0 & 0 \\ 0 & 27 & 0 \\ 0 & 0 & 27 \end{bmatrix} = 27I_3 \end{aligned}$$

Thus, $A(\text{adj } A) = |A|I_3 = (\text{adj } A)A$ is verified for part (i).

Part (ii): Let $A = \begin{bmatrix} 1 & -1 & 2 \\ 3 & 0 & -2 \\ 1 & 0 & 3 \end{bmatrix}$. We find the cofactors C_{ij} of each element a_{ij} :

$$C_{11} = (-1)^{1+1} \begin{vmatrix} 0 & -2 \\ 0 & 3 \end{vmatrix} = 1(0 - 0) = 0$$

$$C_{12} = (-1)^{1+2} \begin{vmatrix} 3 & -2 \\ 1 & 3 \end{vmatrix} = -1(9 - (-2)) = -11$$

$$C_{13} = (-1)^{1+3} \begin{vmatrix} 3 & 0 \\ 1 & 0 \end{vmatrix} = 1(0 - 0) = 0$$

$$C_{21} = (-1)^{2+1} \begin{vmatrix} -1 & 2 \\ 0 & 3 \end{vmatrix} = -1(-3 - 0) = 3$$

$$C_{22} = (-1)^{2+2} \begin{vmatrix} 1 & 2 \\ 1 & 3 \end{vmatrix} = 1(3 - 2) = 1$$

$$C_{23} = (-1)^{2+3} \begin{vmatrix} 1 & -1 \\ 1 & 0 \end{vmatrix} = -1(0 - (-1)) = -1$$

$$C_{31} = (-1)^{3+1} \begin{vmatrix} -1 & 2 \\ 0 & -2 \end{vmatrix} = 1(2 - 0) = 2$$

$$C_{32} = (-1)^{3+2} \begin{vmatrix} 1 & 2 \\ 3 & -2 \end{vmatrix} = -1(-2 - 6) = 8$$

$$C_{33} = (-1)^{3+3} \begin{vmatrix} 1 & -1 \\ 3 & 0 \end{vmatrix} = 1(0 - (-3)) = 3$$

The cofactor matrix C is:

$$C = \begin{bmatrix} 0 & -11 & 0 \\ 3 & 1 & -1 \\ 2 & 8 & 3 \end{bmatrix}$$

Taking the transpose of C to find $\text{adj } A$:

$$\text{adj } A = C' = \begin{bmatrix} 0 & 3 & 2 \\ -11 & 1 & 8 \\ 0 & -1 & 3 \end{bmatrix}$$

Now, we calculate the determinant $|A|$:

$$\begin{aligned} |A| &= a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} \\ &= 1(0) + (-1)(-11) + 2(0) \\ &= 11 \end{aligned}$$

Next, we verify $A(\text{adj } A)$:

$$\begin{aligned}
 A(\text{adj } A) &= \begin{bmatrix} 1 & -1 & 2 \\ 3 & 0 & -2 \\ 1 & 0 & 3 \end{bmatrix} \begin{bmatrix} 0 & 3 & 2 \\ -11 & 1 & 8 \\ 0 & -1 & 3 \end{bmatrix} \\
 &= \begin{bmatrix} 1(0) + (-1)(-11) + 2(0) & 1(3) + (-1)(1) + 2(-1) & 1(2) + (-1)(8) + 2(3) \\ 3(0) + 0(-11) + (-2)(0) & 3(3) + 0(1) + (-2)(-1) & 3(2) + 0(8) + (-2)(3) \\ 1(0) + 0(-11) + 3(0) & 1(3) + 0(1) + 3(-1) & 1(2) + 0(8) + 3(3) \end{bmatrix} \\
 &= \begin{bmatrix} 11 & 3 - 1 - 2 & 2 - 8 + 6 \\ 0 & 9 + 2 & 6 - 6 \\ 0 & 3 - 3 & 2 + 9 \end{bmatrix} \\
 &= \begin{bmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{bmatrix} = 11I_3
 \end{aligned}$$

Now, we verify $(\text{adj } A)A$:

$$\begin{aligned}
 (\text{adj } A)A &= \begin{bmatrix} 0 & 3 & 2 \\ -11 & 1 & 8 \\ 0 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 & -1 & 2 \\ 3 & 0 & -2 \\ 1 & 0 & 3 \end{bmatrix} \\
 &= \begin{bmatrix} 0(1) + 3(3) + 2(1) & 0(-1) + 3(0) + 2(0) & 0(2) + 3(-2) + 2(3) \\ -11(1) + 1(3) + 8(1) & -11(-1) + 1(0) + 8(0) & -11(2) + 1(-2) + 8(3) \\ 0(1) + (-1)(3) + 3(1) & 0(-1) + (-1)(0) + 3(0) & 0(2) + (-1)(-2) + 3(3) \end{bmatrix} \\
 &= \begin{bmatrix} 9 + 2 & 0 & -6 + 6 \\ -11 + 3 + 8 & 11 & -22 - 2 + 24 \\ -3 + 3 & 0 & 2 + 9 \end{bmatrix} \\
 &= \begin{bmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{bmatrix} = 11I_3
 \end{aligned}$$

Thus, $A(\text{adj } A) = |A|I_3 = (\text{adj } A)A$ is verified for part (ii).

Final Answer

$$(i) \text{ adj } A = \begin{bmatrix} 3 & 1 & -11 \\ -12 & 5 & -1 \\ 6 & 2 & 5 \end{bmatrix}$$

$$(ii) \text{ adj } A = \begin{bmatrix} 0 & 3 & 2 \\ -11 & 1 & 8 \\ 0 & -1 & 3 \end{bmatrix}$$

Question 14

If $A = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$, find $\text{adj } A$ and verify that $A(\text{adj } A) = (\text{adj } A)A = |A|I_3$.

TYPE 19 Verification of Matrix Adjoint Properties

Given: The matrix $A = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$

To Find: $\text{adj } A$ and verification of the identity $A(\text{adj } A) = (\text{adj } A)A = |A|I_3$

Concept / Formula Used: $A \quad \text{adj } A = C'$

Solution:

We first find the cofactors C_{ij} of each element a_{ij} :

$$C_{11} = (-1)^{1+1} \begin{vmatrix} \cos \alpha & 0 \\ 0 & 1 \end{vmatrix} = 1(\cos \alpha - 0) = \cos \alpha$$

$$C_{12} = (-1)^{1+2} \begin{vmatrix} \sin \alpha & 0 \\ 0 & 1 \end{vmatrix} = -1(\sin \alpha - 0) = -\sin \alpha$$

$$C_{13} = (-1)^{1+3} \begin{vmatrix} \sin \alpha & \cos \alpha \\ 0 & 0 \end{vmatrix} = 1(0 - 0) = 0$$

$$C_{21} = (-1)^{2+1} \begin{vmatrix} -\sin \alpha & 0 \\ 0 & 1 \end{vmatrix} = -1(-\sin \alpha - 0) = \sin \alpha$$

$$C_{22} = (-1)^{2+2} \begin{vmatrix} \cos \alpha & 0 \\ 0 & 1 \end{vmatrix} = 1(\cos \alpha - 0) = \cos \alpha$$

$$C_{23} = (-1)^{2+3} \begin{vmatrix} \cos \alpha & -\sin \alpha \\ 0 & 0 \end{vmatrix} = -1(0 - 0) = 0$$

$$C_{31} = (-1)^{3+1} \begin{vmatrix} -\sin \alpha & 0 \\ \cos \alpha & 0 \end{vmatrix} = 1(0 - 0) = 0$$

$$C_{32} = (-1)^{3+2} \begin{vmatrix} \cos \alpha & 0 \\ \sin \alpha & 0 \end{vmatrix} = -1(0 - 0) = 0$$

$$C_{33} = (-1)^{3+3} \begin{vmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{vmatrix} = 1(\cos^2 \alpha - (-\sin^2 \alpha)) = \cos^2 \alpha + \sin^2 \alpha = 1$$

The cofactor matrix C is:

$$C = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Taking the transpose of C to find $\text{adj } A$:

$$\text{adj } A = C' = \begin{bmatrix} \cos \alpha & \sin \alpha & 0 \\ -\sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Now, we calculate the determinant $|A|$:

$$\begin{aligned}
 |A| &= a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} \\
 &= \cos \alpha (\cos \alpha) + (-\sin \alpha)(-\sin \alpha) + 0 \\
 &= \cos^2 \alpha + \sin^2 \alpha \\
 &= 1
 \end{aligned}$$

Next, we verify $A(\text{adj } A)$:

$$\begin{aligned}
 A(\text{adj } A) &= \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \alpha & \sin \alpha & 0 \\ -\sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} \cos^2 \alpha + \sin^2 \alpha & \cos \alpha \sin \alpha - \sin \alpha \cos \alpha & 0 \\ \sin \alpha \cos \alpha - \cos \alpha \sin \alpha & \sin^2 \alpha + \cos^2 \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3
 \end{aligned}$$

Now, we verify $(\text{adj } A)A$:

$$\begin{aligned}
 (\text{adj } A)A &= \begin{bmatrix} \cos \alpha & \sin \alpha & 0 \\ -\sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} \cos^2 \alpha + \sin^2 \alpha & -\cos \alpha \sin \alpha + \sin \alpha \cos \alpha & 0 \\ -\sin \alpha \cos \alpha + \cos \alpha \sin \alpha & \sin^2 \alpha + \cos^2 \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3
 \end{aligned}$$

Since $|A| = 1$, we have $|A|I_3 = I_3$. Thus, $A(\text{adj } A) = (\text{adj } A)A = |A|I_3$ is verified.

Final Answer

$$\text{adj } A = \begin{bmatrix} \cos \alpha & \sin \alpha & 0 \\ -\sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Question 15

For the matrix $A = \begin{bmatrix} 2 & -1 & 3 \\ 4 & 2 & 5 \\ 0 & 4 & -1 \end{bmatrix}$, verify that $A(\text{adj } A) = O$.

TYPE 19 Verification of Matrix Adjoint Properties

Given: The matrix $A = \begin{bmatrix} 2 & -1 & 3 \\ 4 & 2 & 5 \\ 0 & 4 & -1 \end{bmatrix}$

To Prove: $A(\text{adj } A) = O$

Concept / Formula Used: $A \quad A(\text{adj } A) = |A|I_n \quad |A| = 0 \quad A(\text{adj } A) = O$

Solution:

First, we calculate the determinant $|A|$:

$$\begin{aligned} |A| &= 2 \begin{vmatrix} 2 & 5 \\ 4 & -1 \end{vmatrix} - (-1) \begin{vmatrix} 4 & 5 \\ 0 & -1 \end{vmatrix} + 3 \begin{vmatrix} 4 & 2 \\ 0 & 4 \end{vmatrix} \\ &= 2(2(-1) - 5(4)) + 1(4(-1) - 5(0)) + 3(4(4) - 2(0)) \\ &= 2(-2 - 20) + 1(-4 - 0) + 3(16 - 0) \\ &= 2(-22) - 4 + 48 \\ &= -44 - 4 + 48 \\ &= 0 \end{aligned}$$

Since $|A| = 0$, the matrix is singular. By the standard result $A(\text{adj } A) = |A|I_3$, we must have $A(\text{adj } A) = O$.

Let us verify this explicitly by calculating $\text{adj } A$: Cofactors of A :

$$\begin{aligned}
 C_{11} &= (-1)^{1+1} \begin{vmatrix} 2 & 5 \\ 4 & -1 \end{vmatrix} = 1(-2 - 20) = -22 \\
 C_{12} &= (-1)^{1+2} \begin{vmatrix} 4 & 5 \\ 0 & -1 \end{vmatrix} = -1(-4 - 0) = 4 \\
 C_{13} &= (-1)^{1+3} \begin{vmatrix} 4 & 2 \\ 0 & 4 \end{vmatrix} = 1(16 - 0) = 16 \\
 C_{21} &= (-1)^{2+1} \begin{vmatrix} -1 & 3 \\ 4 & -1 \end{vmatrix} = -1(1 - 12) = 11 \\
 C_{22} &= (-1)^{2+2} \begin{vmatrix} 2 & 3 \\ 0 & -1 \end{vmatrix} = 1(-2 - 0) = -2 \\
 C_{23} &= (-1)^{2+3} \begin{vmatrix} 2 & -1 \\ 0 & 4 \end{vmatrix} = -1(8 - 0) = -8 \\
 C_{31} &= (-1)^{3+1} \begin{vmatrix} -1 & 3 \\ 2 & 5 \end{vmatrix} = 1(-5 - 6) = -11 \\
 C_{32} &= (-1)^{3+2} \begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix} = -1(10 - 12) = 2 \\
 C_{33} &= (-1)^{3+3} \begin{vmatrix} 2 & -1 \\ 4 & 2 \end{vmatrix} = 1(4 - (-4)) = 8
 \end{aligned}$$

The cofactor matrix C is:

$$C = \begin{bmatrix} -22 & 4 & 16 \\ 11 & -2 & -8 \\ -11 & 2 & 8 \end{bmatrix}$$

Taking the transpose of C to find $\text{adj } A$:

$$\text{adj } A = C' = \begin{bmatrix} -22 & 11 & -11 \\ 4 & -2 & 2 \\ 16 & -8 & 8 \end{bmatrix}$$

Now, we compute $A(\text{adj } A)$:

$$\begin{aligned}
 A(\text{adj } A) &= \begin{bmatrix} 2 & -1 & 3 \\ 4 & 2 & 5 \\ 0 & 4 & -1 \end{bmatrix} \begin{bmatrix} -22 & 11 & -11 \\ 4 & -2 & 2 \\ 16 & -8 & 8 \end{bmatrix} \\
 &= \begin{bmatrix} 2(-22) + (-1)(4) + 3(16) & 2(11) + (-1)(-2) + 3(-8) & 2(-11) + (-1)(2) + 3(8) \\ 4(-22) + 2(4) + 5(16) & 4(11) + 2(-2) + 5(-8) & 4(-11) + 2(2) + 5(8) \\ 0(-22) + 4(4) + (-1)(16) & 0(11) + 4(-2) + (-1)(-8) & 0(-11) + 4(2) + (-1)(8) \end{bmatrix} \\
 &= \begin{bmatrix} -44 - 4 + 48 & 22 + 2 - 24 & -22 - 2 + 24 \\ -88 + 8 + 80 & 44 - 4 - 40 & -44 + 4 + 40 \\ 0 + 16 - 16 & 0 - 8 + 8 & 0 + 8 - 8 \end{bmatrix} \\
 &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O
 \end{aligned}$$

LHS = RHS, hence proved.

Hence Proved*LHS = RHS, hence proved.*

Question 16

For the matrix $A = \begin{bmatrix} 5 & 2 \\ -3 & -1 \end{bmatrix}$, verify that $\text{adj } A' = (\text{adj } A)'$.

TYPE 19 Verification of Matrix Adjoint Properties

Given: The matrix $A = \begin{bmatrix} 5 & 2 \\ -3 & -1 \end{bmatrix}$

To Prove: $\text{adj } A' = (\text{adj } A)'$

Concept / Formula Used: 2×2 $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $\text{adj } M = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Solution:

First, we find the transpose of A , denoted by A' :

$$A' = \begin{bmatrix} 5 & -3 \\ 2 & -1 \end{bmatrix}$$

Using the standard formula for the adjoint of a 2×2 matrix:

$$\text{adj } A' = \begin{bmatrix} -1 & -(-3) \\ -2 & 5 \end{bmatrix} = \begin{bmatrix} -1 & 3 \\ -2 & 5 \end{bmatrix}$$

Now, we find the adjoint of A :

$$\text{adj } A = \begin{bmatrix} -1 & -2 \\ -(-3) & 5 \end{bmatrix} = \begin{bmatrix} -1 & -2 \\ 3 & 5 \end{bmatrix}$$

Taking the transpose of $\text{adj } A$:

$$(\text{adj } A)' = \begin{bmatrix} -1 & 3 \\ -2 & 5 \end{bmatrix}$$

Comparing the two results:

$$\text{adj } A' = (\text{adj } A)' = \begin{bmatrix} -1 & 3 \\ -2 & 5 \end{bmatrix}$$

LHS = RHS, hence proved.

Hence Proved

LHS = RHS, hence proved.

Question 17

- (i) Given $A = \begin{bmatrix} 2 & -3 \\ -4 & 7 \end{bmatrix}$, compute A^{-1} and show that $2A^{-1} = 9I - A$.
- (ii) If $A = \begin{bmatrix} 4 & 5 \\ 2 & 1 \end{bmatrix}$, show that $A - 3I = 2(I + 3A^{-1})$.

TYPE 18 Matrix Inverse Properties and Proofs

Given: The matrices (i) $A = \begin{bmatrix} 2 & -3 \\ -4 & 7 \end{bmatrix}$ and (ii) $A = \begin{bmatrix} 4 & 5 \\ 2 & 1 \end{bmatrix}$

To Prove: (i) $2A^{-1} = 9I - A$
(ii) $A - 3I = 2(I + 3A^{-1})$

Concept / Formula Used: 2×2 $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Solution:

Part (i): Let $A = \begin{bmatrix} 2 & -3 \\ -4 & 7 \end{bmatrix}$. First, we calculate the determinant $|A|$:

$$|A| = (2)(7) - (-3)(-4) = 14 - 12 = 2$$

Since $|A| \neq 0$, A^{-1} exists. The adjoint of A is:

$$\text{adj } A = \begin{bmatrix} 7 & 3 \\ 4 & 2 \end{bmatrix}$$

Thus, the inverse A^{-1} is:

$$A^{-1} = \frac{1}{|A|} \text{adj } A = \frac{1}{2} \begin{bmatrix} 7 & 3 \\ 4 & 2 \end{bmatrix}$$

Now, we compute the LHS: $2A^{-1}$:

$$2A^{-1} = 2 \left(\frac{1}{2} \begin{bmatrix} 7 & 3 \\ 4 & 2 \end{bmatrix} \right) = \begin{bmatrix} 7 & 3 \\ 4 & 2 \end{bmatrix}$$

Next, we compute the RHS: $9I - A$:

$$\begin{aligned} 9I - A &= 9 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 2 & -3 \\ -4 & 7 \end{bmatrix} \\ &= \begin{bmatrix} 9 & 0 \\ 0 & 9 \end{bmatrix} - \begin{bmatrix} 2 & -3 \\ -4 & 7 \end{bmatrix} \\ &= \begin{bmatrix} 9-2 & 0-(-3) \\ 0-(-4) & 9-7 \end{bmatrix} \\ &= \begin{bmatrix} 7 & 3 \\ 4 & 2 \end{bmatrix} \end{aligned}$$

Comparing LHS and RHS:

$$2A^{-1} = 9I - A = \begin{bmatrix} 7 & 3 \\ 4 & 2 \end{bmatrix}$$

LHS = RHS, hence proved.

Part (ii): Let $A = \begin{bmatrix} 4 & 5 \\ 2 & 1 \end{bmatrix}$. First, we calculate the determinant $|A|$:

$$|A| = (4)(1) - (5)(2) = 4 - 10 = -6$$

The adjoint of A is:

$$\text{adj } A = \begin{bmatrix} 1 & -5 \\ -2 & 4 \end{bmatrix}$$

Thus, the inverse A^{-1} is:

$$A^{-1} = -\frac{1}{6} \begin{bmatrix} 1 & -5 \\ -2 & 4 \end{bmatrix}$$

Now, we compute the RHS: $2(I + 3A^{-1})$:

$$\begin{aligned} 3A^{-1} &= 3 \left(-\frac{1}{6} \begin{bmatrix} 1 & -5 \\ -2 & 4 \end{bmatrix} \right) = -\frac{1}{2} \begin{bmatrix} 1 & -5 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} -1/2 & 5/2 \\ 1 & -2 \end{bmatrix} \\ I + 3A^{-1} &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} -1/2 & 5/2 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} 1/2 & 5/2 \\ 1 & -1 \end{bmatrix} \\ 2(I + 3A^{-1}) &= 2 \begin{bmatrix} 1/2 & 5/2 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 5 \\ 2 & -2 \end{bmatrix} \end{aligned}$$

Next, we compute the LHS: $A - 3I$:

$$\begin{aligned} A - 3I &= \begin{bmatrix} 4 & 5 \\ 2 & 1 \end{bmatrix} - 3 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 4-3 & 5-0 \\ 2-0 & 1-3 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 5 \\ 2 & -2 \end{bmatrix} \end{aligned}$$

Comparing LHS and RHS:

$$A - 3I = 2(I + 3A^{-1}) = \begin{bmatrix} 1 & 5 \\ 2 & -2 \end{bmatrix}$$

LHS = RHS, hence proved.

Hence Proved

Both parts (i) and (ii) are verified and proved.

Question 18

Find the inverse of the matrix $A = \begin{bmatrix} a & b \\ c & \frac{1+bc}{a} \end{bmatrix}$ and show that $aA^{-1} = (a^2 + bc + 1)I - aA$.

TYPE 18 Matrix Inverse Properties and Proofs

Given: The matrix $A = \begin{bmatrix} a & b \\ c & \frac{1+bc}{a} \end{bmatrix}$

To Prove: $aA^{-1} = (a^2 + bc + 1)I - aA$

Concept / Formula Used: 2×2 $A = \begin{bmatrix} w & x \\ y & z \end{bmatrix}$ $A^{-1} = \frac{1}{wz - xy} \begin{bmatrix} z & -x \\ -y & w \end{bmatrix}$

Solution:

First, we calculate the determinant $|A|$:

$$\begin{aligned} |A| &= a \left(\frac{1+bc}{a} \right) - b(c) \\ &= 1 + bc - bc \\ &= 1 \end{aligned}$$

Since $|A| = 1$, the inverse A^{-1} is equal to the adjoint of A :

$$A^{-1} = \begin{bmatrix} \frac{1+bc}{a} & -b \\ -c & a \end{bmatrix}$$

Now, we compute the LHS: aA^{-1} :

$$aA^{-1} = a \begin{bmatrix} \frac{1+bc}{a} & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} 1+bc & -ab \\ -ac & a^2 \end{bmatrix}$$

Next, we compute the RHS: $(a^2 + bc + 1)I - aA$:

$$\begin{aligned} (a^2 + bc + 1)I &= \begin{bmatrix} a^2 + bc + 1 & 0 \\ 0 & a^2 + bc + 1 \end{bmatrix} \\ aA &= a \begin{bmatrix} a & b \\ c & \frac{1+bc}{a} \end{bmatrix} = \begin{bmatrix} a^2 & ab \\ ac & 1+bc \end{bmatrix} \\ (a^2 + bc + 1)I - aA &= \begin{bmatrix} (a^2 + bc + 1) - a^2 & 0 - ab \\ 0 - ac & (a^2 + bc + 1) - (1 + bc) \end{bmatrix} \\ &= \begin{bmatrix} 1 + bc & -ab \\ -ac & a^2 \end{bmatrix} \end{aligned}$$

Comparing LHS and RHS:

$$aA^{-1} = (a^2 + bc + 1)I - aA = \begin{bmatrix} 1 + bc & -ab \\ -ac & a^2 \end{bmatrix}$$

LHS = RHS, hence proved.

Hence Proved

LHS = RHS, hence proved.

Question 19

If $A = \begin{bmatrix} 0 & 1 & 3 \\ 1 & 2 & x \\ 2 & 3 & 1 \end{bmatrix}$ and $A^{-1} = \frac{1}{2} \begin{bmatrix} 1 & -8 & 5 \\ -1 & 6 & -3 \\ 1 & 2y & 1 \end{bmatrix}$, find the value(s) of x and y .

TYPE 18 Matrix Inverse Properties and Proofs

Given: The matrix $A = \begin{bmatrix} 0 & 1 & 3 \\ 1 & 2 & x \\ 2 & 3 & 1 \end{bmatrix}$ and its inverse $A^{-1} = \frac{1}{2} \begin{bmatrix} 1 & -8 & 5 \\ -1 & 6 & -3 \\ 1 & 2y & 1 \end{bmatrix}$

To Find: The values of x and y

Concept / Formula Used: $AA^{-1} = I_3$

Solution:

Using the definition $AA^{-1} = I_3$:

$$\begin{bmatrix} 0 & 1 & 3 \\ 1 & 2 & x \\ 2 & 3 & 1 \end{bmatrix} \left(\frac{1}{2} \begin{bmatrix} 1 & -8 & 5 \\ -1 & 6 & -3 \\ 1 & 2y & 1 \end{bmatrix} \right) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Multiplying both sides by 2:

$$\begin{bmatrix} 0 & 1 & 3 \\ 1 & 2 & x \\ 2 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & -8 & 5 \\ -1 & 6 & -3 \\ 1 & 2y & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Let us compute the product of the two matrices on the LHS and equate it to the RHS

matrix: Let $P = \begin{bmatrix} 0 & 1 & 3 \\ 1 & 2 & x \\ 2 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & -8 & 5 \\ -1 & 6 & -3 \\ 1 & 2y & 1 \end{bmatrix}$.

Equating the element at position (1, 2):

$$\begin{aligned} P_{12} &= 0(-8) + 1(6) + 3(2y) = 0 \\ 6 + 6y &= 0 \\ 6y &= -6 \\ y &= -1 \end{aligned}$$

Equating the element at position (2, 1):

$$\begin{aligned} P_{21} &= 1(1) + 2(-1) + x(1) = 0 \\ 1 - 2 + x &= 0 \\ x - 1 &= 0 \\ x &= 1 \end{aligned}$$

Let us verify these values using other elements of the product matrix P :

$$\begin{aligned}P_{22} &= 1(-8) + 2(6) + x(2y) \\&= -8 + 12 + 1(2(-1)) \\&= 4 - 2 = 2 \quad (\text{matches the RHS element at } (2, 2))\end{aligned}$$

$$\begin{aligned}P_{23} &= 1(5) + 2(-3) + x(1) \\&= 5 - 6 + 1(1) \\&= 0 \quad (\text{matches the RHS element at } (2, 3))\end{aligned}$$

$$\begin{aligned}P_{32} &= 2(-8) + 3(6) + 1(2y) \\&= -16 + 18 + 2(-1) \\&= 2 - 2 = 0 \quad (\text{matches the RHS element at } (3, 2))\end{aligned}$$

All elements match perfectly.

Final Answer

$$x = 1, \quad y = -1$$

Question 20

If $A = \begin{bmatrix} 2 & -3 \\ 5 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 3 \\ 4 & 0 \end{bmatrix}$, verify that $\text{adj}(AB) = (\text{adj } B)(\text{adj } A)$.

TYPE 21 Reversal Law for Matrix Inverse and Adjoint

Given: The matrices $A = \begin{bmatrix} 2 & -3 \\ 5 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 3 \\ 4 & 0 \end{bmatrix}$

To Prove: $\text{adj}(AB) = (\text{adj } B)(\text{adj } A)$

Concept / Formula Used: $A \quad B \quad \text{adj}(AB) = (\text{adj } B)(\text{adj } A)$

Solution:

First, we compute the product AB :

$$\begin{aligned} AB &= \begin{bmatrix} 2 & -3 \\ 5 & 1 \end{bmatrix} \begin{bmatrix} -1 & 3 \\ 4 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 2(-1) + (-3)(4) & 2(3) + (-3)(0) \\ 5(-1) + 1(4) & 5(3) + 1(0) \end{bmatrix} \\ &= \begin{bmatrix} -2 - 12 & 6 + 0 \\ -5 + 4 & 15 + 0 \end{bmatrix} \\ &= \begin{bmatrix} -14 & 6 \\ -1 & 15 \end{bmatrix} \end{aligned}$$

Now, we find $\text{adj}(AB)$ using the 2×2 adjoint formula:

$$\text{adj}(AB) = \begin{bmatrix} 15 & -6 \\ -(-1) & -14 \end{bmatrix} = \begin{bmatrix} 15 & -6 \\ 1 & -14 \end{bmatrix}$$

Next, we find $\text{adj } A$ and $\text{adj } B$:

$$\text{adj } A = \begin{bmatrix} 1 & -(-3) \\ -5 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ -5 & 2 \end{bmatrix}$$

$$\text{adj } B = \begin{bmatrix} 0 & -3 \\ -4 & -1 \end{bmatrix}$$

Now, we compute the product $(\text{adj } B)(\text{adj } A)$:

$$\begin{aligned} (\text{adj } B)(\text{adj } A) &= \begin{bmatrix} 0 & -3 \\ -4 & -1 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ -5 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 0(1) + (-3)(-5) & 0(3) + (-3)(2) \\ -4(1) + (-1)(-5) & -4(3) + (-1)(2) \end{bmatrix} \\ &= \begin{bmatrix} 0 + 15 & 0 - 6 \\ -4 + 5 & -12 - 2 \end{bmatrix} \\ &= \begin{bmatrix} 15 & -6 \\ 1 & -14 \end{bmatrix} \end{aligned}$$

Comparing the two results:

$$\operatorname{adj}(AB) = (\operatorname{adj} B)(\operatorname{adj} A) = \begin{bmatrix} 15 & -6 \\ 1 & -14 \end{bmatrix}$$

LHS = RHS, hence proved.

Hence Proved

LHS = RHS, hence proved.

Question 21

Find the inverse of each of the following matrices:

$$(i) \ A = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 3 & 0 \\ 5 & 2 & -1 \end{bmatrix}$$

$$(ii) \ A = \begin{bmatrix} -1 & 1 & 2 \\ 3 & -1 & 1 \\ -1 & 3 & 4 \end{bmatrix}$$

Also verify that $A^{-1}A = I_3$.

TYPE 16 Inverse of a Matrix Using Adjoint

Given: The matrices (i) $A = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 3 & 0 \\ 5 & 2 & -1 \end{bmatrix}$ and (ii) $A = \begin{bmatrix} -1 & 1 & 2 \\ 3 & -1 & 1 \\ -1 & 3 & 4 \end{bmatrix}$

To Find: The inverse of each matrix and verification of $A^{-1}A = I_3$

Concept / Formula Used: $A^{-1} = \frac{1}{|A|} \text{adj } A$

Solution:

Part (i): Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 3 & 0 \\ 5 & 2 & -1 \end{bmatrix}$. Since A is a lower triangular matrix, its determinant is the product of its diagonal elements:

$$|A| = 1 \times 3 \times (-1) = -3$$

Cofactors C_{ij} of A :

$$\begin{aligned} C_{11} &= (-1)^{1+1} \begin{vmatrix} 3 & 0 \\ 2 & -1 \end{vmatrix} = 1(-3 - 0) = -3 \\ C_{12} &= (-1)^{1+2} \begin{vmatrix} 3 & 0 \\ 5 & -1 \end{vmatrix} = -1(-3 - 0) = 3 \\ C_{13} &= (-1)^{1+3} \begin{vmatrix} 3 & 3 \\ 5 & 2 \end{vmatrix} = 1(6 - 15) = -9 \\ C_{21} &= (-1)^{2+1} \begin{vmatrix} 0 & 0 \\ 2 & -1 \end{vmatrix} = -1(0 - 0) = 0 \\ C_{22} &= (-1)^{2+2} \begin{vmatrix} 1 & 0 \\ 5 & -1 \end{vmatrix} = 1(-1 - 0) = -1 \\ C_{23} &= (-1)^{2+3} \begin{vmatrix} 1 & 0 \\ 5 & 2 \end{vmatrix} = -1(2 - 0) = -2 \\ C_{31} &= (-1)^{3+1} \begin{vmatrix} 0 & 0 \\ 3 & 0 \end{vmatrix} = 1(0 - 0) = 0 \\ C_{32} &= (-1)^{3+2} \begin{vmatrix} 1 & 0 \\ 3 & 0 \end{vmatrix} = -1(0 - 0) = 0 \\ C_{33} &= (-1)^{3+3} \begin{vmatrix} 1 & 0 \\ 3 & 3 \end{vmatrix} = 1(3 - 0) = 3 \end{aligned}$$

The cofactor matrix C is:

$$C = \begin{bmatrix} -3 & 3 & -9 \\ 0 & -1 & -2 \\ 0 & 0 & 3 \end{bmatrix}$$

Taking the transpose of C to find $\text{adj } A$:

$$\text{adj } A = C' = \begin{bmatrix} -3 & 0 & 0 \\ 3 & -1 & 0 \\ -9 & -2 & 3 \end{bmatrix}$$

Thus, the inverse A^{-1} is:

$$A^{-1} = \frac{1}{|A|} \text{adj } A = -\frac{1}{3} \begin{bmatrix} -3 & 0 & 0 \\ 3 & -1 & 0 \\ -9 & -2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1/3 & 0 \\ 3 & 2/3 & -1 \end{bmatrix}$$

Verification of $A^{-1}A = I_3$:

$$\begin{aligned} A^{-1}A &= \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1/3 & 0 \\ 3 & 2/3 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 3 & 3 & 0 \\ 5 & 2 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 1(1) + 0 + 0 & 1(0) + 0 + 0 & 1(0) + 0 + 0 \\ -1(1) + (1/3)(3) + 0 & -1(0) + (1/3)(3) + 0 & -1(0) + 0 + 0 \\ 3(1) + (2/3)(3) - 1(5) & 3(0) + (2/3)(2) - 1(2) & 3(0) + 0 - 1(-1) \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ -1 + 1 & 1 & 0 \\ 3 + 2 - 5 & 4/3 - 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3 \end{aligned}$$

Hence verified.

Part (ii): Let $A = \begin{bmatrix} -1 & 1 & 2 \\ 3 & -1 & 1 \\ -1 & 3 & 4 \end{bmatrix}$. First, we calculate the determinant $|A|$:

$$\begin{aligned} |A| &= -1 \begin{vmatrix} -1 & 1 \\ 3 & 4 \end{vmatrix} - 1 \begin{vmatrix} 3 & 1 \\ -1 & 4 \end{vmatrix} + 2 \begin{vmatrix} 3 & -1 \\ -1 & 3 \end{vmatrix} \\ &= -1(-4 - 3) - 1(12 - (-1)) + 2(9 - 1) \\ &= -1(-7) - 1(13) + 2(8) \\ &= 7 - 13 + 16 \\ &= 10 \end{aligned}$$

Cofactors C_{ij} of A :

$$\begin{aligned} C_{11} &= (-1)^{1+1} \begin{vmatrix} -1 & 1 \\ 3 & 4 \end{vmatrix} = 1(-4 - 3) = -7 \\ C_{12} &= (-1)^{1+2} \begin{vmatrix} 3 & 1 \\ -1 & 4 \end{vmatrix} = -1(12 - (-1)) = -13 \\ C_{13} &= (-1)^{1+3} \begin{vmatrix} 3 & -1 \\ -1 & 3 \end{vmatrix} = 1(9 - 1) = 8 \\ C_{21} &= (-1)^{2+1} \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = -1(4 - 6) = 2 \\ C_{22} &= (-1)^{2+2} \begin{vmatrix} -1 & 2 \\ -1 & 4 \end{vmatrix} = 1(-4 - (-2)) = -2 \\ C_{23} &= (-1)^{2+3} \begin{vmatrix} -1 & 1 \\ -1 & 3 \end{vmatrix} = -1(-3 - (-1)) = 2 \\ C_{31} &= (-1)^{3+1} \begin{vmatrix} 1 & 2 \\ -1 & 1 \end{vmatrix} = 1(1 - (-2)) = 3 \\ C_{32} &= (-1)^{3+2} \begin{vmatrix} -1 & 2 \\ 3 & 1 \end{vmatrix} = -1(-1 - 6) = 7 \\ C_{33} &= (-1)^{3+3} \begin{vmatrix} -1 & 1 \\ 3 & -1 \end{vmatrix} = 1(1 - 3) = -2 \end{aligned}$$

The cofactor matrix C is:

$$C = \begin{bmatrix} -7 & -13 & 8 \\ 2 & -2 & 2 \\ 3 & 7 & -2 \end{bmatrix}$$

Taking the transpose of C to find $\text{adj } A$:

$$\text{adj } A = C' = \begin{bmatrix} -7 & 2 & 3 \\ -13 & -2 & 7 \\ 8 & 2 & -2 \end{bmatrix}$$

Thus, the inverse A^{-1} is:

$$A^{-1} = \frac{1}{|A|} \text{adj } A = \frac{1}{10} \begin{bmatrix} -7 & 2 & 3 \\ -13 & -2 & 7 \\ 8 & 2 & -2 \end{bmatrix}$$

Verification of $A^{-1}A = I_3$:

$$\begin{aligned}
 A^{-1}A &= \frac{1}{10} \begin{bmatrix} -7 & 2 & 3 \\ -13 & -2 & 7 \\ 8 & 2 & -2 \end{bmatrix} \begin{bmatrix} -1 & 1 & 2 \\ 3 & -1 & 1 \\ -1 & 3 & 4 \end{bmatrix} \\
 &= \frac{1}{10} \begin{bmatrix} -7(-1) + 2(3) + 3(-1) & -7(1) + 2(-1) + 3(3) & -7(2) + 2(1) + 3(4) \\ -13(-1) + (-2)(3) + 7(-1) & -13(1) + (-2)(-1) + 7(3) & -13(2) + (-2)(1) + 7(4) \\ 8(-1) + 2(3) + (-2)(-1) & 8(1) + 2(-1) + (-2)(3) & 8(2) + 2(1) + (-2)(4) \end{bmatrix} \\
 &= \frac{1}{10} \begin{bmatrix} 7 + 6 - 3 & -7 - 2 + 9 & -14 + 2 + 12 \\ 13 - 6 - 7 & -13 + 2 + 21 & -26 - 2 + 28 \\ -8 + 6 + 2 & 8 - 2 - 6 & 16 + 2 - 8 \end{bmatrix} \\
 &= \frac{1}{10} \begin{bmatrix} 10 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 10 \end{bmatrix} = I_3
 \end{aligned}$$

Hence verified.

Final Answer

$$(i) A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1/3 & 0 \\ 3 & 2/3 & -1 \end{bmatrix}$$

$$(ii) A^{-1} = \frac{1}{10} \begin{bmatrix} -7 & 2 & 3 \\ -13 & -2 & 7 \\ 8 & 2 & -2 \end{bmatrix}$$

Question 22

Given $A = \begin{bmatrix} 5 & 0 & 4 \\ 2 & 3 & 2 \\ 1 & 2 & 1 \end{bmatrix}$, $B^{-1} = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$. Compute $(AB)^{-1}$.

TYPE 21 Reversal Law for Matrix Inverse and Adjoint

Given: The matrix $A = \begin{bmatrix} 5 & 0 & 4 \\ 2 & 3 & 2 \\ 1 & 2 & 1 \end{bmatrix}$ and $B^{-1} = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$

To Find: The matrix $(AB)^{-1}$

Concept / Formula Used: $(AB)^{-1} = B^{-1}A^{-1}$

Solution:

First, we need to find A^{-1} . We calculate the determinant $|A|$:

$$\begin{aligned} |A| &= 5 \begin{vmatrix} 3 & 2 \\ 2 & 1 \end{vmatrix} - 0 + 4 \begin{vmatrix} 2 & 3 \\ 1 & 2 \end{vmatrix} \\ &= 5(3 - 4) + 4(4 - 3) \\ &= 5(-1) + 4(1) \\ &= -1 \end{aligned}$$

Cofactors C_{ij} of A :

$$\begin{aligned} C_{11} &= (-1)^{1+1} \begin{vmatrix} 3 & 2 \\ 2 & 1 \end{vmatrix} = 1(3 - 4) = -1 \\ C_{12} &= (-1)^{1+2} \begin{vmatrix} 2 & 2 \\ 1 & 1 \end{vmatrix} = -1(2 - 2) = 0 \\ C_{13} &= (-1)^{1+3} \begin{vmatrix} 2 & 3 \\ 1 & 2 \end{vmatrix} = 1(4 - 3) = 1 \\ C_{21} &= (-1)^{2+1} \begin{vmatrix} 0 & 4 \\ 2 & 1 \end{vmatrix} = -1(0 - 8) = 8 \\ C_{22} &= (-1)^{2+2} \begin{vmatrix} 5 & 4 \\ 1 & 1 \end{vmatrix} = 1(5 - 4) = 1 \\ C_{23} &= (-1)^{2+3} \begin{vmatrix} 5 & 0 \\ 1 & 2 \end{vmatrix} = -1(10 - 0) = -10 \\ C_{31} &= (-1)^{3+1} \begin{vmatrix} 0 & 4 \\ 3 & 2 \end{vmatrix} = 1(0 - 12) = -12 \\ C_{32} &= (-1)^{3+2} \begin{vmatrix} 5 & 4 \\ 2 & 2 \end{vmatrix} = -1(10 - 8) = -2 \\ C_{33} &= (-1)^{3+3} \begin{vmatrix} 5 & 0 \\ 2 & 3 \end{vmatrix} = 1(15 - 0) = 15 \end{aligned}$$

The cofactor matrix C is:

$$C = \begin{bmatrix} -1 & 0 & 1 \\ 8 & 1 & -10 \\ -12 & -2 & 15 \end{bmatrix}$$

Taking the transpose of C to find $\text{adj } A$:

$$\text{adj } A = C' = \begin{bmatrix} -1 & 8 & -12 \\ 0 & 1 & -2 \\ 1 & -10 & 15 \end{bmatrix}$$

Thus, the inverse A^{-1} is:

$$A^{-1} = \frac{1}{|A|} \text{adj } A = \frac{1}{-1} \begin{bmatrix} -1 & 8 & -12 \\ 0 & 1 & -2 \\ 1 & -10 & 15 \end{bmatrix} = \begin{bmatrix} 1 & -8 & 12 \\ 0 & -1 & 2 \\ -1 & 10 & -15 \end{bmatrix}$$

Now, we compute $(AB)^{-1} = B^{-1}A^{-1}$:

$$\begin{aligned} (AB)^{-1} &= \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & -8 & 12 \\ 0 & -1 & 2 \\ -1 & 10 & -15 \end{bmatrix} \\ &= \begin{bmatrix} 1(1) + 3(0) + 3(-1) & 1(-8) + 3(-1) + 3(10) & 1(12) + 3(2) + 3(-15) \\ 1(1) + 4(0) + 3(-1) & 1(-8) + 4(-1) + 3(10) & 1(12) + 4(2) + 3(-15) \\ 1(1) + 3(0) + 4(-1) & 1(-8) + 3(-1) + 4(10) & 1(12) + 3(2) + 4(-15) \end{bmatrix} \\ &= \begin{bmatrix} 1-3 & -8-3+30 & 12+6-45 \\ 1-3 & -8-4+30 & 12+8-45 \\ 1-4 & -8-3+40 & 12+6-60 \end{bmatrix} \\ &= \begin{bmatrix} -2 & 19 & -27 \\ -2 & 18 & -25 \\ -3 & 29 & -42 \end{bmatrix} \end{aligned}$$

Final Answer

$$(AB)^{-1} = \begin{bmatrix} -2 & 19 & -27 \\ -2 & 18 & -25 \\ -3 & 29 & -42 \end{bmatrix}$$

Question 23

- (i) Show that $A = \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix}$ satisfies the equation $A^2 - 3A - 7I = O$ and hence, find A^{-1} .
- (ii) If $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ and I is the identity matrix of order 2, then show that $A^2 = 4A - 3I$. Hence, find A^{-1} .
- (iii) If $A = \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix}$, then find the value of λ so that $A^2 = \lambda A - 2I$. Hence, find A^{-1} .
- (iv) If $A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix}$, then show that $A^3 - 4A^2 - 3A + 11I = O$. Hence, find A^{-1} .

TYPE 20 Algebraic Matrix Equations and Inverse**Given:** The matrices defined in parts (i), (ii), (iii), and (iv)**To Prove:** The respective matrix equations and finding A^{-1} in each case

Concept / Formula Used: $c_k A^k + \dots + c_1 A + c_0 I = O \quad c_0 \neq 0 \quad A^{-1} = -\frac{1}{c_0}(c_k A^{k-1} + \dots + c_1 I)$

Solution:

Part (i): Let $A = \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix}$. First, we compute A^2 :

$$A^2 = \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix} = \begin{bmatrix} 5(5) + 3(-1) & 5(3) + 3(-2) \\ -1(5) + (-2)(-1) & -1(3) + (-2)(-2) \end{bmatrix} = \begin{bmatrix} 22 & 9 \\ -3 & 1 \end{bmatrix}$$

Now, we evaluate $A^2 - 3A - 7I$:

$$\begin{aligned} A^2 - 3A - 7I &= \begin{bmatrix} 22 & 9 \\ -3 & 1 \end{bmatrix} - 3 \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix} - 7 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 22 & 9 \\ -3 & 1 \end{bmatrix} - \begin{bmatrix} 15 & 9 \\ -3 & -6 \end{bmatrix} - \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} \\ &= \begin{bmatrix} 22 - 15 - 7 & 9 - 9 - 0 \\ -3 - (-3) - 0 & 1 - (-6) - 7 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O \end{aligned}$$

Thus, A satisfies the equation. To find A^{-1} , we multiply the equation $A^2 - 3A - 7I = O$ by A^{-1} :

$$\begin{aligned} A^{-1}(A^2 - 3A - 7I) &= O \\ A - 3I - 7A^{-1} &= O \\ 7A^{-1} &= A - 3I \\ A^{-1} &= \frac{1}{7}(A - 3I) \end{aligned}$$

Substituting A and I :

$$A^{-1} = \frac{1}{7} \left(\begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \right) = \frac{1}{7} \begin{bmatrix} 2 & 3 \\ -1 & -5 \end{bmatrix}$$

Part (ii): Let $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$. First, we compute A^2 :

$$A^2 = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 2(2) + (-1)(-1) & 2(-1) + (-1)(2) \\ -1(2) + 2(-1) & -1(-1) + 2(2) \end{bmatrix} = \begin{bmatrix} 5 & -4 \\ -4 & 5 \end{bmatrix}$$

Now, we compute $4A - 3I$:

$$4A - 3I = 4 \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} - 3 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 8 & -4 \\ -4 & 8 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 5 & -4 \\ -4 & 5 \end{bmatrix}$$

Clearly, $A^2 = 4A - 3I$. To find A^{-1} , we multiply the equation $A^2 = 4A - 3I$ by A^{-1} :

$$\begin{aligned} A &= 4I - 3A^{-1} \\ 3A^{-1} &= 4I - A \\ A^{-1} &= \frac{1}{3}(4I - A) \end{aligned}$$

Substituting A and I :

$$A^{-1} = \frac{1}{3} \left(\begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} - \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \right) = \frac{1}{3} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Part (iii): Let $A = \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix}$. First, we compute A^2 :

$$A^2 = \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix} = \begin{bmatrix} 3(3) + (-2)(4) & 3(-2) + (-2)(-2) \\ 4(3) + (-2)(4) & 4(-2) + (-2)(-2) \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 4 & -4 \end{bmatrix}$$

We are given $A^2 = \lambda A - 2I$:

$$\begin{bmatrix} 1 & -2 \\ 4 & -4 \end{bmatrix} = \lambda \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 3\lambda - 2 & -2\lambda \\ 4\lambda & -2\lambda - 2 \end{bmatrix}$$

Comparing the corresponding elements at position (1, 2):

$$-2 = -2\lambda \implies \lambda = 1$$

Let us verify with other elements: - (1, 1): $3(1) - 2 = 1$ (verified) - (2, 1): $4(1) = 4$ (verified) - (2, 2): $-2(1) - 2 = -4$ (verified) Thus, $\lambda = 1$. The equation is $A^2 = A - 2I$. To find A^{-1} , we multiply by A^{-1} :

$$\begin{aligned} A &= I - 2A^{-1} \\ 2A^{-1} &= I - A \\ A^{-1} &= \frac{1}{2}(I - A) \end{aligned}$$

Substituting A and I :

$$A^{-1} = \frac{1}{2} \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix} \right) = \frac{1}{2} \begin{bmatrix} -2 & 2 \\ -4 & 3 \end{bmatrix}$$

Part (iv): Let $A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix}$. First, we compute A^2 :

$$\begin{aligned} A^2 &= \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 1(1) + 3(2) + 2(1) & 1(3) + 3(0) + 2(2) & 1(2) + 3(-1) + 2(3) \\ 2(1) + 0(2) - 1(1) & 2(3) + 0(0) - 1(2) & 2(2) + 0(-1) - 1(3) \\ 1(1) + 2(2) + 3(1) & 1(3) + 2(0) + 3(2) & 1(2) + 2(-1) + 3(3) \end{bmatrix} \\ &= \begin{bmatrix} 9 & 7 & 5 \\ 1 & 4 & 1 \\ 8 & 9 & 9 \end{bmatrix} \end{aligned}$$

Now, we compute $A^3 = A^2A$:

$$\begin{aligned} A^3 &= \begin{bmatrix} 9 & 7 & 5 \\ 1 & 4 & 1 \\ 8 & 9 & 9 \end{bmatrix} \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 9(1) + 7(2) + 5(1) & 9(3) + 7(0) + 5(2) & 9(2) + 7(-1) + 5(3) \\ 1(1) + 4(2) + 1(1) & 1(3) + 4(0) + 1(2) & 1(2) + 4(-1) + 1(3) \\ 8(1) + 9(2) + 9(1) & 8(3) + 9(0) + 9(2) & 8(2) + 9(-1) + 9(3) \end{bmatrix} \\ &= \begin{bmatrix} 28 & 37 & 26 \\ 10 & 5 & 1 \\ 35 & 42 & 34 \end{bmatrix} \end{aligned}$$

Now, we evaluate $A^3 - 4A^2 - 3A + 11I$:

$$\begin{aligned} A^3 - 4A^2 - 3A + 11I &= \begin{bmatrix} 28 & 37 & 26 \\ 10 & 5 & 1 \\ 35 & 42 & 34 \end{bmatrix} - 4 \begin{bmatrix} 9 & 7 & 5 \\ 1 & 4 & 1 \\ 8 & 9 & 9 \end{bmatrix} - 3 \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix} + 11 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 28 & 37 & 26 \\ 10 & 5 & 1 \\ 35 & 42 & 34 \end{bmatrix} - \begin{bmatrix} 36 & 28 & 20 \\ 4 & 16 & 4 \\ 32 & 36 & 36 \end{bmatrix} - \begin{bmatrix} 3 & 9 & 6 \\ 6 & 0 & -3 \\ 3 & 6 & 9 \end{bmatrix} + \begin{bmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{bmatrix} \\ &= \begin{bmatrix} 28 - 36 - 3 + 11 & 37 - 28 - 9 + 0 & 26 - 20 - 6 + 0 \\ 10 - 4 - 6 + 0 & 5 - 16 - 0 + 11 & 1 - 4 - (-3) + 0 \\ 35 - 32 - 3 + 0 & 42 - 36 - 6 + 0 & 34 - 36 - 9 + 11 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O \end{aligned}$$

Thus, the equation is satisfied. To find A^{-1} , we multiply the equation by A^{-1} :

$$\begin{aligned} A^2 - 4A - 3I + 11A^{-1} &= O \\ 11A^{-1} &= 4A + 3I - A^2 \\ A^{-1} &= \frac{1}{11}(4A + 3I - A^2) \end{aligned}$$

First, we compute $4A + 3I$:

$$4A + 3I = 4 \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix} + 3 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 12 & 8 \\ 8 & 3 & -4 \\ 4 & 8 & 15 \end{bmatrix}$$

Now, we subtract A^2 :

$$A^{-1} = \frac{1}{11} \left(\begin{bmatrix} 7 & 12 & 8 \\ 8 & 3 & -4 \\ 4 & 8 & 15 \end{bmatrix} - \begin{bmatrix} 9 & 7 & 5 \\ 1 & 4 & 1 \\ 8 & 9 & 9 \end{bmatrix} \right) = \frac{1}{11} \begin{bmatrix} -2 & 5 & 3 \\ 7 & -1 & -5 \\ -4 & -1 & 6 \end{bmatrix}$$

Final Answer

$$(i) A^{-1} = \frac{1}{7} \begin{bmatrix} 2 & 3 \\ -1 & -5 \end{bmatrix}$$

$$(ii) A^{-1} = \frac{1}{3} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$(iii) \lambda = 1, \quad A^{-1} = \frac{1}{2} \begin{bmatrix} -2 & 2 \\ -4 & 3 \end{bmatrix}$$

$$(iv) A^{-1} = \frac{1}{11} \begin{bmatrix} -2 & 5 & 3 \\ 7 & -1 & -5 \\ -4 & -1 & 6 \end{bmatrix}$$

Question 24

(i) If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$, prove that $A^2 - 4A - 5I = O$. Hence, find A^{-1} .

(ii) Find A^{-1} if $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$. Also show that $A^{-1} = \frac{A^2 - 3I}{2}$.

TYPE 20 Algebraic Matrix Equations and Inverse

Given: The matrices (i) $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$ and (ii) $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

To Prove: (i) $A^2 - 4A - 5I = O$ and finding A^{-1}

(ii) Finding A^{-1} and showing $A^{-1} = \frac{A^2 - 3I}{2}$

Concept / Formula Used: $A^2 - c_1A - c_0I = O$ $A^{-1} = \frac{1}{c_0}(A - c_1I)$

Solution:

Part (i): Let $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$. First, we compute A^2 :

$$\begin{aligned} A^2 &= \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1(1) + 2(2) + 2(2) & 1(2) + 2(1) + 2(2) & 1(2) + 2(2) + 2(1) \\ 2(1) + 1(2) + 2(2) & 2(2) + 1(1) + 2(2) & 2(2) + 1(2) + 2(1) \\ 2(1) + 2(2) + 1(2) & 2(2) + 2(1) + 1(2) & 2(2) + 2(2) + 1(1) \end{bmatrix} \\ &= \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix} \end{aligned}$$

Now, we evaluate $A^2 - 4A - 5I$:

$$\begin{aligned} A^2 - 4A - 5I &= \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix} - 4 \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} - 5 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix} - \begin{bmatrix} 4 & 8 & 8 \\ 8 & 4 & 8 \\ 8 & 8 & 4 \end{bmatrix} - \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix} \\ &= \begin{bmatrix} 9-4-5 & 8-8-0 & 8-8-0 \\ 8-8-0 & 9-4-5 & 8-8-0 \\ 8-8-0 & 8-8-0 & 9-4-5 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O \end{aligned}$$

Thus, $A^2 - 4A - 5I = O$ is proved. To find A^{-1} , we multiply the equation by A^{-1} :

$$\begin{aligned} A - 4I - 5A^{-1} &= O \\ 5A^{-1} &= A - 4I \\ A^{-1} &= \frac{1}{5}(A - 4I) \end{aligned}$$

Substituting A and I :

$$A^{-1} = \frac{1}{5} \left(\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} - \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} \right) = \frac{1}{5} \begin{bmatrix} -3 & 2 & 2 \\ 2 & -3 & 2 \\ 2 & 2 & -3 \end{bmatrix}$$

Part (ii): Let $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$. First, we find A^{-1} using the standard cofactor method.

We calculate the determinant $|A|$:

$$|A| = 0(0 - 1) - 1(0 - 1) + 1(1 - 0) = 1 + 1 = 2$$

Cofactors C_{ij} of A :

$$\begin{aligned} C_{11} &= (-1)^{1+1} \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = 1(0 - 1) = -1 \\ C_{12} &= (-1)^{1+2} \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} = -1(0 - 1) = 1 \\ C_{13} &= (-1)^{1+3} \begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix} = 1(1 - 0) = 1 \\ C_{21} &= (-1)^{2+1} \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} = -1(0 - 1) = 1 \\ C_{22} &= (-1)^{2+2} \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = 1(0 - 1) = -1 \\ C_{23} &= (-1)^{2+3} \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} = -1(0 - 1) = 1 \\ C_{31} &= (-1)^{3+1} \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} = 1(1 - 0) = 1 \\ C_{32} &= (-1)^{3+2} \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} = -1(0 - 1) = 1 \\ C_{33} &= (-1)^{3+3} \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = 1(0 - 1) = -1 \end{aligned}$$

The cofactor matrix C is:

$$C = \begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$$

Since C is symmetric, $\text{adj } A = C' = C$. Thus, the inverse A^{-1} is:

$$A^{-1} = \frac{1}{|A|} \text{adj } A = \frac{1}{2} \begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$$

Now, let us show that $A^{-1} = \frac{A^2 - 3I}{2}$. First, we compute A^2 :

$$A^2 = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

Now, we compute $\frac{A^2 - 3I}{2}$:

$$A^2 - 3I = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} - \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$$

$$\frac{A^2 - 3I}{2} = \frac{1}{2} \begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$$

This is exactly equal to A^{-1} . Hence proved.

Final Answer

$$(i) A^{-1} = \frac{1}{5} \begin{bmatrix} -3 & 2 & 2 \\ 2 & -3 & 2 \\ 2 & 2 & -3 \end{bmatrix}$$

$$(ii) A^{-1} = \frac{1}{2} \begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$$

Question 25

- (i) If $A = \frac{1}{9} \begin{bmatrix} -8 & 1 & 4 \\ 4 & 4 & 7 \\ 1 & -8 & 4 \end{bmatrix}$, prove that $A^{-1} = A'$.
- (ii) If $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$, prove that $A^{-1} = A^2$.
- (iii) If $F(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$, prove that $(F(x))^{-1} = F(-x)$.

TYPE 18 Matrix Inverse Properties and Proofs**Given:** The matrices defined in parts (i), (ii), and (iii)**To Prove:** The respective inverse relations**Concept / Formula Used:** $M \quad N \quad MN = NM = I$ **Solution:****Part (i):** To prove $A^{-1} = A'$, we show that $AA' = I_3$. First, we write the transpose A' :

$$A' = \frac{1}{9} \begin{bmatrix} -8 & 4 & 1 \\ 1 & 4 & -8 \\ 4 & 7 & 4 \end{bmatrix}$$

Now, we compute AA' :

$$\begin{aligned} AA' &= \left(\frac{1}{9} \begin{bmatrix} -8 & 1 & 4 \\ 4 & 4 & 7 \\ 1 & -8 & 4 \end{bmatrix} \right) \left(\frac{1}{9} \begin{bmatrix} -8 & 4 & 1 \\ 1 & 4 & -8 \\ 4 & 7 & 4 \end{bmatrix} \right) \\ &= \frac{1}{81} \begin{bmatrix} -8 & 1 & 4 \\ 4 & 4 & 7 \\ 1 & -8 & 4 \end{bmatrix} \begin{bmatrix} -8 & 4 & 1 \\ 1 & 4 & -8 \\ 4 & 7 & 4 \end{bmatrix} \\ &= \frac{1}{81} \begin{bmatrix} (-8)(-8) + 1(1) + 4(4) & (-8)(4) + 1(4) + 4(7) & (-8)(1) + 1(-8) + 4(4) \\ 4(-8) + 4(1) + 7(4) & 4(4) + 4(4) + 7(7) & 4(1) + 4(-8) + 7(4) \\ 1(-8) + (-8)(1) + 4(4) & 1(4) + (-8)(4) + 4(7) & 1(1) + (-8)(-8) + 4(4) \end{bmatrix} \\ &= \frac{1}{81} \begin{bmatrix} 64 + 1 + 16 & -32 + 4 + 28 & -8 - 8 + 16 \\ -32 + 4 + 28 & 16 + 16 + 49 & 4 - 32 + 28 \\ -8 - 8 + 16 & 4 - 32 + 28 & 1 + 64 + 16 \end{bmatrix} \\ &= \frac{1}{81} \begin{bmatrix} 81 & 0 & 0 \\ 0 & 81 & 0 \\ 0 & 0 & 81 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3 \end{aligned}$$

Since $AA' = I_3$, it follows that $A^{-1} = A'$. Hence proved.

Part (ii): To prove $A^{-1} = A^2$, we show that $A^3 = I_3$. First, we compute A^2 :

$$\begin{aligned} A^2 &= \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1(1) + (-1)(2) + 1(1) & 1(-1) + (-1)(-1) + 1(0) & 1(1) + (-1)(0) + 1(0) \\ 2(1) + (-1)(2) + 0(1) & 2(-1) + (-1)(-1) + 0(0) & 2(1) + (-1)(0) + 0(0) \\ 1(1) + 0(2) + 0(1) & 1(-1) + 0(-1) + 0(0) & 1(1) + 0(0) + 0(0) \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 2 \\ 1 & -1 & 1 \end{bmatrix} \end{aligned}$$

Now, we compute $A^3 = A^2A$:

$$\begin{aligned} A^3 &= \begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 2 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0(1) + 0(2) + 1(1) & 0(-1) + 0(-1) + 1(0) & 0(1) + 0(0) + 1(0) \\ 0(1) + (-1)(2) + 2(1) & 0(-1) + (-1)(-1) + 2(0) & 0(1) + (-1)(0) + 2(0) \\ 1(1) + (-1)(2) + 1(1) & 1(-1) + (-1)(-1) + 1(0) & 1(1) + (-1)(0) + 1(0) \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3 \end{aligned}$$

Since $A^3 = I_3$, multiplying both sides by A^{-1} gives $A^2 = A^{-1}$. Hence proved.

Part (iii): To prove $(F(x))^{-1} = F(-x)$, we show that $F(x)F(-x) = I_3$. First, we write $F(-x)$:

$$F(-x) = \begin{bmatrix} \cos(-x) & -\sin(-x) & 0 \\ \sin(-x) & \cos(-x) & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos x & \sin x & 0 \\ -\sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Now, we compute $F(x)F(-x)$:

$$\begin{aligned} F(x)F(-x) &= \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos x & \sin x & 0 \\ -\sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \cos^2 x + \sin^2 x & \cos x \sin x - \sin x \cos x & 0 \\ \sin x \cos x - \cos x \sin x & \sin^2 x + \cos^2 x & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3 \end{aligned}$$

Since $F(x)F(-x) = I_3$, it follows that $(F(x))^{-1} = F(-x)$. Hence proved.

Hence Proved

All parts (i), (ii), and (iii) are proved.

Question 26

Show that for the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix}$, $A^3 - 6A^2 + 5A + 11I = O$. Hence, find A^{-1} .

TYPE 20 Algebraic Matrix Equations and Inverse

Given: The matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix}$

To Prove: $A^3 - 6A^2 + 5A + 11I = O$ and finding A^{-1}

Concept / Formula Used: $A^3 - c_2A^2 - c_1A - c_0I = O$ $A^{-1} = \frac{1}{c_0}(A^2 - c_2A - c_1I)$

Solution:

First, we compute A^2 :

$$\begin{aligned} A^2 &= \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 1(1) + 1(1) + 1(2) & 1(1) + 1(2) + 1(-1) & 1(1) + 1(-3) + 1(3) \\ 1(1) + 2(1) - 3(2) & 1(1) + 2(2) - 3(-1) & 1(1) + 2(-3) - 3(3) \\ 2(1) - 1(1) + 3(2) & 2(1) - 1(2) + 3(-1) & 2(1) - 1(-3) + 3(3) \end{bmatrix} \\ &= \begin{bmatrix} 4 & 2 & 1 \\ -3 & 8 & -14 \\ 7 & -3 & 14 \end{bmatrix} \end{aligned}$$

Now, we compute $A^3 = A^2A$:

$$\begin{aligned} A^3 &= \begin{bmatrix} 4 & 2 & 1 \\ -3 & 8 & -14 \\ 7 & -3 & 14 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 4(1) + 2(1) + 1(2) & 4(1) + 2(2) + 1(-1) & 4(1) + 2(-3) + 1(3) \\ -3(1) + 8(1) - 14(2) & -3(1) + 8(2) - 14(-1) & -3(1) + 8(-3) - 14(3) \\ 7(1) - 3(1) + 14(2) & 7(1) - 3(2) + 14(-1) & 7(1) - 3(-3) + 14(3) \end{bmatrix} \\ &= \begin{bmatrix} 8 & 7 & 1 \\ -23 & 27 & -69 \\ 32 & -13 & 58 \end{bmatrix} \end{aligned}$$

Now, we evaluate $A^3 - 6A^2 + 5A + 11I$:

$$\begin{aligned}
 A^3 - 6A^2 + 5A + 11I &= \begin{bmatrix} 8 & 7 & 1 \\ -23 & 27 & -69 \\ 32 & -13 & 58 \end{bmatrix} - 6 \begin{bmatrix} 4 & 2 & 1 \\ -3 & 8 & -14 \\ 7 & -3 & 14 \end{bmatrix} + 5 \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix} + 11 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 8 & 7 & 1 \\ -23 & 27 & -69 \\ 32 & -13 & 58 \end{bmatrix} - \begin{bmatrix} 24 & 12 & 6 \\ -18 & 48 & -84 \\ 42 & -18 & 84 \end{bmatrix} + \begin{bmatrix} 5 & 5 & 5 \\ 5 & 10 & -15 \\ 10 & -5 & 15 \end{bmatrix} + \begin{bmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{bmatrix} \\
 &= \begin{bmatrix} 8 - 24 + 5 + 11 & 7 - 12 + 5 + 0 & 1 - 6 + 5 + 0 \\ -23 - (-18) + 5 + 0 & 27 - 48 + 10 + 11 & -69 - (-84) - 15 + 0 \\ 32 - 42 + 10 + 0 & -13 - (-18) - 5 + 0 & 58 - 84 + 15 + 11 \end{bmatrix} \\
 &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O
 \end{aligned}$$

Thus, the equation is satisfied. To find A^{-1} , we multiply the equation by A^{-1} :

$$\begin{aligned}
 A^2 - 6A + 5I + 11A^{-1} &= O \\
 11A^{-1} &= 6A - 5I - A^2 \\
 A^{-1} &= \frac{1}{11}(6A - 5I - A^2)
 \end{aligned}$$

First, we compute $6A - 5I$:

$$6A - 5I = 6 \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix} - 5 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 6 & 6 \\ 6 & 7 & -18 \\ 12 & -6 & 13 \end{bmatrix}$$

Now, we subtract A^2 :

$$A^{-1} = \frac{1}{11} \left(\begin{bmatrix} 1 & 6 & 6 \\ 6 & 7 & -18 \\ 12 & -6 & 13 \end{bmatrix} - \begin{bmatrix} 4 & 2 & 1 \\ -3 & 8 & -14 \\ 7 & -3 & 14 \end{bmatrix} \right) = \frac{1}{11} \begin{bmatrix} -3 & 4 & 5 \\ 9 & -1 & -4 \\ 5 & -3 & -1 \end{bmatrix}$$

Final Answer

$$A^{-1} = \frac{1}{11} \begin{bmatrix} -3 & 4 & 5 \\ 9 & -1 & -4 \\ 5 & -3 & -1 \end{bmatrix}$$

Question 27

- (i) Find a 2×2 matrix B such that $B \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}$.
- (ii) Solve the matrix equation $\begin{bmatrix} 5 & 4 \\ 1 & 1 \end{bmatrix} A = \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}$.

TYPE 20 Algebraic Matrix Equations and Inverse

Given: The matrix equations (i) $B \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}$ and (ii) $\begin{bmatrix} 5 & 4 \\ 1 & 1 \end{bmatrix} A = \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}$

To Find: The matrices B and A

Concept / Formula Used: $MX = N \quad M = NX^{-1} \quad X^{-1} \quad YA = Z \quad A = Y^{-1}Z \quad Y^{-1}$

Solution:

Part (i): Let $X = \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix}$. The equation is $BX = 6I_2$. Thus, $B = 6X^{-1}$. First, we find X^{-1} :

$$|X| = (1)(4) - (-2)(1) = 4 + 2 = 6$$

The adjoint of X is:

$$\text{adj } X = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$$

Thus:

$$X^{-1} = \frac{1}{6} \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$$

Therefore:

$$B = 6 \left(\frac{1}{6} \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix} \right) = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$$

Part (ii): Let $Y = \begin{bmatrix} 5 & 4 \\ 1 & 1 \end{bmatrix}$ and $Z = \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}$. The equation is $YA = Z$. Thus, $A = Y^{-1}Z$. First, we find Y^{-1} :

$$|Y| = (5)(1) - (4)(1) = 5 - 4 = 1$$

The adjoint of Y is:

$$\text{adj } Y = \begin{bmatrix} 1 & -4 \\ -1 & 5 \end{bmatrix}$$

Since $|Y| = 1$, we have:

$$Y^{-1} = \begin{bmatrix} 1 & -4 \\ -1 & 5 \end{bmatrix}$$

Now, we compute $A = Y^{-1}Z$:

$$\begin{aligned} A &= \begin{bmatrix} 1 & -4 \\ -1 & 5 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 1(1) + (-4)(1) & 1(-2) + (-4)(3) \\ -1(1) + 5(1) & -1(-2) + 5(3) \end{bmatrix} \\ &= \begin{bmatrix} 1 - 4 & -2 - 12 \\ -1 + 5 & 2 + 15 \end{bmatrix} \\ &= \begin{bmatrix} -3 & -14 \\ 4 & 17 \end{bmatrix} \end{aligned}$$

Final Answer

(i) $B = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$

(ii) $A = \begin{bmatrix} -3 & -14 \\ 4 & 17 \end{bmatrix}$

Question 28

- (i) Find the matrix A satisfying the matrix equation $\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} A \begin{bmatrix} 4 & 7 \\ 3 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.
- (ii) Find the matrix P satisfying the matrix equation $\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} P \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$.

TYPE 20 Algebraic Matrix Equations and Inverse

Given: The matrix equations (i) $\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} A \begin{bmatrix} 4 & 7 \\ 3 & 5 \end{bmatrix} = I_2$ and (ii) $\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} P \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$

To Find: The matrices A and P

Concept / Formula Used: $XY = Z \implies X = ZY^{-1}$ $Y = X^{-1}ZY^{-1} \implies X = ZY^{-1}$ $X^{-1} = Y^{-1}Z^{-1}$

Solution:

Part (i): Let $X = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$, $Y = \begin{bmatrix} 4 & 7 \\ 3 & 5 \end{bmatrix}$, and $Z = I_2$. The equation is $XY = Z$. Thus, $A = X^{-1}Y^{-1}$. First, we find X^{-1} :

$$|X| = (1)(3) - (2)(2) = 3 - 4 = -1$$

$$\text{adj } X = \begin{bmatrix} 3 & -2 \\ -2 & 1 \end{bmatrix}$$

$$X^{-1} = \frac{1}{-1} \begin{bmatrix} 3 & -2 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} -3 & 2 \\ 2 & -1 \end{bmatrix}$$

Next, we find Y^{-1} :

$$|Y| = (4)(5) - (7)(3) = 20 - 21 = -1$$

$$\text{adj } Y = \begin{bmatrix} 5 & -7 \\ -3 & 4 \end{bmatrix}$$

$$Y^{-1} = \frac{1}{-1} \begin{bmatrix} 5 & -7 \\ -3 & 4 \end{bmatrix} = \begin{bmatrix} -5 & 7 \\ 3 & -4 \end{bmatrix}$$

Now, we compute $A = X^{-1}Y^{-1}$:

$$\begin{aligned} A &= \begin{bmatrix} -3 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} -5 & 7 \\ 3 & -4 \end{bmatrix} \\ &= \begin{bmatrix} -3(-5) + 2(3) & -3(7) + 2(-4) \\ 2(-5) + (-1)(3) & 2(7) + (-1)(-4) \end{bmatrix} \\ &= \begin{bmatrix} 15 + 6 & -21 - 8 \\ -10 - 3 & 14 + 4 \end{bmatrix} \\ &= \begin{bmatrix} 21 & -29 \\ -13 & 18 \end{bmatrix} \end{aligned}$$

Part (ii): Let $X = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$, $Y = \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix}$, and $Z = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$. The equation is $XPY = Z$. Thus, $P = X^{-1}ZY^{-1}$. First, we find X^{-1} :

$$|X| = (2)(2) - (1)(3) = 4 - 3 = 1$$

$$\text{adj } X = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$$

$$X^{-1} = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$$

Next, we find Y^{-1} :

$$|Y| = (-3)(-3) - (2)(5) = 9 - 10 = -1$$

$$\text{adj } Y = \begin{bmatrix} -3 & -2 \\ -5 & -3 \end{bmatrix}$$

$$Y^{-1} = \frac{1}{-1} \begin{bmatrix} -3 & -2 \\ -5 & -3 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix}$$

Now, we compute $X^{-1}Z$:

$$\begin{aligned} X^{-1}Z &= \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 2(1) + (-1)(2) & 2(2) + (-1)(-1) \\ -3(1) + 2(2) & -3(2) + 2(-1) \end{bmatrix} \\ &= \begin{bmatrix} 2 - 2 & 4 + 1 \\ -3 + 4 & -6 - 2 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 5 \\ 1 & -8 \end{bmatrix} \end{aligned}$$

Finally, we compute $P = (X^{-1}Z)Y^{-1}$:

$$\begin{aligned} P &= \begin{bmatrix} 0 & 5 \\ 1 & -8 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 0(3) + 5(5) & 0(2) + 5(3) \\ 1(3) + (-8)(5) & 1(2) + (-8)(3) \end{bmatrix} \\ &= \begin{bmatrix} 0 + 25 & 0 + 15 \\ 3 - 40 & 2 - 24 \end{bmatrix} \\ &= \begin{bmatrix} 25 & 15 \\ -37 & -22 \end{bmatrix} \end{aligned}$$

Final Answer

$$(i) A = \begin{bmatrix} 21 & -29 \\ -13 & 18 \end{bmatrix}$$

$$(ii) P = \begin{bmatrix} 25 & 15 \\ -37 & -22 \end{bmatrix}$$

4.5 – Solving Systems of Linear Equations Using Matrices**Question 1 (i)**

Use matrix method to solve the following system of equations:

$$5x + 2y = 4$$

$$7x + 3y = 5$$

TYPE 22 Solving Linear Equations Using Matrix Method

Given: $5x + 2y = 4$ and $7x + 3y = 5$

Concept / Formula Used: $AX = B$ $X = A^{-1}B$ $A^{-1} = \frac{1}{|A|}\text{adj}(A)$

Solution:

Write the system in matrix form $AX = B$:

$$A = \begin{bmatrix} 5 & 2 \\ 7 & 3 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

Calculate the determinant $|A|$:

$$\begin{aligned} |A| &= (5)(3) - (2)(7) \\ &= 15 - 14 \\ &= 1 \neq 0 \end{aligned}$$

Since $|A| \neq 0$, A^{-1} exists and the system has a unique solution.

Find the cofactors of matrix A :

$$\begin{aligned} C_{11} &= 3 \\ C_{12} &= -7 \\ C_{21} &= -2 \\ C_{22} &= 5 \end{aligned}$$

The adjugate matrix $\text{adj}(A)$ is:

$$\text{adj}(A) = \begin{bmatrix} 3 & -2 \\ -7 & 5 \end{bmatrix}$$

The inverse matrix A^{-1} is:

$$A^{-1} = \frac{1}{1} \begin{bmatrix} 3 & -2 \\ -7 & 5 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ -7 & 5 \end{bmatrix}$$

Now, calculate $X = A^{-1}B$:

$$\begin{aligned} \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} 3 & -2 \\ -7 & 5 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix} \\ &= \begin{bmatrix} (3)(4) + (-2)(5) \\ (-7)(4) + (5)(5) \end{bmatrix} \\ &= \begin{bmatrix} 12 - 10 \\ -28 + 25 \end{bmatrix} \\ &= \begin{bmatrix} 2 \\ -3 \end{bmatrix} \end{aligned}$$

Equating corresponding entries:

$$x = 2, \quad y = -3$$

Final Answer

$$x = 2, \quad y = -3$$

Question 1 (ii)

Use matrix method to solve the following system of equations:

$$4x - 3y = 3$$

$$3x - 5y = 7$$

TYPE 22 Solving Linear Equations Using Matrix Method

Given: $4x - 3y = 3$ and $3x - 5y = 7$

Concept / Formula Used: $AX = B$ $X = A^{-1}B$ $A^{-1} = \frac{1}{|A|}\text{adj}(A)$

Solution:

Write the system in matrix form $AX = B$:

$$A = \begin{bmatrix} 4 & -3 \\ 3 & -5 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$$

Calculate the determinant $|A|$:

$$\begin{aligned}|A| &= (4)(-5) - (-3)(3) \\ &= -20 + 9 \\ &= -11 \neq 0\end{aligned}$$

Find the cofactors of matrix A :

$$\begin{aligned}C_{11} &= -5 \\ C_{12} &= -3 \\ C_{21} &= 3 \\ C_{22} &= 4\end{aligned}$$

The adjugate matrix $\text{adj}(A)$ is:

$$\text{adj}(A) = \begin{bmatrix} -5 & 3 \\ -3 & 4 \end{bmatrix}$$

The inverse matrix A^{-1} is:

$$A^{-1} = -\frac{1}{11} \begin{bmatrix} -5 & 3 \\ -3 & 4 \end{bmatrix}$$

Now, calculate $X = A^{-1}B$:

$$\begin{aligned}\begin{bmatrix} x \\ y \end{bmatrix} &= -\frac{1}{11} \begin{bmatrix} -5 & 3 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} 3 \\ 7 \end{bmatrix} \\ &= -\frac{1}{11} \begin{bmatrix} (-5)(3) + (3)(7) \\ (-3)(3) + (4)(7) \end{bmatrix} \\ &= -\frac{1}{11} \begin{bmatrix} -15 + 21 \\ -9 + 28 \end{bmatrix} \\ &= -\frac{1}{11} \begin{bmatrix} 6 \\ 19 \end{bmatrix} \\ &= \begin{bmatrix} -\frac{6}{11} \\ -\frac{19}{11} \end{bmatrix}\end{aligned}$$

Equating corresponding entries:

$$x = -\frac{6}{11}, \quad y = -\frac{19}{11}$$

Final Answer

$$x = -\frac{6}{11}, \quad y = -\frac{19}{11}$$

Question 2 (i)

Using matrix method, determine whether the following system of equations is consistent or inconsistent:

$$x + 2y = 2$$

$$2x + 3y = 3$$

TYPE 23 Consistency of System of Linear Equations

★ Likely Exam Question

Given: $x + 2y = 2$ and $2x + 3y = 3$

Concept / Formula Used: $AX = B$ $|A| \neq 0$

Solution:

Write the system in matrix form $AX = B$:

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

Calculate the determinant $|A|$:

$$\begin{aligned} |A| &= (1)(3) - (2)(2) \\ &= 3 - 4 \\ &= -1 \end{aligned}$$

Since $|A| = -1 \neq 0$, matrix A is non-singular and A^{-1} exists. Therefore, the given system of linear equations is consistent and has a unique solution.

Final Answer

The system of equations is consistent.

Question 2 (ii)

Using matrix method, determine whether the following system of equations is consistent or inconsistent:

$$x + 2y = 9$$

$$2x + 4y = 7$$

TYPE 23 Consistency of System of Linear Equations

Given: $x + 2y = 9$ and $2x + 4y = 7$

Concept / Formula Used: $|A| = 0$ $(\text{adj } A)B \neq O$ $AX = B$

Solution:

Write the system in matrix form $AX = B$:

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} 9 \\ 7 \end{bmatrix}$$

Calculate the determinant $|A|$:

$$\begin{aligned} |A| &= (1)(4) - (2)(2) \\ &= 4 - 4 \\ &= 0 \end{aligned}$$

Since $|A| = 0$, calculate $(\text{adj } A)B$.

The cofactors of A are:

$$C_{11} = 4, \quad C_{12} = -2, \quad C_{21} = -2, \quad C_{22} = 1$$

$$\text{adj}(A) = \begin{bmatrix} 4 & -2 \\ -2 & 1 \end{bmatrix}$$

Calculate the product $(\text{adj } A)B$:

$$\begin{aligned} (\text{adj } A)B &= \begin{bmatrix} 4 & -2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 9 \\ 7 \end{bmatrix} \\ &= \begin{bmatrix} 4(9) + (-2)(7) \\ -2(9) + 1(7) \end{bmatrix} \\ &= \begin{bmatrix} 36 - 14 \\ -18 + 7 \end{bmatrix} \\ &= \begin{bmatrix} 22 \\ -11 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix} \end{aligned}$$

Since $|A| = 0$ and $(\text{adj } A)B \neq O$, the system has no solution.

Final Answer

The system of equations is inconsistent.

Question 2 (iii)

Using matrix method, determine whether the following system of equations is consistent or inconsistent:

$$3x - y + 2z = 3$$

$$2x + y + 3z = 5$$

$$x - 2y - z = 1$$

TYPE 23 Consistency of System of Linear Equations

Given: $3x - y + 2z = 3$, $2x + y + 3z = 5$, and $x - 2y - z = 1$

Concept / Formula Used: $|A| = 0$ $(\text{adj } A)B \neq O$ $AX = B$

Solution:

Write the system in matrix form $AX = B$:

$$A = \begin{bmatrix} 3 & -1 & 2 \\ 2 & 1 & 3 \\ 1 & -2 & -1 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 3 \\ 5 \\ 1 \end{bmatrix}$$

Calculate the determinant $|A|$ by expanding along Row 1:

$$\begin{aligned} |A| &= 3 \begin{vmatrix} 1 & 3 \\ -2 & -1 \end{vmatrix} - (-1) \begin{vmatrix} 2 & 3 \\ 1 & -1 \end{vmatrix} + 2 \begin{vmatrix} 2 & 1 \\ 1 & -2 \end{vmatrix} \\ &= 3(-1 - (-6)) + 1(-2 - 3) + 2(-4 - 1) \\ &= 3(5) + 1(-5) + 2(-5) \\ &= 15 - 5 - 10 \\ &= 0 \end{aligned}$$

Since $|A| = 0$, calculate $(\text{adj } A)B$.

Find the cofactors of A :

$$\begin{aligned} C_{11} &= +(-1 - (-6)) = 5, & C_{12} &= -(-2 - 3) = 5, & C_{13} &= +(-4 - 1) = -5 \\ C_{21} &= -(1 - (-4)) = -5, & C_{22} &= +(-3 - 2) = -5, & C_{23} &= -(-6 - (-1)) = 5 \\ C_{31} &= +(-3 - 2) = -5, & C_{32} &= -(9 - 4) = -5, & C_{33} &= +(3 - (-2)) = 5 \end{aligned}$$

$$\text{adj}(A) = \begin{bmatrix} 5 & -5 & -5 \\ 5 & -5 & -5 \\ -5 & 5 & 5 \end{bmatrix}$$

Calculate $(\text{adj } A)B$:

$$\begin{aligned} (\text{adj } A)B &= \begin{bmatrix} 5 & -5 & -5 \\ 5 & -5 & -5 \\ -5 & 5 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 5(3) - 5(5) - 5(1) \\ 5(3) - 5(5) - 5(1) \\ -5(3) + 5(5) + 5(1) \end{bmatrix} \\ &= \begin{bmatrix} 15 - 25 - 5 \\ 15 - 25 - 5 \\ -15 + 25 + 5 \end{bmatrix} \\ &= \begin{bmatrix} -15 \\ -15 \\ 15 \end{bmatrix} \neq O \end{aligned}$$

Since $|A| = 0$ and $(\text{adj } A)B \neq O$, the system has no solution.

Final Answer

The system of equations is inconsistent.

Question 2 (iv)

Using matrix method, determine whether the following system of equations is consistent or inconsistent:

$$\begin{aligned} 5x - y + 4z &= 5 \\ 2x + 3y + 5z &= 2 \\ 5x - 2y + 6z &= -1 \end{aligned}$$

TYPE 23 Consistency of System of Linear Equations

★ Likely Exam Question

Given: $5x - y + 4z = 5$, $2x + 3y + 5z = 2$, and $5x - 2y + 6z = -1$

Concept / Formula Used: $AX = B$ $|A| \neq 0$

Solution:

Write the system in matrix form $AX = B$:

$$A = \begin{bmatrix} 5 & -1 & 4 \\ 2 & 3 & 5 \\ 5 & -2 & 6 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix}$$

Calculate $|A|$ by expanding along Row 1:

$$\begin{aligned} |A| &= 5(18 - (-10)) - (-1)(12 - 25) + 4(-4 - 15) \\ &= 5(28) + 1(-13) + 4(-19) \\ &= 140 - 13 - 76 \\ &= 51 \neq 0 \end{aligned}$$

Since $|A| = 51 \neq 0$, A^{-1} exists. Therefore, the given system of linear equations is consistent and has a unique solution.

Final Answer

The system of equations is consistent.

Question 3

Solve the following system of equations by using matrix method:

$$x + y = 1$$

$$7x + 7y = 7$$

TYPE 24 Solving Dependent System of Linear Equations

Given: $x + y = 1$ and $7x + 7y = 7$

Concept / Formula Used: $|A| = 0$ $(\text{adj } A)B = O$ $AX = B$

Solution:

Write the system in matrix form $AX = B$:

$$A = \begin{bmatrix} 1 & 1 \\ 7 & 7 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 7 \end{bmatrix}$$

Calculate the determinant $|A|$:

$$\begin{aligned} |A| &= (1)(7) - (1)(7) \\ &= 7 - 7 \\ &= 0 \end{aligned}$$

Since $|A| = 0$, compute $(\text{adj } A)B$.

The cofactors of matrix A are:

$$C_{11} = 7, \quad C_{12} = -7, \quad C_{21} = -1, \quad C_{22} = 1$$

$$\text{adj}(A) = \begin{bmatrix} 7 & -1 \\ -7 & 1 \end{bmatrix}$$

Calculate $(\text{adj } A)B$:

$$\begin{aligned} (\text{adj } A)B &= \begin{bmatrix} 7 & -1 \\ -7 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 7 \end{bmatrix} \\ &= \begin{bmatrix} 7(1) + (-1)(7) \\ -7(1) + 1(7) \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} = O \end{aligned}$$

Since $|A| = 0$ and $(\text{adj } A)B = O$, the system has infinitely many solutions.

Let $y = k$, where $k \in \mathbb{R}$. Substitute $y = k$ into the first equation $x + y = 1$:

$$x + k = 1 \implies x = 1 - k$$

Final Answer

$x = 1 - k, \quad y = k, \quad \text{where } k \text{ is any real number.}$

Question 4 (i)

Using matrix method, solve the following system of equations:

$$x - 2y + 3z = 6$$

$$x + 4y + z = 12$$

$$x - 3y + 2z = 1$$

TYPE 22 Solving Linear Equations Using Matrix Method

Given: $x - 2y + 3z = 6$, $x + 4y + z = 12$, and $x - 3y + 2z = 1$

Concept / Formula Used: $AX = B \quad X = A^{-1}B = \frac{1}{|A|}(\text{adj } A)B$

Solution:

Write the system in matrix form $AX = B$:

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 1 & 4 & 1 \\ 1 & -3 & 2 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 6 \\ 12 \\ 1 \end{bmatrix}$$

Calculate $|A|$ by expanding along Row 1:

$$\begin{aligned} |A| &= 1(8 - (-3)) - (-2)(2 - 1) + 3(-3 - 4) \\ &= 1(11) + 2(1) + 3(-7) \\ &= 11 + 2 - 21 \\ &= -8 \neq 0 \end{aligned}$$

Find cofactors of matrix A :

$$\begin{aligned} C_{11} &= +(8 - (-3)) = 11, & C_{12} &= -(2 - 1) = -1, & C_{13} &= +(-3 - 4) = -7 \\ C_{21} &= -(-4 - (-9)) = -5, & C_{22} &= +(2 - 3) = -1, & C_{23} &= -(-3 - (-2)) = 1 \\ C_{31} &= +(-2 - 12) = -14, & C_{32} &= -(1 - 3) = 2, & C_{33} &= +(4 - (-2)) = 6 \end{aligned}$$

$$\text{adj}(A) = \begin{bmatrix} 11 & -5 & -14 \\ -1 & -1 & 2 \\ -7 & 1 & 6 \end{bmatrix}$$

Calculate A^{-1} :

$$A^{-1} = -\frac{1}{8} \begin{bmatrix} 11 & -5 & -14 \\ -1 & -1 & 2 \\ -7 & 1 & 6 \end{bmatrix}$$

Now, compute $X = A^{-1}B$:

$$\begin{aligned}
 \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= -\frac{1}{8} \begin{bmatrix} 11 & -5 & -14 \\ -1 & -1 & 2 \\ -7 & 1 & 6 \end{bmatrix} \begin{bmatrix} 6 \\ 12 \\ 1 \end{bmatrix} \\
 &= -\frac{1}{8} \begin{bmatrix} 11(6) - 5(12) - 14(1) \\ -1(6) - 1(12) + 2(1) \\ -7(6) + 1(12) + 6(1) \end{bmatrix} \\
 &= -\frac{1}{8} \begin{bmatrix} 66 - 60 - 14 \\ -6 - 12 + 2 \\ -42 + 12 + 6 \end{bmatrix} \\
 &= -\frac{1}{8} \begin{bmatrix} -8 \\ -16 \\ -24 \end{bmatrix} \\
 &= \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}
 \end{aligned}$$

Final Answer

$$x = 1, \quad y = 2, \quad z = 3$$

Question 4 (ii)

Using matrix method, solve the following system of equations:

$$x + y + z = 6$$

$$x - y + z = 2$$

$$2x + y - z = 1$$

TYPE 22 Solving Linear Equations Using Matrix Method

Given: $x + y + z = 6$, $x - y + z = 2$, and $2x + y - z = 1$

Concept / Formula Used: $AX = B \quad X = A^{-1}B = \frac{1}{|A|}(\text{adj } A)B$

Solution:

Write the system in matrix form $AX = B$:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix}$$

Calculate $|A|$ by expanding along Row 1:

$$\begin{aligned}
 |A| &= 1(1 - 1) - 1(-1 - 2) + 1(1 - (-2)) \\
 &= 1(0) - 1(-3) + 1(3) \\
 &= 0 + 3 + 3 \\
 &= 6 \neq 0
 \end{aligned}$$

Find cofactors of matrix A :

$$\begin{aligned}
 C_{11} &= +(1 - 1) = 0, & C_{12} &= -(-1 - 2) = 3, & C_{13} &= +(1 - (-2)) = 3 \\
 C_{21} &= -(-1 - 1) = 2, & C_{22} &= +(-1 - 2) = -3, & C_{23} &= -(1 - 2) = 1 \\
 C_{31} &= +(1 - (-1)) = 2, & C_{32} &= -(1 - 1) = 0, & C_{33} &= +(-1 - 1) = -2
 \end{aligned}$$

$$\text{adj}(A) = \begin{bmatrix} 0 & 2 & 2 \\ 3 & -3 & 0 \\ 3 & 1 & -2 \end{bmatrix}$$

Calculate A^{-1} :

$$A^{-1} = \frac{1}{6} \begin{bmatrix} 0 & 2 & 2 \\ 3 & -3 & 0 \\ 3 & 1 & -2 \end{bmatrix}$$

Now, compute $X = A^{-1}B$:

$$\begin{aligned}
 \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \frac{1}{6} \begin{bmatrix} 0 & 2 & 2 \\ 3 & -3 & 0 \\ 3 & 1 & -2 \end{bmatrix} \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix} \\
 &= \frac{1}{6} \begin{bmatrix} 0(6) + 2(2) + 2(1) \\ 3(6) - 3(2) + 0(1) \\ 3(6) + 1(2) - 2(1) \end{bmatrix} \\
 &= \frac{1}{6} \begin{bmatrix} 0 + 4 + 2 \\ 18 - 6 + 0 \\ 18 + 2 - 2 \end{bmatrix} \\
 &= \frac{1}{6} \begin{bmatrix} 6 \\ 12 \\ 18 \end{bmatrix} \\
 &= \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}
 \end{aligned}$$

Final Answer

$$x = 1, \quad y = 2, \quad z = 3$$

Question 4 (iii)

Using matrix method, solve the following system of equations:

$$x + y + z = 9$$

$$2x + 5y + 7z = 52$$

$$2x + y - z = 0$$

TYPE 22 Solving Linear Equations Using Matrix Method

Given: $x + y + z = 9$, $2x + 5y + 7z = 52$, and $2x + y - z = 0$

Concept / Formula Used: $AX = B \quad X = A^{-1}B = \frac{1}{|A|}(\text{adj } A)B$

Solution:

Write the system in matrix form $AX = B$:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 5 & 7 \\ 2 & 1 & -1 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 9 \\ 52 \\ 0 \end{bmatrix}$$

Calculate $|A|$ by expanding along Row 1:

$$\begin{aligned} |A| &= 1(-5 - 7) - 1(-2 - 14) + 1(2 - 10) \\ &= 1(-12) - 1(-16) + 1(-8) \\ &= -12 + 16 - 8 \\ &= -4 \neq 0 \end{aligned}$$

Find cofactors of matrix A :

$$\begin{aligned} C_{11} &= +(-5 - 7) = -12, & C_{12} &= -(-2 - 14) = 16, & C_{13} &= +(2 - 10) = -8 \\ C_{21} &= -(-1 - 1) = 2, & C_{22} &= +(-1 - 2) = -3, & C_{23} &= -(1 - 2) = 1 \\ C_{31} &= +(7 - 5) = 2, & C_{32} &= -(7 - 2) = -5, & C_{33} &= +(5 - 2) = 3 \end{aligned}$$

$$\text{adj}(A) = \begin{bmatrix} -12 & 2 & 2 \\ 16 & -3 & -5 \\ -8 & 1 & 3 \end{bmatrix}$$

Calculate A^{-1} :

$$A^{-1} = -\frac{1}{4} \begin{bmatrix} -12 & 2 & 2 \\ 16 & -3 & -5 \\ -8 & 1 & 3 \end{bmatrix}$$

Now, compute $X = A^{-1}B$:

$$\begin{aligned}
 \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= -\frac{1}{4} \begin{bmatrix} -12 & 2 & 2 \\ 16 & -3 & -5 \\ -8 & 1 & 3 \end{bmatrix} \begin{bmatrix} 9 \\ 52 \\ 0 \end{bmatrix} \\
 &= -\frac{1}{4} \begin{bmatrix} -12(9) + 2(52) + 2(0) \\ 16(9) - 3(52) - 5(0) \\ -8(9) + 1(52) + 3(0) \end{bmatrix} \\
 &= -\frac{1}{4} \begin{bmatrix} -108 + 104 + 0 \\ 144 - 156 - 0 \\ -72 + 52 + 0 \end{bmatrix} \\
 &= -\frac{1}{4} \begin{bmatrix} -4 \\ -12 \\ -20 \end{bmatrix} \\
 &= \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}
 \end{aligned}$$

Final Answer

$$x = 1, \quad y = 3, \quad z = 5$$

Question 4 (iv)

Using matrix method, solve the following system of equations:

$$x - 2y = 10$$

$$2x - y - z = 8$$

$$-2y + z = 7$$

TYPE 22 Solving Linear Equations Using Matrix Method

★ Likely Exam Question

Given: $x - 2y + 0z = 10$, $2x - y - z = 8$, and $0x - 2y + z = 7$

Concept / Formula Used: $AX = B \quad X = A^{-1}B = \frac{1}{|A|}(\text{adj } A)B$

Solution:

Write the system in matrix form $AX = B$:

$$A = \begin{bmatrix} 1 & -2 & 0 \\ 2 & -1 & -1 \\ 0 & -2 & 1 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 10 \\ 8 \\ 7 \end{bmatrix}$$

Calculate $|A|$ by expanding along Row 1:

$$\begin{aligned}|A| &= 1(-1 - 2) - (-2)(2 - 0) + 0(-4 - 0) \\&= 1(-3) + 2(2) + 0 \\&= -3 + 4 \\&= 1 \neq 0\end{aligned}$$

Find cofactors of matrix A :

$$\begin{aligned}C_{11} &= +(-1 - 2) = -3, & C_{12} &= -(2 - 0) = -2, & C_{13} &= +(-4 - 0) = -4 \\C_{21} &= -(-2 - 0) = 2, & C_{22} &= +(1 - 0) = 1, & C_{23} &= -(-2 - 0) = 2 \\C_{31} &= +(2 - 0) = 2, & C_{32} &= -(-1 - 0) = 1, & C_{33} &= +(-1 - (-4)) = 3\end{aligned}$$

$$\text{adj}(A) = \begin{bmatrix} -3 & 2 & 2 \\ -2 & 1 & 1 \\ -4 & 2 & 3 \end{bmatrix}$$

Since $|A| = 1$:

$$A^{-1} = \text{adj}(A) = \begin{bmatrix} -3 & 2 & 2 \\ -2 & 1 & 1 \\ -4 & 2 & 3 \end{bmatrix}$$

Now, compute $X = A^{-1}B$:

$$\begin{aligned}\begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \begin{bmatrix} -3 & 2 & 2 \\ -2 & 1 & 1 \\ -4 & 2 & 3 \end{bmatrix} \begin{bmatrix} 10 \\ 8 \\ 7 \end{bmatrix} \\&= \begin{bmatrix} -3(10) + 2(8) + 2(7) \\ -2(10) + 1(8) + 1(7) \\ -4(10) + 2(8) + 3(7) \end{bmatrix} \\&= \begin{bmatrix} -30 + 16 + 14 \\ -20 + 8 + 7 \\ -40 + 16 + 21 \end{bmatrix} \\&= \begin{bmatrix} 0 \\ -5 \\ -3 \end{bmatrix}\end{aligned}$$

Final Answer

$$x = 0, \quad y = -5, \quad z = -3$$

Question 4 (v)

Using matrix method, solve the following system of equations:

$$3x + y + z = 1$$

$$2x + 2z = 0$$

$$5x + y + 2z = 2$$

TYPE 22 Solving Linear Equations Using Matrix Method

★ Likely Exam Question

Given: $3x + y + z = 1$, $2x + 0y + 2z = 0$, and $5x + y + 2z = 2$

Concept / Formula Used: $AX = B \quad X = A^{-1}B = \frac{1}{|A|}(\text{adj } A)B$

Solution:

Write the system in matrix form $AX = B$:

$$A = \begin{bmatrix} 3 & 1 & 1 \\ 2 & 0 & 2 \\ 5 & 1 & 2 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$$

Calculate $|A|$ by expanding along Row 1:

$$\begin{aligned} |A| &= 3(0 - 2) - 1(4 - 10) + 1(2 - 0) \\ &= 3(-2) - 1(-6) + 1(2) \\ &= -6 + 6 + 2 \\ &= 2 \neq 0 \end{aligned}$$

Find cofactors of matrix A :

$$\begin{array}{lll} C_{11} = +(0 - 2) = -2, & C_{12} = -(4 - 10) = 6, & C_{13} = +(2 - 0) = 2 \\ C_{21} = -(2 - 1) = -1, & C_{22} = +(6 - 5) = 1, & C_{23} = -(3 - 5) = 2 \\ C_{31} = +(2 - 0) = 2, & C_{32} = -(6 - 2) = -4, & C_{33} = +(0 - 2) = -2 \end{array}$$

$$\text{adj}(A) = \begin{bmatrix} -2 & -1 & 2 \\ 6 & 1 & -4 \\ 2 & 2 & -2 \end{bmatrix}$$

Calculate A^{-1} :

$$A^{-1} = \frac{1}{2} \begin{bmatrix} -2 & -1 & 2 \\ 6 & 1 & -4 \\ 2 & 2 & -2 \end{bmatrix}$$

Now, compute $X = A^{-1}B$:

$$\begin{aligned}
 \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \frac{1}{2} \begin{bmatrix} -2 & -1 & 2 \\ 6 & 1 & -4 \\ 2 & 2 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} \\
 &= \frac{1}{2} \begin{bmatrix} -2(1) - 1(0) + 2(2) \\ 6(1) + 1(0) - 4(2) \\ 2(1) + 2(0) - 2(2) \end{bmatrix} \\
 &= \frac{1}{2} \begin{bmatrix} -2 + 0 + 4 \\ 6 + 0 - 8 \\ 2 + 0 - 4 \end{bmatrix} \\
 &= \frac{1}{2} \begin{bmatrix} 2 \\ -2 \\ -2 \end{bmatrix} \\
 &= \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}
 \end{aligned}$$

Final Answer

$$x = 1, \quad y = -1, \quad z = -1$$

Question 4 (vi)

Using matrix method, solve the following system of equations:

$$2x - 3y + 5z = 11$$

$$3x + 2y - 4z = -5$$

$$x + y - 2z = 3$$

TYPE 22 Solving Linear Equations Using Matrix Method

Given: $2x - 3y + 5z = 11$, $3x + 2y - 4z = -5$, and $x + y - 2z = 3$

Concept / Formula Used: $AX = B \quad X = A^{-1}B = \frac{1}{|A|}(\text{adj } A)B$

Solution:

Write the system in matrix form $AX = B$:

$$A = \begin{bmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 11 \\ -5 \\ 3 \end{bmatrix}$$

Calculate $|A|$ by expanding along Row 1:

$$\begin{aligned}|A| &= 2(-4 - (-4)) - (-3)(-6 - (-4)) + 5(3 - 2) \\&= 2(0) + 3(-2) + 5(1) \\&= 0 - 6 + 5 \\&= -1 \neq 0\end{aligned}$$

Find cofactors of matrix A :

$$\begin{aligned}C_{11} &= +(-4 - (-4)) = 0, & C_{12} &= -(-6 - (-4)) = 2, & C_{13} &= +(3 - 2) = 1 \\C_{21} &= -(6 - 5) = -1, & C_{22} &= +(-4 - 5) = -9, & C_{23} &= -(2 - (-3)) = -5 \\C_{31} &= +(12 - 10) = 2, & C_{32} &= -(-8 - 15) = 23, & C_{33} &= +(4 - (-9)) = 13\end{aligned}$$

$$\text{adj}(A) = \begin{bmatrix} 0 & -1 & 2 \\ 2 & -9 & 23 \\ 1 & -5 & 13 \end{bmatrix}$$

Calculate A^{-1} :

$$A^{-1} = \frac{1}{-1} \begin{bmatrix} 0 & -1 & 2 \\ 2 & -9 & 23 \\ 1 & -5 & 13 \end{bmatrix} = \begin{bmatrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{bmatrix}$$

Now, compute $X = A^{-1}B$:

$$\begin{aligned}\begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \begin{bmatrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{bmatrix} \begin{bmatrix} 11 \\ -5 \\ 3 \end{bmatrix} \\&= \begin{bmatrix} 0(11) + 1(-5) - 2(3) \\ -2(11) + 9(-5) - 23(3) \\ -1(11) + 5(-5) - 13(3) \end{bmatrix} \\&= \begin{bmatrix} 0 - 5 - 6 \\ -22 - 45 - 69 \\ -11 - 25 - 39 \end{bmatrix} \\&= \begin{bmatrix} -11 \\ -136 \\ -75 \end{bmatrix}\end{aligned}$$

Final Answer

$$x = -11, \quad y = -136, \quad z = -75$$

Question 5

Solve the following system of linear equations using matrix method:

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 9$$

$$\frac{2}{x} + \frac{5}{y} + \frac{7}{z} = 52$$

$$\frac{2}{x} + \frac{1}{y} - \frac{1}{z} = 0$$

TYPE 25 Solving Reducible System of Linear Equations

★ Likely Exam Question

Given: $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 9$, $\frac{2}{x} + \frac{5}{y} + \frac{7}{z} = 52$, and $\frac{2}{x} + \frac{1}{y} - \frac{1}{z} = 0$

Concept / Formula Used: $u = \frac{1}{x}, v = \frac{1}{y}, w = \frac{1}{z}$ $AU = B \implies U = A^{-1}B$

Solution:

Let:

$$u = \frac{1}{x}, \quad v = \frac{1}{y}, \quad w = \frac{1}{z}$$

Substitute u, v, w into the system of equations:

$$\begin{aligned} u + v + w &= 9 \\ 2u + 5v + 7w &= 52 \\ 2u + v - w &= 0 \end{aligned}$$

Write in matrix form $AU = B$:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 5 & 7 \\ 2 & 1 & -1 \end{bmatrix}, \quad U = \begin{bmatrix} u \\ v \\ w \end{bmatrix}, \quad B = \begin{bmatrix} 9 \\ 52 \\ 0 \end{bmatrix}$$

Calculate $|A|$:

$$\begin{aligned} |A| &= 1(-5 - 7) - 1(-2 - 14) + 1(2 - 10) \\ &= -12 + 16 - 8 \\ &= -4 \neq 0 \end{aligned}$$

Find cofactors of matrix A :

$$\begin{array}{lll} C_{11} = -12, & C_{12} = 16, & C_{13} = -8 \\ C_{21} = 2, & C_{22} = -3, & C_{23} = 1 \\ C_{31} = 2, & C_{32} = -5, & C_{33} = 3 \end{array}$$

$$\text{adj}(A) = \begin{bmatrix} -12 & 2 & 2 \\ 16 & -3 & -5 \\ -8 & 1 & 3 \end{bmatrix}$$

Calculate A^{-1} :

$$A^{-1} = -\frac{1}{4} \begin{bmatrix} -12 & 2 & 2 \\ 16 & -3 & -5 \\ -8 & 1 & 3 \end{bmatrix}$$

Compute $U = A^{-1}B$:

$$\begin{aligned} \begin{bmatrix} u \\ v \\ w \end{bmatrix} &= -\frac{1}{4} \begin{bmatrix} -12 & 2 & 2 \\ 16 & -3 & -5 \\ -8 & 1 & 3 \end{bmatrix} \begin{bmatrix} 9 \\ 52 \\ 0 \end{bmatrix} \\ &= -\frac{1}{4} \begin{bmatrix} -12(9) + 2(52) + 2(0) \\ 16(9) - 3(52) - 5(0) \\ -8(9) + 1(52) + 3(0) \end{bmatrix} \\ &= -\frac{1}{4} \begin{bmatrix} -108 + 104 + 0 \\ 144 - 156 - 0 \\ -72 + 52 + 0 \end{bmatrix} \\ &= -\frac{1}{4} \begin{bmatrix} -4 \\ -12 \\ -20 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} \end{aligned}$$

So $u = 1$, $v = 3$, $w = 5$.

Perform back-substitution:

$$\begin{aligned} x &= \frac{1}{u} = \frac{1}{1} = 1 \\ y &= \frac{1}{v} = \frac{1}{3} \\ z &= \frac{1}{w} = \frac{1}{5} \end{aligned}$$

Final Answer

$$x = 1, \quad y = \frac{1}{3}, \quad z = \frac{1}{5}$$

Common Mistake: Forgetting to back-substitute $x = \frac{1}{u}$, $y = \frac{1}{v}$, $z = \frac{1}{w}$ and leaving u, v, w as the final answer.

Question 6

State the condition under which the following equations have a unique solution:

$$x + 2y - 2z + 5 = 0$$

$$-x + 3y + 4 = 0$$

$$-2y + z - 4 = 0$$

Using Martin's rule, find the unique solution of the above system of linear equations.

TYPE 22 Solving Linear Equations Using Matrix Method

Given: $x + 2y - 2z = -5$, $-x + 3y + 0z = -4$, and $0x - 2y + z = 4$

Concept / Formula Used: $AX = B$ $|A| \neq 0$ $X = A^{-1}B$

Solution:

Rearrange the given equations into standard form $AX = B$:

$$\begin{aligned}x + 2y - 2z &= -5 \\ -x + 3y + 0z &= -4 \\ 0x - 2y + z &= 4\end{aligned}$$

Matrix representation:

$$A = \begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} -5 \\ -4 \\ 4 \end{bmatrix}$$

Condition for Unique Solution: The system of equations has a unique solution if and only if $|A| \neq 0$.

Calculate determinant $|A|$:

$$\begin{aligned}|A| &= 1(3 - 0) - 2(-1 - 0) + (-2)(2 - 0) \\ &= 1(3) - 2(-1) - 2(2) \\ &= 3 + 2 - 4 \\ &= 1 \neq 0\end{aligned}$$

Since $|A| = 1 \neq 0$, the system possesses a unique solution.

Find cofactors of matrix A :

$$\begin{aligned}C_{11} &= +(3 - 0) = 3, & C_{12} &= -(-1 - 0) = 1, & C_{13} &= +(2 - 0) = 2 \\ C_{21} &= -(2 - 4) = 2, & C_{22} &= +(1 - 0) = 1, & C_{23} &= -(-2 - 0) = 2 \\ C_{31} &= +(0 - (-6)) = 6, & C_{32} &= -(0 - 2) = 2, & C_{33} &= +(3 - (-2)) = 5\end{aligned}$$

$$\text{adj}(A) = \begin{bmatrix} 3 & 2 & 6 \\ 1 & 1 & 2 \\ 2 & 2 & 5 \end{bmatrix}$$

Since $|A| = 1$, $A^{-1} = \text{adj}(A)$.

Using Martin's Rule $X = A^{-1}B$:

$$\begin{aligned}\begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \begin{bmatrix} 3 & 2 & 6 \\ 1 & 1 & 2 \\ 2 & 2 & 5 \end{bmatrix} \begin{bmatrix} -5 \\ -4 \\ 4 \end{bmatrix} \\ &= \begin{bmatrix} 3(-5) + 2(-4) + 6(4) \\ 1(-5) + 1(-4) + 2(4) \\ 2(-5) + 2(-4) + 5(4) \end{bmatrix} \\ &= \begin{bmatrix} -15 - 8 + 24 \\ -5 - 4 + 8 \\ -10 - 8 + 20 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}\end{aligned}$$

Final Answer

Condition: $|A| \neq 0$.

Solution: $x = 1, \quad y = -1, \quad z = 2$

Question 7

The sum of three numbers is 6. If we multiply third number by 3 and add second to it we get 11. By adding first and third numbers, we get double of the second number. Represent it algebraically and find the numbers using matrix method.

TYPE 26 Word Problems on System of Linear Equations

★ Likely Exam Question

Given: Let x, y, z be the first, second, and third numbers respectively

Concept / Formula Used: $AX = B \quad X = A^{-1}B = \frac{1}{|A|}(\text{adj } A)B$

Solution:

Formulate equations from the given conditions: 1. "The sum of three numbers is 6":

$$x + y + z = 6$$

2. "Multiply third number by 3 and add second to it we get 11":

$$y + 3z = 11 \implies 0x + y + 3z = 11$$

3. "Adding first and third numbers, we get double of the second number":

$$x + z = 2y \implies x - 2y + z = 0$$

Write system in matrix form $AX = B$:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 1 & -2 & 1 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 6 \\ 11 \\ 0 \end{bmatrix}$$

Calculate $|A|$ by expanding along Row 1:

$$\begin{aligned} |A| &= 1(1 - (-6)) - 1(0 - 3) + 1(0 - 1) \\ &= 1(7) - 1(-3) + 1(-1) \\ &= 7 + 3 - 1 \\ &= 9 \neq 0 \end{aligned}$$

Find cofactors of matrix A :

$$\begin{aligned} C_{11} &= +(1 - (-6)) = 7, & C_{12} &= -(0 - 3) = 3, & C_{13} &= +(0 - 1) = -1 \\ C_{21} &= -(1 - (-2)) = -3, & C_{22} &= +(1 - 1) = 0, & C_{23} &= -(-2 - 1) = 3 \\ C_{31} &= +(3 - 1) = 2, & C_{32} &= -(3 - 0) = -3, & C_{33} &= +(1 - 0) = 1 \end{aligned}$$

$$\text{adj}(A) = \begin{bmatrix} 7 & -3 & 2 \\ 3 & 0 & -3 \\ -1 & 3 & 1 \end{bmatrix}$$

Now, compute $X = A^{-1}B$:

$$\begin{aligned} \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \frac{1}{9} \begin{bmatrix} 7 & -3 & 2 \\ 3 & 0 & -3 \\ -1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 6 \\ 11 \\ 0 \end{bmatrix} \\ &= \frac{1}{9} \begin{bmatrix} 7(6) - 3(11) + 2(0) \\ 3(6) + 0(11) - 3(0) \\ -1(6) + 3(11) + 1(0) \end{bmatrix} \\ &= \frac{1}{9} \begin{bmatrix} 42 - 33 + 0 \\ 18 + 0 - 0 \\ -6 + 33 + 0 \end{bmatrix} \\ &= \frac{1}{9} \begin{bmatrix} 9 \\ 18 \\ 27 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \end{aligned}$$

Final Answer

Algebraic representation:

$$x + y + z = 6$$

$$y + 3z = 11$$

$$x - 2y + z = 0$$

Numbers: First number $x = 1$, Second number $y = 2$, Third number $z = 3$

Question 8

Garima owns a cafe that serves burgers, samosas, and juice bottles. On a particular day, 100 items in total were sold. The number of burgers sold was thrice the number of samosas sold. Also, the number of samosas sold was 10 more than the number of juice bottles sold.

- (i) Form a set of simultaneous equations for the above information.
- (ii) Solve the set of equations formed in (i) by matrix method.

TYPE 26 Word Problems on System of Linear Equations

★ **Likely Exam Question**

Given: Let x, y, z be the number of burgers, samosas, and juice bottles sold respectively

Concept / Formula Used: $AX = B \quad X = A^{-1}B = \frac{1}{|A|}(\text{adj } A)B$

Solution:

(i) Formulating simultaneous equations: 1. Total items sold:

$$x + y + z = 100$$

2. Burgers sold was thrice samosas sold:

$$x = 3y \implies x - 3y + 0z = 0$$

3. Samosas sold was 10 more than juice bottles sold:

$$y = z + 10 \implies 0x + y - z = 10$$

System of equations:

$$\begin{aligned} x + y + z &= 100 \\ x - 3y + 0z &= 0 \\ 0x + y - z &= 10 \end{aligned}$$

(ii) Solving by matrix method: Write in matrix form $AX = B$:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -3 & 0 \\ 0 & 1 & -1 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 100 \\ 0 \\ 10 \end{bmatrix}$$

Calculate $|A|$:

$$\begin{aligned} |A| &= 1(3 - 0) - 1(-1 - 0) + 1(1 - 0) \\ &= 1(3) - 1(-1) + 1(1) \\ &= 3 + 1 + 1 \\ &= 5 \neq 0 \end{aligned}$$

Find cofactors of matrix A :

$$\begin{aligned} C_{11} &= +(3 - 0) = 3, & C_{12} &= -(-1 - 0) = 1, & C_{13} &= +(1 - 0) = 1 \\ C_{21} &= -(-1 - 1) = 2, & C_{22} &= +(-1 - 0) = -1, & C_{23} &= -(1 - 0) = -1 \\ C_{31} &= +(0 - (-3)) = 3, & C_{32} &= -(0 - 1) = 1, & C_{33} &= +(-3 - 1) = -4 \end{aligned}$$

$$\text{adj}(A) = \begin{bmatrix} 3 & 2 & 3 \\ 1 & -1 & 1 \\ 1 & -1 & -4 \end{bmatrix}$$

Now, compute $X = A^{-1}B$:

$$\begin{aligned} \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \frac{1}{5} \begin{bmatrix} 3 & 2 & 3 \\ 1 & -1 & 1 \\ 1 & -1 & -4 \end{bmatrix} \begin{bmatrix} 100 \\ 0 \\ 10 \end{bmatrix} \\ &= \frac{1}{5} \begin{bmatrix} 3(100) + 2(0) + 3(10) \\ 1(100) - 1(0) + 1(10) \\ 1(100) - 1(0) - 4(10) \end{bmatrix} \\ &= \frac{1}{5} \begin{bmatrix} 300 + 0 + 30 \\ 100 - 0 + 10 \\ 100 - 0 - 40 \end{bmatrix} \\ &= \frac{1}{5} \begin{bmatrix} 330 \\ 110 \\ 60 \end{bmatrix} \\ &= \begin{bmatrix} 66 \\ 22 \\ 12 \end{bmatrix} \end{aligned}$$

Final Answer

(i) Equations:

$$x + y + z = 100$$

$$x - 3y = 0$$

$$y - z = 10$$

(ii) Solution: Burgers $x = 66$, Samosas $y = 22$, Juice bottles $z = 12$

Question 9

If $A = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 3 & 2 \\ 3 & -3 & -4 \end{bmatrix}$, find A^{-1} . Using A^{-1} , solve the system of linear equations:

$$x + 2y - 3z = -4$$

$$2x + 3y + 2z = 2$$

$$3x - 3y - 4z = 11$$

TYPE 27 Solving Linear Equations Using Matrix Inverse

★ Likely Exam Question

Given: $A = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 3 & 2 \\ 3 & -3 & -4 \end{bmatrix}$ and system equations $x + 2y - 3z = -4$, $2x + 3y + 2z = 2$, $3x - 3y - 4z = 11$

Concept / Formula Used: $A^{-1} = \frac{1}{|A|} \text{adj}(A)$ $X = A^{-1}B$

Solution:

First, calculate determinant $|A|$:

$$\begin{aligned} |A| &= 1(-12 - (-6)) - 2(-8 - 6) + (-3)(-6 - 9) \\ &= 1(-6) - 2(-14) - 3(-15) \\ &= -6 + 28 + 45 \\ &= 67 \neq 0 \end{aligned}$$

Find cofactors of matrix A :

$$\begin{aligned} C_{11} &= +(-12 - (-6)) = -6, & C_{12} &= -(-8 - 6) = 14, & C_{13} &= +(-6 - 9) = -15 \\ C_{21} &= -(-8 - 9) = 17, & C_{22} &= +(-4 - (-9)) = 5, & C_{23} &= -(-3 - 6) = 9 \\ C_{31} &= +(4 - (-9)) = 13, & C_{32} &= -(2 - (-6)) = -8, & C_{33} &= +(3 - 4) = -1 \end{aligned}$$

$$\text{adj}(A) = \begin{bmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{bmatrix}$$

Calculate A^{-1} :

$$A^{-1} = \frac{1}{67} \begin{bmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{bmatrix}$$

The given system of linear equations in matrix form is $AX = B$:

$$\begin{bmatrix} 1 & 2 & -3 \\ 2 & 3 & 2 \\ 3 & -3 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -4 \\ 2 \\ 11 \end{bmatrix}$$

Now, compute $X = A^{-1}B$:

$$\begin{aligned}\begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \frac{1}{67} \begin{bmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{bmatrix} \begin{bmatrix} -4 \\ 2 \\ 11 \end{bmatrix} \\ &= \frac{1}{67} \begin{bmatrix} -6(-4) + 17(2) + 13(11) \\ 14(-4) + 5(2) - 8(11) \\ -15(-4) + 9(2) - 1(11) \end{bmatrix} \\ &= \frac{1}{67} \begin{bmatrix} 24 + 34 + 143 \\ -56 + 10 - 88 \\ 60 + 18 - 11 \end{bmatrix} \\ &= \frac{1}{67} \begin{bmatrix} 201 \\ -134 \\ 67 \end{bmatrix} \\ &= \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}\end{aligned}$$

Final Answer

$$A^{-1} = \frac{1}{67} \begin{bmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{bmatrix}$$

$x = 3, \quad y = -2, \quad z = 1$

Question 10

If $A = \begin{bmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix}$, find A^{-1} . Using A^{-1} , solve the following system of equations:

$$2x - 3y + 5z = 11$$

$$3x + 2y - 4z = -5$$

$$x + y - 2z = -3$$

TYPE 27 Solving Linear Equations Using Matrix Inverse

★ Likely Exam Question

Given: $A = \begin{bmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix}$ and system equations $2x - 3y + 5z = 11, 3x + 2y - 4z = -5, x + y - 2z = -3$

Concept / Formula Used: $A^{-1} = \frac{1}{|A|} \text{adj}(A) \quad X = A^{-1}B$

Solution:

Calculate determinant $|A|$:

$$\begin{aligned}
 |A| &= 2(-4 - (-4)) - (-3)(-6 - (-4)) + 5(3 - 2) \\
 &= 2(0) + 3(-2) + 5(1) \\
 &= 0 - 6 + 5 \\
 &= -1 \neq 0
 \end{aligned}$$

Find cofactors of matrix A :

$$\begin{aligned}
 C_{11} &= +(-4 - (-4)) = 0, & C_{12} &= -(-6 - (-4)) = 2, & C_{13} &= +(3 - 2) = 1 \\
 C_{21} &= -(6 - 5) = -1, & C_{22} &= +(-4 - 5) = -9, & C_{23} &= -(2 - (-3)) = -5 \\
 C_{31} &= +(12 - 10) = 2, & C_{32} &= -(-8 - 15) = 23, & C_{33} &= +(4 - (-9)) = 13
 \end{aligned}$$

$$\text{adj}(A) = \begin{bmatrix} 0 & -1 & 2 \\ 2 & -9 & 23 \\ 1 & -5 & 13 \end{bmatrix}$$

Calculate A^{-1} :

$$A^{-1} = \frac{1}{-1} \begin{bmatrix} 0 & -1 & 2 \\ 2 & -9 & 23 \\ 1 & -5 & 13 \end{bmatrix} = \begin{bmatrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{bmatrix}$$

Write the system of equations in matrix form $AX = B$:

$$\begin{bmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 11 \\ -5 \\ -3 \end{bmatrix}$$

Now, compute $X = A^{-1}B$:

$$\begin{aligned}
 \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \begin{bmatrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{bmatrix} \begin{bmatrix} 11 \\ -5 \\ -3 \end{bmatrix} \\
 &= \begin{bmatrix} 0(11) + 1(-5) - 2(-3) \\ -2(11) + 9(-5) - 23(-3) \\ -1(11) + 5(-5) - 13(-3) \end{bmatrix} \\
 &= \begin{bmatrix} 0 - 5 + 6 \\ -22 - 45 + 69 \\ -11 - 25 + 39 \end{bmatrix} \\
 &= \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}
 \end{aligned}$$

Final Answer

$$A^{-1} = \begin{bmatrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{bmatrix}$$

$$x = 1, \quad y = 2, \quad z = 3$$

Question 11

If $A = \begin{bmatrix} 3 & 5 \\ -2 & 3 \end{bmatrix}$, find A^{-1} and use it to solve the system of equations:

$$3x - 2y = 7$$

$$5x + 3y = 1$$

TYPE 28 Solving Linear Equations Using Transpose Matrix Inverse

Given: $A = \begin{bmatrix} 3 & 5 \\ -2 & 3 \end{bmatrix}$ and system equations $3x - 2y = 7, 5x + 3y = 1$

Concept / Formula Used: $C = A^T \quad C^{-1} = (A^T)^{-1} = (A^{-1})^T$

Solution:

Calculate determinant $|A|$:

$$\begin{aligned} |A| &= (3)(3) - (5)(-2) \\ &= 9 + 10 \\ &= 19 \neq 0 \end{aligned}$$

Find cofactors of matrix A :

$$C_{11} = 3, \quad C_{12} = 2, \quad C_{21} = -5, \quad C_{22} = 3$$

$$\text{adj}(A) = \begin{bmatrix} 3 & -5 \\ 2 & 3 \end{bmatrix}$$

Calculate A^{-1} :

$$A^{-1} = \frac{1}{19} \begin{bmatrix} 3 & -5 \\ 2 & 3 \end{bmatrix}$$

Notice that the coefficient matrix C of the system $3x - 2y = 7, 5x + 3y = 1$ is:

$$C = \begin{bmatrix} 3 & -2 \\ 5 & 3 \end{bmatrix} = A^T$$

Using the property $(A^T)^{-1} = (A^{-1})^T$:

$$C^{-1} = (A^{-1})^T = \frac{1}{19} \begin{bmatrix} 3 & 2 \\ -5 & 3 \end{bmatrix}$$

Solve $CX = B$ where $B = \begin{bmatrix} 7 \\ 1 \end{bmatrix}$:

$$\begin{aligned} \begin{bmatrix} x \\ y \end{bmatrix} &= C^{-1}B \\ &= \frac{1}{19} \begin{bmatrix} 3 & 2 \\ -5 & 3 \end{bmatrix} \begin{bmatrix} 7 \\ 1 \end{bmatrix} \\ &= \frac{1}{19} \begin{bmatrix} 3(7) + 2(1) \\ -5(7) + 3(1) \end{bmatrix} \\ &= \frac{1}{19} \begin{bmatrix} 21 + 2 \\ -35 + 3 \end{bmatrix} \\ &= \frac{1}{19} \begin{bmatrix} 23 \\ -32 \end{bmatrix} \\ &= \begin{bmatrix} \frac{23}{19} \\ -\frac{32}{19} \end{bmatrix} \end{aligned}$$

Final Answer

$$\begin{aligned} A^{-1} &= \frac{1}{19} \begin{bmatrix} 3 & -5 \\ 2 & 3 \end{bmatrix} \\ x &= \frac{23}{19}, \quad y = -\frac{32}{19} \end{aligned}$$

Question 12

If $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix}$, find A^{-1} and use it to solve the system of equations:

$$x + 2y + z = 4$$

$$-x + y + z = 0$$

$$x - 3y + z = 2$$

TYPE 28 Solving Linear Equations Using Transpose Matrix Inverse

Given: $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix}$ and system equations $x + 2y + z = 4, -x + y + z = 0, x - 3y + z = 2$

Concept / Formula Used: $C = A^T \quad C^{-1} = (A^T)^{-1} = (A^{-1})^T$

Solution:

Calculate determinant $|A|$:

$$\begin{aligned}
 |A| &= 1(1 - (-3)) - (-1)(2 - (-3)) + 1(2 - 1) \\
 &= 1(4) + 1(5) + 1(1) \\
 &= 4 + 5 + 1 \\
 &= 10 \neq 0
 \end{aligned}$$

Find cofactors of matrix A :

$$\begin{aligned}
 C_{11} &= +(1 - (-3)) = 4, & C_{12} &= -(2 - (-3)) = -5, & C_{13} &= +(2 - 1) = 1 \\
 C_{21} &= -(-1 - 1) = 2, & C_{22} &= +(1 - 1) = 0, & C_{23} &= -(1 - (-1)) = -2 \\
 C_{31} &= +(3 - 1) = 2, & C_{32} &= -(-3 - 2) = 5, & C_{33} &= +(1 - (-2)) = 3
 \end{aligned}$$

$$\text{adj}(A) = \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{bmatrix}$$

Calculate A^{-1} :

$$A^{-1} = \frac{1}{10} \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{bmatrix}$$

Observe the coefficient matrix C of the given system:

$$C = \begin{bmatrix} 1 & 2 & 1 \\ -1 & 1 & 1 \\ 1 & -3 & 1 \end{bmatrix} = A^T$$

Therefore:

$$C^{-1} = (A^T)^{-1} = (A^{-1})^T = \frac{1}{10} \begin{bmatrix} 4 & -5 & 1 \\ 2 & 0 & -2 \\ 2 & 5 & 3 \end{bmatrix}$$

Now solve $CX = B$ where $B = \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}$:

$$\begin{aligned} \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \frac{1}{10} \begin{bmatrix} 4 & -5 & 1 \\ 2 & 0 & -2 \\ 2 & 5 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix} \\ &= \frac{1}{10} \begin{bmatrix} 4(4) - 5(0) + 1(2) \\ 2(4) + 0(0) - 2(2) \\ 2(4) + 5(0) + 3(2) \end{bmatrix} \\ &= \frac{1}{10} \begin{bmatrix} 16 + 0 + 2 \\ 8 + 0 - 4 \\ 8 + 0 + 6 \end{bmatrix} \\ &= \frac{1}{10} \begin{bmatrix} 18 \\ 4 \\ 14 \end{bmatrix} \\ &= \begin{bmatrix} \frac{9}{5} \\ \frac{2}{5} \\ \frac{7}{5} \end{bmatrix} \end{aligned}$$

Final Answer

$$A^{-1} = \frac{1}{10} \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{bmatrix}$$

$$x = \frac{9}{5}, \quad y = \frac{2}{5}, \quad z = \frac{7}{5}$$

Question 13

Given two matrices A and B where $A = \begin{bmatrix} 1 & -2 & 3 \\ 1 & 4 & 1 \\ 1 & -3 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 11 & -5 & -14 \\ -1 & -1 & 2 \\ -7 & 1 & 6 \end{bmatrix}$, find AB and use this result to solve the following system of equations:

$$x - 2y + 3z = 6$$

$$x + 4y + z = 12$$

$$x - 3y + 2z = 1$$

TYPE 29 Solving Linear Equations Using Matrix Product

★ Likely Exam Question

Given: $A = \begin{bmatrix} 1 & -2 & 3 \\ 1 & 4 & 1 \\ 1 & -3 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 11 & -5 & -14 \\ -1 & -1 & 2 \\ -7 & 1 & 6 \end{bmatrix}$

Concept / Formula Used: $AB = kI \quad A^{-1} = \frac{1}{k}B$

Solution:

First, compute the product AB :

$$\begin{aligned}
 AB &= \begin{bmatrix} 1 & -2 & 3 \\ 1 & 4 & 1 \\ 1 & -3 & 2 \end{bmatrix} \begin{bmatrix} 11 & -5 & -14 \\ -1 & -1 & 2 \\ -7 & 1 & 6 \end{bmatrix} \\
 &= \begin{bmatrix} 1(11) + (-2)(-1) + 3(-7) & 1(-5) + (-2)(-1) + 3(1) & 1(-14) + (-2)(2) + 3(6) \\ 1(11) + 4(-1) + 1(-7) & 1(-5) + 4(-1) + 1(1) & 1(-14) + 4(2) + 1(6) \\ 1(11) + (-3)(-1) + 2(-7) & 1(-5) + (-3)(-1) + 2(1) & 1(-14) + (-3)(2) + 2(6) \end{bmatrix} \\
 &= \begin{bmatrix} 11 + 2 - 21 & -5 + 2 + 3 & -14 - 4 + 18 \\ 11 - 4 - 7 & -5 - 4 + 1 & -14 + 8 + 6 \\ 11 + 3 - 14 & -5 + 3 + 2 & -14 - 6 + 12 \end{bmatrix} \\
 &= \begin{bmatrix} -8 & 0 & 0 \\ 0 & -8 & 0 \\ 0 & 0 & -8 \end{bmatrix} = -8I
 \end{aligned}$$

Since $AB = -8I$, we have:

$$A \left(-\frac{1}{8}B \right) = I \implies A^{-1} = -\frac{1}{8}B$$

Write the given system of linear equations in matrix form $AX = C$:

$$\begin{bmatrix} 1 & -2 & 3 \\ 1 & 4 & 1 \\ 1 & -3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 12 \\ 1 \end{bmatrix}$$

Now, compute $X = A^{-1}C = -\frac{1}{8}BC$:

$$\begin{aligned}\begin{bmatrix} x \\ y \\ z \end{bmatrix} &= -\frac{1}{8} \begin{bmatrix} 11 & -5 & -14 \\ -1 & -1 & 2 \\ -7 & 1 & 6 \end{bmatrix} \begin{bmatrix} 6 \\ 12 \\ 1 \end{bmatrix} \\ &= -\frac{1}{8} \begin{bmatrix} 11(6) - 5(12) - 14(1) \\ -1(6) - 1(12) + 2(1) \\ -7(6) + 1(12) + 6(1) \end{bmatrix} \\ &= -\frac{1}{8} \begin{bmatrix} 66 - 60 - 14 \\ -6 - 12 + 2 \\ -42 + 12 + 6 \end{bmatrix} \\ &= -\frac{1}{8} \begin{bmatrix} -8 \\ -16 \\ -24 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}\end{aligned}$$

Final Answer

$$AB = -8I$$

$$x = 1, \quad y = 2, \quad z = 3$$

Question 14

If $A = \begin{bmatrix} -3 & -2 & -4 \\ 2 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 0 \\ -2 & -1 & -2 \\ 0 & -1 & 1 \end{bmatrix}$, then find AB and use it to solve the following system of equations:

$$x - 2y = 3$$

$$2x - y - z = 2$$

$$-2y + z = 3$$

TYPE 29 Solving Linear Equations Using Matrix Product

★ Likely Exam Question

Given: $A = \begin{bmatrix} -3 & -2 & -4 \\ 2 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 & 0 \\ -2 & -1 & -2 \\ 0 & -1 & 1 \end{bmatrix}$, and system $x - 2y = 3$, $2x - y - z = 2$, $-2y + z = 3$.

Concept / Formula Used: $AB = I$ $B^{-1} = A$ $(B^T)^{-1} = A^T$ $CX = D$ $X = C^{-1}D$

Solution:

First, compute the matrix product AB :

$$\begin{aligned}
 AB &= \begin{bmatrix} -3 & -2 & -4 \\ 2 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ -2 & -1 & -2 \\ 0 & -1 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} (-3)(1) + (-2)(-2) + (-4)(0) & (-3)(2) + (-2)(-1) + (-4)(-1) & (-3)(0) + (-2)(-2) + (-4)(1) \\ (2)(1) + (1)(-2) + (2)(0) & (2)(2) + (1)(-1) + (2)(-1) & (2)(0) + (1)(-2) + (2)(1) \\ (2)(1) + (1)(-2) + (3)(0) & (2)(2) + (1)(-1) + (3)(-1) & (2)(0) + (1)(-2) + (3)(1) \end{bmatrix} \\
 &= \begin{bmatrix} -3 + 4 + 0 & -6 + 2 + 4 & 0 + 4 - 4 \\ 2 - 2 + 0 & 4 - 1 - 2 & 0 - 2 + 2 \\ 2 - 2 + 0 & 4 - 1 - 3 & 0 - 2 + 3 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I
 \end{aligned}$$

Since $AB = I$, by definition of matrix inverse:

$$B^{-1} = A$$

Now, write the given system of linear equations in matrix form $CX = D$:

$$\begin{bmatrix} 1 & -2 & 0 \\ 2 & -1 & -1 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 3 \end{bmatrix}$$

Observe the coefficient matrix C :

$$C = \begin{bmatrix} 1 & -2 & 0 \\ 2 & -1 & -1 \\ 0 & -2 & 1 \end{bmatrix}$$

Comparing matrix C with matrix B :

$$B^T = \begin{bmatrix} 1 & 2 & 0 \\ -2 & -1 & -2 \\ 0 & -1 & 1 \end{bmatrix}^T = \begin{bmatrix} 1 & -2 & 0 \\ 2 & -1 & -1 \\ 0 & -2 & 1 \end{bmatrix} = C$$

Thus, $C = B^T$.

Using the identity $(B^T)^{-1} = (B^{-1})^T$:

$$C^{-1} = (B^T)^{-1} = (B^{-1})^T = A^T$$

Taking the transpose of matrix A :

$$A^T = \begin{bmatrix} -3 & 2 & 2 \\ -2 & 1 & 1 \\ -4 & 2 & 3 \end{bmatrix}$$

The solution vector X is given by $X = C^{-1}D = A^T D$:

$$\begin{aligned} \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \begin{bmatrix} -3 & 2 & 2 \\ -2 & 1 & 1 \\ -4 & 2 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 3 \end{bmatrix} \\ &= \begin{bmatrix} (-3)(3) + 2(2) + 2(3) \\ (-2)(3) + 1(2) + 1(3) \\ (-4)(3) + 2(2) + 3(3) \end{bmatrix} \\ &= \begin{bmatrix} -9 + 4 + 6 \\ -6 + 2 + 3 \\ -12 + 4 + 9 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \end{aligned}$$

Equating corresponding elements:

$$x = 1, \quad y = -1, \quad z = 1$$

Final Answer

$$AB = I; \quad x = 1, \quad y = -1, \quad z = 1$$

Common Mistake: Confusing the coefficient matrix C with B directly instead of recognizing $C = B^T$, which requires using $C^{-1} = A^T$ rather than A .

Question 15 (i)

Using matrices, solve the following system of homogeneous equations:

$$5x + 5y + 2z = 0$$

$$2x + 5y + 4z = 0$$

$$4x + 5y + 2z = 0$$

TYPE 30 Solving System of Homogeneous Linear Equations

Given: System of homogeneous equations: $5x + 5y + 2z = 0$, $2x + 5y + 4z = 0$, and $4x + 5y + 2z = 0$.

Concept / Formula Used: $AX = O \quad x = 0, y = 0, z = 0 \quad |A| \neq 0$

Solution:

Expressing the given system in matrix form $AX = O$:

$$\begin{bmatrix} 5 & 5 & 2 \\ 2 & 5 & 4 \\ 4 & 5 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Where the coefficient matrix is:

$$A = \begin{bmatrix} 5 & 5 & 2 \\ 2 & 5 & 4 \\ 4 & 5 & 2 \end{bmatrix}$$

Evaluating the determinant of A :

$$\begin{aligned} |A| &= 5 \begin{vmatrix} 5 & 4 \\ 5 & 2 \end{vmatrix} - 5 \begin{vmatrix} 2 & 4 \\ 4 & 2 \end{vmatrix} + 2 \begin{vmatrix} 2 & 5 \\ 4 & 5 \end{vmatrix} \\ &= 5[(5)(2) - (4)(5)] - 5[(2)(2) - (4)(4)] + 2[(2)(5) - (5)(4)] \\ &= 5[10 - 20] - 5[4 - 16] + 2[10 - 20] \\ &= 5(-10) - 5(-12) + 2(-10) \\ &= -50 + 60 - 20 \\ &= -10 \end{aligned}$$

Since $|A| = -10 \neq 0$, the coefficient matrix A is non-singular.

Using the standard result that if $|A| \neq 0$, the system $AX = O$ has only the trivial solution:

$$x = 0, \quad y = 0, \quad z = 0$$

Final Answer

$$x = 0, \quad y = 0, \quad z = 0$$

Question 15 (ii)

Using matrices, solve the following system of homogeneous equations:

$$3x + y - 2z = 0$$

$$x + y + z = 0$$

$$x - 2y + z = 0$$

TYPE 30 Solving System of Homogeneous Linear Equations

Given: System of homogeneous equations: $3x + y - 2z = 0$, $x + y + z = 0$, and $x - 2y + z = 0$.

Concept / Formula Used: $AX = O \quad x = 0, y = 0, z = 0 \quad |A| \neq 0$

Solution:

Expressing the given system in matrix form $AX = O$:

$$\begin{bmatrix} 3 & 1 & -2 \\ 1 & 1 & 1 \\ 1 & -2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Where the coefficient matrix is:

$$A = \begin{bmatrix} 3 & 1 & -2 \\ 1 & 1 & 1 \\ 1 & -2 & 1 \end{bmatrix}$$

Evaluating the determinant of A :

$$\begin{aligned} |A| &= 3 \begin{vmatrix} 1 & 1 \\ -2 & 1 \end{vmatrix} - 1 \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} + (-2) \begin{vmatrix} 1 & 1 \\ 1 & -2 \end{vmatrix} \\ &= 3[(1)(1) - (1)(-2)] - 1[(1)(1) - (1)(1)] - 2[(1)(-2) - (1)(1)] \\ &= 3[1 + 2] - 1[1 - 1] - 2[-2 - 1] \\ &= 3(3) - 1(0) - 2(-3) \\ &= 9 - 0 + 6 \\ &= 15 \end{aligned}$$

Since $|A| = 15 \neq 0$, the coefficient matrix A is non-singular.

Using the standard result that if $|A| \neq 0$, the system $AX = O$ has only the trivial solution:

$$x = 0, \quad y = 0, \quad z = 0$$

Final Answer

$$x = 0, \quad y = 0, \quad z = 0$$

Question 16 (i)

Using matrices, solve the following system of homogeneous equations:

$$2x - 3y - z = 0$$

$$x + 3y - 2z = 0$$

$$x - 3y = 0$$

TYPE 30 Solving System of Homogeneous Linear Equations

★ Likely Exam Question

Given: System of homogeneous equations: $2x - 3y - z = 0$, $x + 3y - 2z = 0$, and $x - 3y + 0z = 0$.

Concept / Formula Used: $|A| = 0 \quad AX = O$

Solution:

Expressing the given system in matrix form $AX = O$:

$$\begin{bmatrix} 2 & -3 & -1 \\ 1 & 3 & -2 \\ 1 & -3 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Where the coefficient matrix is:

$$A = \begin{bmatrix} 2 & -3 & -1 \\ 1 & 3 & -2 \\ 1 & -3 & 0 \end{bmatrix}$$

Evaluating the determinant of A :

$$\begin{aligned} |A| &= 2 \begin{vmatrix} 3 & -2 \\ -3 & 0 \end{vmatrix} - (-3) \begin{vmatrix} 1 & -2 \\ 1 & 0 \end{vmatrix} + (-1) \begin{vmatrix} 1 & 3 \\ 1 & -3 \end{vmatrix} \\ &= 2[(3)(0) - (-2)(-3)] + 3[(1)(0) - (-2)(1)] - 1[(1)(-3) - (3)(1)] \\ &= 2[0 - 6] + 3[0 + 2] - 1[-3 - 3] \\ &= 2(-6) + 3(2) - 1(-6) \\ &= -12 + 6 + 6 \\ &= 0 \end{aligned}$$

Since $|A| = 0$, the system has infinitely many non-trivial solutions.

Let $z = k$, where k is any real parameter ($k \in \mathbb{R}$).

Substituting $z = k$ into the third equation $x - 3y = 0$:

$$x = 3y \quad \text{--- (Equation 1)}$$

Substituting $z = k$ into the second equation $x + 3y - 2z = 0$:

$$x + 3y = 2k \quad \text{--- (Equation 2)}$$

Substituting $x = 3y$ from Equation 1 into Equation 2:

$$3y + 3y = 2k$$

$$6y = 2k$$

$$y = \frac{2k}{6}$$

$$y = \frac{1}{3}k$$

Now substitute $y = \frac{1}{3}k$ back into Equation 1:

$$x = 3 \left(\frac{1}{3}k \right)$$

$$x = k$$

Check these values in the first equation $2x - 3y - z = 0$:

$$2(k) - 3 \left(\frac{1}{3}k \right) - k = 2k - k - k = 0$$

This satisfies all equations.

Final Answer

$$x = k, y = \frac{1}{3}k, z = k, \text{ where } k \text{ is any real number.}$$

Common Mistake: Stopping after finding $|A| = 0$ and stating 'no solution' is incorrect; homogeneous systems with $|A| = 0$ always have infinitely many non-trivial solutions.

Question 16 (ii)

Using matrices, solve the following system of homogeneous equations:

$$x + y - z = 0$$

$$x - 2y + z = 0$$

$$3x + 6y - 5z = 0$$

TYPE 30 Solving System of Homogeneous Linear Equations

Given: System of homogeneous equations: $x + y - z = 0$, $x - 2y + z = 0$, and $3x + 6y - 5z = 0$.

Concept / Formula Used: $|A| = 0$ $AX = O$

Solution:

Expressing the given system in matrix form $AX = O$:

$$\begin{bmatrix} 1 & 1 & -1 \\ 1 & -2 & 1 \\ 3 & 6 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Where the coefficient matrix is:

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 1 & -2 & 1 \\ 3 & 6 & -5 \end{bmatrix}$$

Evaluating the determinant of A :

$$\begin{aligned} |A| &= 1 \begin{vmatrix} -2 & 1 \\ 6 & -5 \end{vmatrix} - 1 \begin{vmatrix} 1 & 1 \\ 3 & -5 \end{vmatrix} + (-1) \begin{vmatrix} 1 & -2 \\ 3 & 6 \end{vmatrix} \\ &= 1[(-2)(-5) - (1)(6)] - 1[(1)(-5) - (1)(3)] - 1[(1)(6) - (-2)(3)] \\ &= 1[10 - 6] - 1[-5 - 3] - 1[6 + 6] \\ &= 1(4) - 1(-8) - 1(12) \\ &= 4 + 8 - 12 \\ &= 0 \end{aligned}$$

Since $|A| = 0$, the system has infinitely many non-trivial solutions.

Let $z = k$, where k is any real parameter ($k \in \mathbb{R}$).

Substitute $z = k$ into the first two equations:

$$x + y = k \quad \text{--- (Equation 1)}$$

$$x - 2y = -k \quad \text{--- (Equation 2)}$$

Subtracting Equation 2 from Equation 1:

$$(x + y) - (x - 2y) = k - (-k)$$

$$x + y - x + 2y = k + k$$

$$3y = 2k$$

$$y = \frac{2}{3}k$$

Substituting $y = \frac{2}{3}k$ into Equation 1:

$$x + \frac{2}{3}k = k$$

$$x = k - \frac{2}{3}k$$

$$x = \frac{1}{3}k$$

Check these values in the third equation $3x + 6y - 5z = 0$:

$$3\left(\frac{1}{3}k\right) + 6\left(\frac{2}{3}k\right) - 5k = k + 4k - 5k = 0$$

This satisfies all equations.

Final Answer

$x = \frac{1}{3}k$, $y = \frac{2}{3}k$, $z = k$, where k is any real number.